

Generalized Hierarchical Bayesian Segmentation with Irregular Designs, Multi-Sequence Hierarchies, and Grouped/Latent-Group Designs

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Bayesian change-point and segmentation models provide uncertainty-aware piecewise-constant representations of ordered data, but exact inference is often tied to narrow likelihood classes, single-sequence settings, or index-uniform designs. We present **BayesBreak**^a, a modular offline Bayesian segmentation framework built around a simple separation: each candidate block contributes a marginal likelihood and any required moment numerators, and a global dynamic program combines those block scores into posterior quantities over segment counts, boundary locations, and latent signals. For weighted exponential-family likelihoods with conjugate priors and a segment-factorized partition prior, block evidences and posterior moments are available in closed form from cumulative sufficient statistics, yielding exact sum-product inference for $p(y \mid k)$, $p(k \mid y)$, boundary marginals, and Bayes regression curves. We also distinguish these quantities from the *joint* MAP segmentation, which is recovered by a separate max-sum backtracking recursion.

The framework extends naturally to design-aware partition priors for irregularly spaced observations, to exact boundary-posterior pooling across replicates with shared boundaries, and to a latent-template mixture for unknown group membership with exact coordinate updates for the stated template-mixture objective. For non-conjugate GLM blocks, the same DP layer is reused with deterministic local approximations such as Laplace, variational (including Jaakkola–Jordan and Pólya–Gamma mean-field), EP, or one-dimensional quadrature block routines; conditional on a uniform per-block log-evidence error ε , we prove a posterior-odds stability bound perturbing k -odds and boundary-odds on the segmentation DP by at most $(k + k')\varepsilon$ and $2k\varepsilon$ respectively. At prediction time, BayesBreak defines posterior-predictive scoring interfaces for new sequences or set-valued units under exported segmentations or Bayes curves. Empirically, the paper reports a completed synthetic recovery, calibration, and scaling suite that validates the core implementation; four real-data case studies (well-log geology, array-CGH copy number, equity-return volatility, and CpG-atlas methylation) are then specified as planned empirical evaluations, with placeholder figures and tables and reproducibility pipelines that the finalized runs populate.

Keywords: Bayesian segmentation; change-point detection; dynamic programming;

product partitions; hierarchical models.

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^aReference implementation: <https://github.com/osolari/bayesbreak>.



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1 Introduction

Piecewise-constant segmentation—also called multiple change-point inference—is a core problem in statistics, signal processing, econometrics, computational biology, and machine learning. In its simplest form, one observes an ordered sequence $y_1, \dots, y_n$ (optionally at locations $x_1 < \dots < x_n$) and seeks to infer the number of regimes, the boundary locations, and the segment-specific parameters governing the observations inside each regime. The appeal of Bayesian formulations is well known: they quantify uncertainty over boundary locations and segment counts, they support principled model comparison through marginal likelihoods, and they naturally propagate uncertainty to downstream tasks such as denoising, regression-curve estimation, and prediction (Barry and Hartigan, 1992, 1993; Fearnhead, 2006; Hutter, 2007).

For offline segmentation, exact Bayesian inference applies whenever single-segment parameters can be integrated out and the partition prior factorizes appropriately. In that case, one can precompute block evidences for all candidate segments and then perform dynamic programming (DP) over contiguous partitions to obtain posterior quantities in polynomial time (Fearnhead, 2006; Hutter, 2007). This observation makes piecewise-constant Bayesian regression unusually modular: once blockwise marginal likelihoods are available, the same global DP machinery delivers evidences, posterior boundary probabilities, Bayes regression curves, and joint MAP segmentations.

Despite this favorable structure, the literature remains fragmented. Exact offline Bayesian treatments are typically developed for one or two concrete likelihood/prior pairs at a time—most commonly Gaussian- or Poisson-conjugate blocks on index-uniform grids—and the corresponding software seldom exposes a clean abstraction over the local block-evidence module. Important exceptions exist: dependence across segments (Fearnhead and Liu, 2011), multi-sequence synchrony (Fan and Mackey, 2017), and multivariate joint partitions (Quinlan et al., 2024) have all been studied. Even so, hierarchical or multi-sequence extensions tend to be specialized to particular data types, based on approximate sampling schemes, or focused on synchrony assumptions that are difficult to combine with a reusable block-evidence interface (Carlin et al., 1992; Fearnhead and Liu, 2011; Fan and Mackey, 2017; Quinlan et al., 2024). At the same time, many modern applications require more than boundary recovery: they need predictive scores for new sequences, group-level pooling, and transparent uncertainty summaries that can be exported to downstream decision pipelines.

This paper develops **BayesBreak**, a unified framework for offline Bayesian segmentation that separates *local block modeling* from *global partition inference*. The framework is exact whenever block evidences are available in closed form and extends to controlled deterministic approximations when the block integral lacks a closed form. The central design principle is simple: if a candidate segment $(i, j]$ can be summarized by sufficient statistics and assigned a block score $A_{ij}^{(0)}$ together with any required moment numerators, then the same DP machinery can be reused across observation models, design priors, and several structured extensions.

The paper uses an explicit status convention. The conjugate-block and DP results are exact mathematical results under their hypotheses; non-conjugate results are exact for the approximate block scores and conditional on stated block-error controls; the synthetic experiments are completed archived validations; and the four real-data examples are planned empiri-

cal evaluations whose visible placeholder figures and tables are populated by the finalized pipelines.

Contributions. The main contributions are as follows.

- **A reusable exponential-family block interface.** For exposure-aware or precision-aware exponential-family models with conjugate priors, we derive closed-form block evidences and posterior moments from cumulative sufficient statistics. This recovers familiar Gaussian, Poisson, and Beta–Binomial formulas as special cases, and isolates all model-specific work inside a block routine.
- **Exact offline Bayesian inference with coherent posterior summaries.** We derive forward/backward DP recursions for marginal evidences, posterior segment-count probabilities, boundary marginals, and Bayes regression curves. We also distinguish these sum-product quantities from the *joint* MAP segmentation, which is recovered by a separate max-sum/backtracking recursion rather than by maximizing boundary marginals independently.
- **Irregular designs via design-aware partition priors.** Irregular spacing is handled at the partition-prior level through segment-length factors $g(\Delta_x(t_{q-1}, t_q))$, where Δ_x denotes the physical length between candidate boundary coordinates. We make explicit that physical spacing and exposure weighting are conceptually distinct: irregular designs do not, by themselves, force likelihood weights, but segment-length priors provide a principled way to encode design geometry.
- **Exact pooling for shared-boundary replicates and known groups.** When multiple sequences share boundaries but retain sequence-specific segment parameters, and given conditional independence across subjects conditional on the shared boundaries, subject-level block evidences multiply and the pooled posterior remains exactly tractable under the same DP recursions. The same construction applies groupwise when memberships are known.
- **A latent-group template mixture with exact coordinate updates.** For unknown group membership, we develop a mixture model in which each group carries a latent *template segmentation*. Responsibilities are updated in the E-step, while the M-step solves an exact responsibility-weighted max-sum DP to update each group template for the stated finite-template objective. This yields a rigorous alternating optimization scheme for the stated finite-template objective, distinct from full Bayesian model averaging over latent-group segmentations.
- **Approximate non-conjugate block routines with stability guarantees.** For non-conjugate GLM blocks we describe Laplace, variational (Jaakkola–Jordan and Pólya–Gamma mean-field), EP, and one-dimensional quadrature approximations. We then show how a verified uniform block-level log-evidence error ε (taken over all candidate blocks reachable by any k -segmentation with $k \leq k_{\max}$) propagates through the DP to perturb global log-evidence and posterior odds in a controlled way.
- **A prediction layer for fitted segmenters.** We formalize posterior-predictive scoring for new pointwise sequences, set-valued units, and vector-valued responses, distinguish-

ing between conditional prediction under an exported segmentation and more expensive resegmentation-based scoring.

1.1 Related work and literature review

Frequentist offline segmentation. Classical segmentation algorithms optimize additive cost functions over contiguous partitions. Segment-neighborhood and optimal-partitioning recursions provide exact solutions for a fixed number of segments or a fixed penalty (Auger and Lawrence, 1989; Jackson et al., 2005). Later work reduced practical cost through pruning: pruned dynamic programming and PELT (Rigaill, 2010; Killick et al., 2012) prune candidate changepoints, with PELT attaining linear *expected* cost under a changepoint-density condition while retaining the $\mathcal{O}(n^2)$ worst case, and functional-pruning algorithms such as FPOP and SNIP (Maidstone et al., 2017) prune in the cost-function space. Convex relaxations such as total variation denoising and the fused lasso trade exact combinatorial optimization for scalable regularized estimation, with multisequence extensions such as the group fused lasso encouraging shared boundaries across signals (Rudin et al., 1992; Tibshirani et al., 2005; Bleakley and Vert, 2011). Multiscale methods, including SMUCE and wild binary segmentation, emphasize detection guarantees and confidence statements across scales rather than full posterior modeling (Frick et al., 2014; Fryzlewicz, 2014). Circular binary segmentation (Olshen et al., 2004) is a widely used recursive detection rule in array-based genomic applications and provides a natural point of comparison for the Gaussian-block instance of BayesBreak. These methods are indispensable baselines, but they do not natively return posterior probabilities over partitions or segment parameters.

Bayesian offline segmentation and product partitions. The product-partition formulation of Barry and Hartigan (1992, 1993), building on the abstract partition-model framework of Hartigan (1990), is one of the most important starting points for Bayesian change-point analysis. The key idea is that if partition probabilities factorize over segments and segment parameters can be marginalized, then posterior inference can be reduced to operations on block scores. Chib (1998) recast multiple change-point inference through a hidden-Markov state representation, an intermediate formulation that remains a common Bayesian baseline. Fearnhead (2006) gave an exact and efficient DP-based treatment for a broad class of offline changepoint models, while Hutter (2007) showed how Bayesian averaging over partitions yields regression curves, boundary probabilities, and evidence-based model selection in the piecewise-constant setting. Wyse et al. (2011) extended the exact-DP viewpoint to models with within-segment dependence. Earlier work by Yao (1988) studied changepoint estimation via Schwarz’s criterion, complementing the evidence-based Bayesian model-selection view. The astronomical *Bayesian Blocks* construction of Scargle et al. (2013) is essentially a Poisson-block instance of the product-partition program and illustrates how domain-specific block routines combine with a single DP recursion. Work by Fearnhead and Liu (2011) and Ruggieri and Lawrence (2014) demonstrated how far the DP viewpoint can be pushed when one adds dependence across segments or richer within-segment regressions. Punskeya et al. (2002) applied a related MCMC-based Bayesian curve-fitting program to noisy signals.

Reviews and surveys. A selective survey of modern offline change-point methods, including penalized, Bayesian, nonparametric, and multiscale variants, is given by [Truong et al. \(2020\)](#); we view BayesBreak as sitting inside the Bayesian exact-offline cell of that taxonomy, broadened across block models.

Bayesian online and approximate methods. When streaming updates are required, Bayesian online changepoint detection provides an online filtering alternative that computes an exact run-length posterior rather than an exact offline partition posterior ([Adams and MacKay, 2007](#)). When exact marginalization is unavailable, reversible-jump MCMC and related sampling methods remain central Bayesian tools ([Green, 1995](#); [Denison et al., 1998](#)). BayesBreak is aimed squarely at the offline regime: we prioritize reusable block marginalization and exact DP whenever the model permits it, and use deterministic local approximations only when the block integral itself lacks a closed form.

Multi-sequence and structured Bayesian segmentation. Many applications exhibit boundary sharing or boundary dependence across related sequences. Hierarchical Bayesian changepoint models appear in early work by [Carlin et al. \(1992\)](#), while more recent multisequence methods such as BASIC exploit cross-sequence co-occurrence of changepoints through hierarchical priors ([Fan and Mackey, 2017](#)). Joint random partition models provide a broader Bayesian nonparametric framework for borrowing information across multiple ordered processes ([Quinlan et al., 2024](#)). Related product-partition literature also studies covariate-aware random partitions and generalized cohesion functions, which are conceptually close to the design-aware partition priors used here ([Müller et al., 2011](#); [Park and Dunson, 2010](#)).

Positioning of BayesBreak. BayesBreak occupies a modular middle ground in this landscape: it retains exact offline DP for the core single-sequence and shared-boundary pooled settings, broadens the menu of admissible block models, clarifies how to incorporate irregular designs through the prior rather than through ad hoc weighting, and exports a prediction interface that is usually missing from segmentation papers. For latent groups, BayesBreak uses a template-mixture extension whose EM updates are exact for the stated finite objective. The closest modular comparators are general-purpose segmentation toolkits such as the `ruptures` Python library ([Truong et al., 2018](#)) and the `changepoint` R package ([Killick and Eckley, 2014](#)), which expose a cost-function-plus-search abstraction; these are optimization-based and frequentist, whereas BayesBreak targets exact marginal-likelihood block evidences within a Bayesian exact-DP engine.

Annotated review. A compact annotated literature review, organized by methodological theme and relation to BayesBreak, is provided in [Appendix A.7](#).

BayesBreak scope and guarantees. *Exactness:* given a conjugate exponential-family block and a segment-factorized partition prior, the DP delivers exact posteriors $p(k \mid y)$, exact boundary marginals $p(t_p = h \mid y, k)$, exact Bayes regression-curve moments conditional on k , and the exact joint MAP segmentation by max-sum backtracking. *Monotone optimization:* for latent groups, EM updates are monotone for the stated finite template-mixture objective under the exact coordinate updates and tie-handling conventions in Section 10. *Stability:* for non-conjugate blocks, a verified uniform per-block log-evidence error ε perturbs segment-count and boundary posterior odds by at most $(k + k')\varepsilon$ and $2k\varepsilon$ respectively. *Scope:* the framework provides exact fully Bayesian inference for the stated conjugate DP models, exact coordinate updates for the finite latent-template objective, and block-error-based stability guarantees for approximations. Latent-template optima are coordinatewise within each restart, and approximation guarantees rely on verified small block errors.

Code and reproduction. A reference implementation of BayesBreak, including the block routines, DP layer, and the reproduction pipelines for the four real-data case studies, is available at <https://github.com/osolari/bayesbreak>. Installation, version pinning, end-to-end commands, and the verification steps for populating the placeholder real-data outputs are collected in Appendix A.12.

Limitations and manuscript-development status. BayesBreak is an offline framework, and its exact guarantees are scoped accordingly. The conjugate-block DP results are exact under the stated product-partition prior; the non-conjugate GLM extension reuses the same DP layer applied to a surrogate block-score matrix, with end-to-end behavior controlled by the local approximation through the posterior-odds stability results of Section 12 rather than by an unconditional guarantee. The latent-template extension optimizes a finite template-mixture objective and inherits standard EM caveats around restart-dependent local optima. The four real-data case studies in Section 15.9 are planned validation pipelines: their placeholder artifacts define required outputs rather than completed empirical findings, and they are populated only after source paths, preprocessing details, baselines, and generated figures and tables have been finalized.

Paper organization. Section 2 formalizes the inferential targets. Section 3 introduces notation, and Section 4 states the Bayesian segmentation model and the class of partition priors considered. Section 5 derives block evidences for conjugate exponential-family models. Section 6 develops exact DP inference, including boundary marginals, Bayes regression curves, and the distinction between sum-product posterior summaries and max-sum MAP segmentations. Section 7 studies design-aware priors, while Sections 8 and 9 treat shared-boundary pooling and known groups. Section 10 develops the latent-group template mixture. Section 11 gives closed-form block formulas for common conjugate families, and Section 12 collects non-conjugate approximations and stability results. Section 13 formalizes prediction for new data. Section 14 collects implementation-centric pseudocode, complexity analysis, and numerical-stability notes complementing the DP theory. Section 15 presents the com-

pleted synthetic validation together with four planned real-data case studies (well-log geology, array-CGH copy number, equity-return volatility, and CpG-atlas methylation) whose figures and tables document the planned outputs, and records the additional external experiments for a larger benchmarking study. We conclude in Section 16.

2 Problem formulation and inferential targets

BayesBreak addresses *offline* Bayesian segmentation of ordered data. Throughout, the fundamental unknown is a contiguous partition of the index set $\{1, \dots, n\}$ together with a segment parameter attached to each block. This section states the objects of interest in a model-agnostic way; Section 3 introduces the detailed notation used later.

Unless stated otherwise, the standing assumptions are: observations are ordered on a fixed set of candidate boundary locations; the admissible segment-count set is finite; block evidences are finite and positive on admissible blocks and are represented as zero or $-\infty$ on inadmissible blocks; segment parameters are conditionally independent across blocks given a segmentation; and the conditional partition prior factorizes over segments as stated in Assumption 4.1.

2.1 Single-sequence segmentation and Bayes regression

We observe ordered pairs $\{(x_i, y_i)\}_{i=1}^n$ with strictly increasing design points $x_1 < \dots < x_n$. The design can be irregular. When the observation model includes known exposures, trial counts, or precision weights, we also allow deterministic nonnegative quantities w_i ; unless explicitly stated otherwise, point-observation models use $w_i \equiv 1$.

A segmentation into k contiguous blocks is represented by a boundary vector

$$0 = t_0 < t_1 < \dots < t_{k-1} < t_k = n, \quad (1)$$

where block $q \in \{1, \dots, k\}$ is the index set

$$I_q(t) := \{t_{q-1} + 1, \dots, t_q\}. \quad (2)$$

The corresponding segment parameter is denoted $\theta_q \in \Theta$. Conditionally on $(k, t, \theta_{1:k})$, observations factorize across blocks:

$$p(y_{1:n} \mid k, t, \theta_{1:k}) = \prod_{q=1}^k \prod_{i \in I_q(t)} p(y_i \mid \theta_q, w_i), \quad (3)$$

where $p(y_i \mid \theta_q, w_i)$ is interpreted according to the chosen family. For example, depending on the family, w_i encodes exposure in Poisson models, trial count in Binomial models, or known precision in Gaussian models. Irregular spacing by itself does not force a non-unit w_i ; design geometry can instead enter through the partition prior.

For any candidate block $(i, j]$ with $0 \leq i < j \leq n$, the key primitive is the integrated single-block evidence

$$\mathcal{L}(i, j; \alpha_0, \beta_0) := p(y_{i+1:j} \mid \alpha_0, \beta_0) = \int \prod_{t=i+1}^j p(y_t \mid \theta, w_t) p(\theta \mid \alpha_0, \beta_0) d\theta, \quad (4)$$

where $p(\theta \mid \alpha_0, \beta_0)$ is the segment prior with hyperparameters (α_0, β_0) (see Section 3). When $\mathcal{L}(i, j)$ is available in closed form, the entire offline posterior can be computed exactly by dynamic programming. When it is only approximated, BayesBreak isolates the approximation at the block level and reuses the same global DP.

The inferential targets are:

- (i) the marginal evidence at fixed segment count,

$$p(y_{1:n} \mid k) = \sum_{t \in \mathcal{T}_k} p(t \mid k) \prod_{q=1}^k \mathcal{L}(t_{q-1}, t_q), \quad (5)$$

where $\mathcal{T}_k := \{(t_0, \dots, t_k) : 0 = t_0 < t_1 < \dots < t_k = n\}$ is the set of all admissible k -segment boundary vectors;

- (ii) the posterior over segment counts and boundaries, $p(k, t \mid y_{1:n})$, together with marginals such as $p(k \mid y_{1:n})$, $p(t_p = h \mid y_{1:n}, k)$, and the *boundary-event marginal* $p(b_i = 1 \mid y_{1:n}, k) = \sum_{p=1}^{k-1} p(t_p = i \mid y_{1:n}, k)$ that interior index i is a boundary (this latter object is what calibration experiments in Section 15 target);
- (iii) posterior summaries of the latent piecewise-constant signal. Writing θ_i for the parameter of the random segment containing observation i , we are interested in quantities such as

$$p(\theta_i \mid y_{1:n}), \quad \mathbb{E}[m(\theta_i) \mid y_{1:n}], \quad (6)$$

where $m(\theta)$ denotes the observation-scale mean parameter. These yield Bayes regression curves and uncertainty bands. We distinguish curves marginalized over k from curves conditional on $k = \hat{k}$; unless stated otherwise, the latter is the default and is what Section 6 computes;

- (iv) point estimators for downstream use. BayesBreak distinguishes *joint* MAP segmentations

$$\hat{t}^{\text{MAP}}(k) \in \arg \max_{t \in \mathcal{T}_k} p(t \mid y_{1:n}, k), \quad (7)$$

from *marginal* boundary summaries such as $\arg \max_h p(t_p = h \mid y_{1:n}, k)$. These are not generally the same object and must not be conflated. The fixed- k MAP objective uses the same length-absorbed block scores as the sum-product DP. Count-prior and normalizer offsets such as $\log p(k) - \log C_k$ enter only when we also optimize over k .

2.2 Multi-sequence structure

Many applications provide multiple related sequences observed on a common ordered grid or on a common set of candidate boundary locations. Let sequence $s \in \{1, \dots, S\}$ be denoted by $y_{1:n}^{(s)}$. BayesBreak treats three progressively richer settings.

Shared-boundary replicates. All sequences share a common segmentation t , but each sequence has its own within-block parameters $\theta_{1:k}^{(s)}$. Conditional on t , the sequences are inde-

pendent:

$$p(\{y^{(s)}\}_{s=1}^S \mid t, \{\theta^{(s)}\}) = \prod_{s=1}^S \prod_{q=1}^k \prod_{i \in I_q(t)} p(y_i^{(s)} \mid \theta_q^{(s)}, w_i^{(s)}). \quad (8)$$

Integrating out the subject-specific segment parameters yields pooled block evidences obtained by multiplying subject-level block evidences.

Known groups. Sequences are partitioned into known groups $g \in \{1, \dots, G\}$. Each group has its own segmentation, while sequences within a group share that segmentation and retain their own segment parameters.

Latent groups. Group labels are unknown. A fully Bayesian formulation averages over both assignments and group-specific segmentations. BayesBreak adopts a tractable *template-mixture* formulation: each group is associated with a single latent boundary template, and EM alternates between soft assignments (exact responsibilities for the current templates) and exact responsibility-weighted max-sum DP updates of those templates. The finite-template objective is monotone non-decreasing under the exact coordinate updates (Theorem 10.1); this is a finite-template optimization model rather than exact Bayesian averaging over latent-group segmentations.

2.3 Computational scope

For fixed $k_{\max}$, the core BayesBreak workflow consists of:

- (a) precomputing all block evidences and required moment numerators;
- (b) running sum-product DP recursions for evidences and posterior marginals; and
- (c) optionally running a max-sum/backtracking recursion for a joint MAP segmentation.

When block evaluation is constant time after prefix-sum preprocessing, this yields $\mathcal{O}(n^2)$ block precomputation and $\mathcal{O}(k_{\max} n^2)$ DP time. These worst-case costs make the exact method most natural for moderate sequence lengths (hundreds to low thousands), where exact posterior quantities are still computationally feasible and often worth the cost. Prediction for new data reuses the exported posterior summaries at negligible additional cost, except for the optional resegmentation scoring mode of Section 13, which reruns the DP on the new data at $\mathcal{O}(k_{\max} m^2)$ per group.

2.4 Prediction as a first-class inferential target

Beyond posterior inference on the training sequence(s), BayesBreak treats *prediction for new data* as an additional inferential target at the same level as evidence, posteriors, and point estimators. Given a fitted model $\mathcal{M}_g$ (one per group) and new observations $(X^{\text{new}}, y^{\text{new}})$, the relevant quantities are (i) group-conditional posterior-predictive likelihoods $p(y^{\text{new}} \mid \mathcal{M}_g)$, (ii) group membership posteriors $p(g \mid y^{\text{new}})$, and (iii) signal predictions $\hat{f}_g^{\text{MAP}}(x^*)$ or $\hat{f}_g^{\text{Bayes}}(x^*)$ at query points x^* . Section 13 develops the prediction layer in detail, covering pointwise,

set-valued, and vector-valued new data, and separates exported-segmentation scoring from optional full resegmentation of new data.

3 Notation and symbols

This section fixes notation used throughout the paper. We use lower-case Roman letters for indices, reserve $t = (t_0, \dots, t_k)$ for boundary vectors, and adopt the half-open block notation $(i, j] := \{i + 1, \dots, j\}$ for $0 \leq i < j \leq n$.

For irregular designs, physical segment lengths are defined using candidate boundary coordinates $u_0 < u_1 < \dots < u_n$ rather than by implicitly referencing an undefined observation location x_0 . The physical length of block $(i, j]$ is $\Delta_x(i, j) := u_j - u_i$. On a regular index grid set $u_i = i$; for interval data the u_i are the observed interval endpoints. Point observations without meaningful physical interval endpoints use the index-uniform prior or provide an external left endpoint u_0 .

Conjugate-prior normalization convention. For exponential-family blocks we use the following (Diaconis–Ylvisaker) convention for the conjugate prior normalizer. For hyperparameters (α, β) ,

$$p(\theta \mid \alpha, \beta) = \exp\{\eta(\theta)^\top \alpha - \beta A(\theta) - \log Z(\alpha, \beta)\}, \quad Z(\alpha, \beta) := \int \exp\{\eta(\theta)^\top \alpha - \beta A(\theta)\} d\theta. \quad (9)$$

With this convention, Z is the integral itself (not its reciprocal) and block evidences take the ratio form $Z(\alpha_0 + S_{ij}, \beta_0 + W_{ij})/Z(\alpha_0, \beta_0)$ (Theorem 5.1 and Appendix A.1).

Symbol	Meaning
n	Number of observations on the ordered design.
x_i	Design point / location of observation y_i .
u_i	Candidate boundary coordinate used for physical segment lengths, with $u_0 < \dots < u_n$.
$\Delta_x(i, j)$	Physical length of block $(i, j]$, defined as $u_j - u_i$.
y_i	Observation at location x_i (single-sequence case).
w_i	Deterministic exposure, trial-count, or precision weight when required by the observation model. Unless otherwise stated, point-observation models use $w_i \equiv 1$.
$k; k_{\max}$	Number of segments; maximum segment count considered by the DP.
$t = (t_0, \dots, t_k)$	Boundary vector, with $0 = t_0 < t_1 < \dots < t_k = n$.
$I_q(t)$	Index set of segment q : $\{t_{q-1} + 1, \dots, t_q\}$.
$(i, j]$	Candidate block $\{i + 1, \dots, j\}$.
$T(y), \eta(\theta), A(\theta), h(y, w)$	Sufficient statistic, natural parameter map, log-partition function, and base measure (in weighted form) for the chosen exponential-family block model.

$m(\theta)$	Observation-scale mean parameter induced by θ , i.e. $m(\theta) = \mathbb{E}[Y \mid \theta]$.
(α, β)	Conjugate hyperparameters with normalizer $Z(\alpha, \beta)$ defined in (9).
(α_0, β_0)	Prior hyperparameters before observing block data.
S_{ij}	Block sufficient-statistic sum $\sum_{t=i+1}^j w_t T(y_t)$.
W_{ij}	Block exposure/weight sum $\sum_{t=i+1}^j w_t$.
H_{ij}	Block base-measure sum $\sum_{t=i+1}^j \log h(y_t, w_t)$.
$A_{ij}^{(0)}$	Block evidence on $(i, j]$.
$A_{ij}^{(r)}$	Block moment numerator $A_{ij}^{(0)} \mathbb{E}[m(\theta)^r \mid y_{i+1:j}]$ (for integer $r \in \{1, 2, \dots\}$; conjugate-case definition). In the non-conjugate Laplace discussion of Section 12, the superscript is a test function rather than a moment order; those objects are denoted $M_{ij}^{[g]}$ to avoid collision.
$\tilde{A}_{ij}^{(r)}$	Length-prior-adjusted block quantity $A_{ij}^{(r)} g(\Delta_x(i, j))$.
$L_{k,j}$	Prefix evidence for $y_{1:j}$ with exactly k segments.
$R_{k,i}$	Suffix evidence for $y_{i+1:n}$ with exactly k segments.
C_k	Normalizer of $p(t \mid k)$ under a segment-factorized partition prior.
$p(k \mid y)$	Posterior over segment counts.
$p(t_p = h \mid y, k)$	Marginal posterior of the p th boundary given k .
$p(b_i = 1 \mid y, k)$	Boundary-event marginal at interior index i , equal to $\sum_{p=1}^{k-1} p(t_p = i \mid y, k)$. This is the calibration target in Section 15.
$\hat{t}^{\text{MAP}}(k)$	Joint MAP segmentation given k , obtained by max-sum DP and backtracking.
$\hat{\mu}_q^{(r)}$	Posterior moment of the segment- q mean parameter conditional on a chosen segmentation.
$\hat{f}_t^{\text{Bayes}}$	Bayes regression curve at index t (posterior mean of the latent piecewise-constant signal).
$s \in \{1, \dots, S\}$	Sequence / subject index in multi-sequence settings.
$g \in \{1, \dots, G\}$	Group index.
r_{sg}	Soft assignment (responsibility) of sequence s to group g in the latent-group template mixture.
$\mathcal{M}_g$	Fitted group-specific model exported for prediction.
ℓ_g	Log predictive score of new data under group g .
π_g	Prior group weight used for prediction or in the latent-group mixture.

Remarks. Two distinctions are important and recur throughout the paper. First, *design irregularity* and *likelihood weighting* are not the same thing: irregular spacing is usually encoded through the partition prior (see Section 7), whereas w_i belongs inside the likelihood only when the observation model itself supplies exposures or known precisions. Second, *boundary marginals* and the *joint MAP segmentation* are different summaries of the posterior

(Section 6). BayesBreak computes both, and they must not be conflated.

4 Problem setup and Bayesian segmentation model

4.1 Warm-up: the equispaced Gaussian BPCR view

To motivate the BayesBreak construction, recall the equispaced Gaussian model studied by Hutter (2007). When $x_i = i$, $w_i \equiv 1$, and

$$y_t \mid \mu_q \sim \mathcal{N}(\mu_q, \sigma^2), \quad \mu_q \sim \mathcal{N}(\nu, \rho^2), \quad t \in I_q(t),$$

(the symbols (ν, ρ^2) here are the Gaussian prior mean and variance; they are reused in Section 11 for the same objects under a weighted Gaussian likelihood) one can integrate out the segment mean on every candidate block and precompute a triangular array of block evidences. All posterior quantities of interest then reduce to sums or maxima over products of those block evidences, which can be computed by DP.

4.2 General segmentation model

For the general BayesBreak model, conditional on a block-specific parameter θ , observations inside a block follow an exponential-family likelihood of the form

$$p(y \mid \theta) = h(y) \exp\{\eta(\theta)^\top T(y) - A(\theta)\}. \quad (10)$$

When the observation model carries known exposure or precision information, we write the weighted version in Section 5. For a candidate block $(i, j]$, BayesBreak stores aggregated summaries

$$S_{ij} = \sum_{t=i+1}^j w_t T(y_t), \quad W_{ij} = \sum_{t=i+1}^j w_t, \quad H_{ij} = \sum_{t=i+1}^j \log h(y_t, w_t),$$

and uses these to evaluate $A_{ij}^{(0)}$ and the moment numerators $A_{ij}^{(r)}$.

Partition prior. We place a prior $p(k)$ on the number of segments and, conditional on k , a prior on boundary vectors t . Two cases are especially important.

- **Index-uniform prior.** For regularly spaced or index-based segmentation, use $p(t \mid k) = \binom{n-1}{k-1}^{-1}$, uniform over all ordered $(k-1)$ -subsets of $\{1, \dots, n-1\}$.
- **Design-aware prior.** For irregular designs, let $\Delta x_q = \Delta_x(t_{q-1}, t_q)$ be the physical length of segment q under the boundary-coordinate convention above. We consider priors of the form

$$p(t \mid k) \propto \prod_{q=1}^k g(\Delta x_q), \quad (11)$$

where g is a nonnegative cohesion function. This is a contiguous product-partition prior and is a natural way to encode physical spacing, minimum-length constraints, or renewal-style hazard assumptions.

The normalizing constant hidden in (11) is

$$C_k := \sum_{0=t_0 < t_1 < \dots < t_k=n} \prod_{q=1}^k g(\Delta_x(t_{q-1}, t_q)), \quad (12)$$

so that $p(t \mid k) = C_k^{-1} \prod_q g(\Delta x_q)$. In the index-uniform case, $g \equiv 1$ and $C_k = \binom{n-1}{k-1}$.

Hazard interpretation. When an unconditioned renewal process on an index grid has iid geometric lengths with hazard $\rho \in (0, 1)$, its unnormalized segment-length factor is proportional to $(1 - \rho)^{m-1}$. After conditioning on a fixed segment count and endpoint, these factors often become constant over all admissible partitions, especially on a regular grid. More generally, g functions here as a product-partition cohesion on physical block lengths; renewal-process language supplies the corresponding process interpretation when the endpoint and conditioning conventions are explicit.

Assumption 4.1 (Factorizing partition prior). *For each $k \in \{1, \dots, k_{\max}\}$, the conditional prior on the boundary vector factorizes as in (11) for some nonnegative g with finite and positive normalizer $C_k \in (0, \infty)$ defined in (12). The admissibility class includes, as special cases, (i) the index-uniform prior ($g \equiv 1$), (ii) renewal-style length priors with g depending only on the physical segment length, (iii) minimum-length truncations ($g(\ell) = 0$ for $\ell < \ell_{\min}$ and $g(\ell) > 0$ otherwise), and (iv) general design-aware cohesion functions that remain multiplicative over segments.*

Assumption 4.1 is the key structural condition enabling exact dynamic programming. The admissibility condition $C_k \in (0, \infty)$ is automatic for $g \equiv 1$ (giving $C_k = \binom{n-1}{k-1}$) and holds whenever the finite set of admissible blocks has at least one positive-weight k -segmentation and all included block weights are finite. Blocks excluded by minimum length, missingness rules, or impossible likelihood constraints are represented by $g(\Delta_x(i, j)) = 0$ or by a block log-score of $-\infty$, both of which remain compatible with the DP recursions.

Block-score contract. For every model family, the block routine returns a finite log-evidence $\log A_{ij}^{(0)}$ on every admissible block, the sentinel $-\infty$ on every inadmissible block, and finite moment numerators on every admissible block for which Bayes-curve moments are requested. The partition-prior normalizer $\log C_k$ is computed under the *same* admissibility mask as the score array, so that no inadmissible block contributes to either side of the DP. The dynamic program of Section 6 treats the supplied score matrix as authoritative: with exact conjugate-block evidences, it returns exact Bayesian posteriors; with surrogate scores from a non-conjugate approximation, it returns exact summaries of the surrogate model and the approximation must be assessed separately through the block-error diagnostics of Section 12. Mixing exact and surrogate scores within the same score matrix without an explicit declaration breaks this contract.

A renewal-process view of this class of priors is developed in Appendix A.2.

5 Exponential–family conjugate single–segment evidence and moments

Plate model (single segment). For a block $(i, j]$ and conjugate hyperparameters (α_0, β_0) , the generative model is

$$\theta \sim p(\theta \mid \alpha_0, \beta_0), \quad y_t \mid \theta \stackrel{\text{ind}}{\sim} p(y_t \mid \theta, w_t) \text{ as in (14)}, \quad t \in \{i+1, \dots, j\}. \quad (13)$$

The block evidence $A_{ij}^{(0)}$ is obtained by marginalizing θ out of (13).

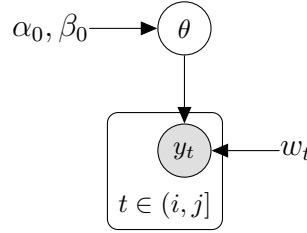


Figure 1 Plate model for a single conjugate EF block. Shaded nodes are observed; unshaded are latent; point nodes are fixed hyperparameters/exposures. Marginalizing θ gives $A_{ij}^{(0)}$.

Recall the generic EF–conjugate specification (Diaconis and Ylvisaker, 1979; Bernardo and Smith, 1994). Conditional on a (possibly low-dimensional) parameter θ , observations y_t in a block $(i, j]$ are independent with density

$$p(y_t \mid \theta) = \exp\{w_t[T(y_t)^\top \eta(\theta) - A(\theta)] + \log h(y_t, w_t)\}, \quad (14)$$

where $w_t \geq 0$ are exposure weights and h collects base measures. The conjugate prior for θ is

$$p(\theta \mid \alpha, \beta) = \exp\{\eta(\theta)^\top \alpha - \beta A(\theta) - \log Z(\alpha, \beta)\},$$

with normalizing integral $Z(\alpha, \beta)$ defined by (9).

For a block $(i, j]$ define the cumulative sums

$$S_{ij} := \sum_{t=i+1}^j w_t T(y_t), \quad W_{ij} := \sum_{t=i+1}^j w_t, \quad H_{ij} := \sum_{t=i+1}^j \log h(y_t, w_t).$$

Theorem 5.1 (EF–conjugate block integration). *Fix a block $(i, j]$ and a conjugate prior $p(\theta \mid \alpha_0, \beta_0)$. Assume the integral $Z(\alpha_0 + S_{ij}, \beta_0 + W_{ij})$ is finite for every $(i, j]$ with $0 \leq i < j \leq n$, and that the moment $\mathbb{E}[m(\theta)^r \mid \alpha_0 + S_{ij}, \beta_0 + W_{ij}]$ is finite for each required $r \geq 1$. Under the weighted EF model (14), the integrated single–segment evidence and the r th moment integral*

of the observation-scale EF mean $m(\theta) := \mathbb{E}[Y \mid \theta]$ are

$$\begin{aligned} A_{ij}^{(0)} &= \int \left[\prod_{t=i+1}^j p(y_t \mid \theta) \right] p(\theta \mid \alpha_0, \beta_0) d\theta \\ &= \exp\{H_{ij}\} \frac{Z(\alpha_0 + S_{ij}, \beta_0 + W_{ij})}{Z(\alpha_0, \beta_0)}, \end{aligned} \quad (15)$$

$$\begin{aligned} A_{ij}^{(r)} &= \int \left[\prod_{t=i+1}^j p(y_t \mid \theta) \right] p(\theta \mid \alpha_0, \beta_0) m(\theta)^r d\theta \\ &= A_{ij}^{(0)} \mathbb{E}[m(\theta)^r \mid \alpha_0 + S_{ij}, \beta_0 + W_{ij}], \quad r = 1, 2, \dots \end{aligned} \quad (16)$$

Proof. **Step 1: factorize the block likelihood.** By independence and (14),

$$\prod_{t=i+1}^j p(y_t \mid \theta) = \exp \left\{ \sum_{t=i+1}^j w_t [T(y_t)^\top \eta(\theta) - A(\theta)] + \sum_{t=i+1}^j \log h(y_t, w_t) \right\}.$$

Collecting block sums gives

$$\prod_{t=i+1}^j p(y_t \mid \theta) = \exp \{ H_{ij} + \eta(\theta)^\top S_{ij} - W_{ij} A(\theta) \}.$$

Step 2: multiply by the conjugate prior and identify the posterior kernel. Multiply by $p(\theta \mid \alpha_0, \beta_0) = \exp \{ \eta(\theta)^\top \alpha_0 - \beta_0 A(\theta) - \log Z(\alpha_0, \beta_0) \}$:

$$\left[\prod_{t=i+1}^j p(y_t \mid \theta) \right] p(\theta \mid \alpha_0, \beta_0) = \frac{\exp\{H_{ij}\}}{Z(\alpha_0, \beta_0)} \exp \{ \eta(\theta)^\top (\alpha_0 + S_{ij}) - (\beta_0 + W_{ij}) A(\theta) \}.$$

Step 3: integrate by using the normalizer definition. By the definition (9),

$$\int \exp \{ \eta(\theta)^\top (\alpha_0 + S_{ij}) - (\beta_0 + W_{ij}) A(\theta) \} d\theta = Z(\alpha_0 + S_{ij}, \beta_0 + W_{ij}).$$

Therefore,

$$A_{ij}^{(0)} = \exp\{H_{ij}\} \frac{Z(\alpha_0 + S_{ij}, \beta_0 + W_{ij})}{Z(\alpha_0, \beta_0)},$$

which is (15).

Step 4: moment integrals. For $r \geq 1$, the integrand defining $A_{ij}^{(r)}$ equals the integrand of $A_{ij}^{(0)}$ times $m(\theta)^r$. Dividing by $A_{ij}^{(0)}$ therefore yields the expectation of $m(\theta)^r$ under the posterior $p(\theta \mid \alpha_0 + S_{ij}, \beta_0 + W_{ij})$, which is finite by the hypothesis of the theorem, giving (16). $\square$

6 Dynamic programming: evidence, boundaries, and moments

For $k \in \{0, \dots, k_{\max}\}$ and $j \in \{0, \dots, n\}$, let L_{kj} denote the evidence of the prefix $y_{1:j}$ under a segmentation into k segments (after integrating out segment parameters), and let R_{ki} denote the evidence of the suffix $y_{i+1:n}$ under k segments. We adopt the conventions

$$L_{0,0} = 1, \quad L_{0,j} = 0 \ (j > 0), \quad R_{0,n} = 1, \quad R_{0,i} = 0 \ (i < n).$$

Remark 6.1 (Exact DP conditional on the score matrix). *The recursions in this section are algebraically exact for any supplied admissible block-score matrix $A_{ij}^{(0)}$. They are exact Bayesian recursions for the original data-generating model only when those entries are the true marginal likelihoods of (15), when the same admissibility convention and prior normalization are used throughout, and when the partition prior is a product-partition prior of the form fixed in Section 4. With surrogate block scores from a non-conjugate approximation (Section 12), the recursions remain algebraically exact for the surrogate score matrix and must be interpreted through the approximation-error analysis of Proposition 12.6 and Corollary 12.8.*

6.1 Forward/backward recursions

Given single-block evidences $A_{ij}^{(0)}$, the forward and backward recursions are

$$L_{k+1,j} = \sum_{h=k}^{j-1} L_{k,h} A_{hj}^{(0)}, \quad R_{k+1,i} = \sum_{h=i+1}^{n-k} A_{ih}^{(0)} R_{k,h}, \quad (17)$$

for $k \geq 0$, $1 \leq j \leq n$, and $0 \leq i \leq n-1$, where sums are restricted to valid indices.

Implementation. Algorithm 1 presents the core forward/backward recursions in a numerically stable log-space form. The routine takes as input a triangular array of block log-evidences $\log \mathcal{L}_{ij}$ (from Section 5 or Sections 11–12) and returns the forward messages, backward messages, and segment-count posteriors needed for boundary and moment extraction.

Algorithm 1: Generic DP for EF segmentation (prefix/suffix form)

Input: Block evidences $\tilde{A}_{ij}^{(0)}$ for $0 \leq i < j \leq n$; $k_{\max}$; C_k (default $C_k = \binom{n-1}{k-1}$)

```

1 Initialize  $\tilde{L}_{0,0} = 1, \tilde{L}_{0,j>0} = 0; \tilde{R}_{0,n} = 1, \tilde{R}_{0,i<n} = 0;$ 
2 for  $k = 0$  to  $k_{\max} - 1$  do
3   for  $j = 1$  to  $n$  do
4      $\tilde{L}_{k+1,j} \leftarrow \sum_{h=k}^{j-1} \tilde{L}_{k,h} \tilde{A}_{hj}^{(0)};$ 
5   for  $i = 0$  to  $n - 1$  do
6      $\tilde{R}_{k+1,i} \leftarrow \sum_{h=i+1}^{n-k} \tilde{A}_{ih}^{(0)} \tilde{R}_{k,h};$ 
7 Compute  $P(k | y) \propto p(k) \tilde{L}_{kn} / C_k$  (cf. (20)) and choose  $\hat{k}$ ;
8 for  $p = 1$  to  $\hat{k} - 1$  do
9   For  $h \in \{p, \dots, n - (\hat{k} - p)\}$  set  $P(t_p = h | y, \hat{k}) \leftarrow \tilde{L}_{p,h} \tilde{R}_{\hat{k}-p,h} / \tilde{L}_{\hat{k}n}$ ;
10  Record the marginal mode  $\arg \max_h P(t_p = h | y, \hat{k})$  as a diagnostic only;
```

Output: $\{\tilde{L}_{kj}\}, \{\tilde{R}_{ki}\}, P(k | y), \hat{k}$, and boundary-marginal diagnostics

6.2 Incorporating segment-factorized boundary priors

Suppose the boundary prior given k factorizes over segment lengths as in (11):

$$p(t | k) = \frac{1}{C_k} \prod_{q=1}^k g(\Delta_x(t_{q-1}, t_q)), \quad C_k := \sum_{t: 0=t_0 < \dots < t_k=n} \prod_{q=1}^k g(\Delta_x(t_{q-1}, t_q)).$$

Then we can absorb g into block evidences by defining

$$\tilde{A}_{ij}^{(0)} := A_{ij}^{(0)} g(\Delta_x(i, j)), \quad \tilde{A}_{ij}^{(r)} := A_{ij}^{(r)} g(\Delta_x(i, j)). \quad (18)$$

Running the DP (17) with $\tilde{A}_{ij}^{(0)}$ computes $\tilde{L}_{kn}$, the *unnormalized* evidence sum over all segmentations weighted by g . The normalizer C_k can be computed by the same DP with $A_{ij}^{(0)} \equiv 1$:

$$C_k = \sum_t \prod_{q=1}^k g(\Delta_x(t_{q-1}, t_q)) = L_{kn}^{(g)}, \quad \text{where} \quad L_{k+1,j}^{(g)} = \sum_{h=k}^{j-1} L_{k,h}^{(g)} g(\Delta_x(h, j)), \quad L_{0,0}^{(g)} = 1. \quad (19)$$

In the index-uniform case $g \equiv 1$, (19) reduces to $C_k = \binom{n-1}{k-1}$.

6.3 Posteriors over k , boundary marginals, and joint MAP segmentations

Let $p(k)$ be a prior on segment count. Under index-uniform $p(t | k)$ we have $C_k = \binom{n-1}{k-1}$ and

$$P(y | k) = \frac{L_{kn}}{C_k}, \quad P(k | y) \propto p(k) \frac{L_{kn}}{C_k}.$$

For general segment-factorized $p(t \mid k)$, replace L by $\tilde{L}$ and C_k by (19). In all cases we can write compactly

$$P(k \mid y) \propto p(k) \frac{\tilde{L}_{kn}}{C_k}, \quad k = 1, \dots, k_{\max}. \quad (20)$$

Let $\hat{k} = \arg \max_k P(k \mid y)$ denote the MAP segment count.

Conditional on k , the marginal posterior for the p th boundary t_p is

$$P(t_p = h \mid y, k) = \frac{\tilde{L}_{p,h} \tilde{R}_{k-p,h}}{\tilde{L}_{kn}}, \quad h \in \{p, \dots, n - (k - p)\}, \quad (21)$$

where $(\tilde{L}, \tilde{R})$ are computed from $\tilde{A}^{(0)}$ as in (17). The denominator is the unnormalized evidence under the length-weighted partition prior; the normalizer C_k cancels in (21) because the same C_k multiplies every boundary vector in the numerator and denominator. (When $g \equiv 1$, tildes can be dropped.)

Remark 6.2 (Boundary-event marginal). *A related object we use repeatedly is the boundary-event marginal*

$$P(b_i = 1 \mid y, k) := \sum_{p=1}^{k-1} P(t_p = i \mid y, k) = \sum_{p=1}^{k-1} \frac{\tilde{L}_{p,i} \tilde{R}_{k-p,i}}{\tilde{L}_{kn}}, \quad 1 \leq i \leq n - 1, \quad (22)$$

the posterior probability that interior index i coincides with some boundary, without conditioning on which boundary. The objects $P(t_p = h \mid y, k)$ and $P(b_i = 1 \mid y, k)$ are distinct: the former is a probability distribution over h for each fixed p (and sums to 1 over h), while the latter is a per-index Bernoulli probability summing to $k - 1$ over i . Calibration experiments in Section 15 target $P(b_i = 1 \mid y, k)$; a full derivation is given in Appendix A.3.

A useful diagnostic is the *marginal boundary mode*

$$\tilde{t}_p^{\text{marg}}(k) \in \arg \max_h P(t_p = h \mid y, k).$$

However, maximizing the marginals independently need not yield an ordered boundary vector and, in general, does *not* recover the joint MAP segmentation.

Remark 6.3 (A small counterexample). *Consider $n = 5$ and $k = 3$, so the two interior boundaries satisfy $1 \leq t_1 < t_2 \leq 4$. For example, let the joint posterior put masses 0.30, 0.28, 0.22, 0.20 on the four ordered configurations (1, 4), (2, 3), (2, 4), (3, 4), respectively. The marginal modes are then $t_1 = 2$ and $t_2 = 4$, so the independently selected marginal-mode vector is (2, 4), even though the joint MAP configuration is (1, 4). This is exactly the kind of discrepancy the max-sum DP resolves.*

Remark 6.4 (When the two summaries agree). *The discrepancy in Remark 6.3 requires posterior dependence among the boundary coordinates. In the degenerate case where the fixed- k posterior over admissible boundary vectors factorizes as a product measure across the coordinates $t_1, \dots, t_{k-1}$ and the resulting coordinate-wise modes already form an ordered vector, maximizing the marginals independently does recover the joint MAP segmentation. This product structure is not generic—the ordering constraint $t_1 < \dots < t_{k-1}$ alone induces*

dependence—so *BayesBreak* computes the joint MAP by max-sum DP rather than relying on marginal modes.

The joint MAP segmentation is the solution of

$$\hat{t}^{\text{MAP}}(k) \in \arg \max_{0=t_0 < \dots < t_k=n} \left\{ \sum_{q=1}^k \log \tilde{A}_{t_{q-1}t_q}^{(0)} \right\}, \quad (23)$$

with the convention that any k -specific prior terms such as $\log p(k) - \log C_k$ are added only when selecting among different values of k . Equation (23) is solved exactly by a max-sum DP with backpointers. Inline, the max-sum recursion reads

$$M_{k+1,j} = \max_{h \in \{k, \dots, j-1\}} \{M_{k,h} + \log \tilde{A}_{hj}^{(0)}\}, \quad h^*(k+1, j) \in \arg \max_h \{M_{k,h} + \log \tilde{A}_{hj}^{(0)}\}, \quad (24)$$

with initial conditions $M_{0,0} = 0$, $M_{0,j} = -\infty$ for $j > 0$, and backtracking from (k, n) as detailed in Section 14. In the sequel, whenever we report an *exported MAP segmentation*, we mean the joint optimizer $\hat{t}^{\text{MAP}} := \hat{t}^{\text{MAP}}(\hat{k})$ rather than a vector of marginal boundary modes.

6.4 Segment-level and regression-curve posterior moments

For segment-wise posterior moments attached to a *fixed* segmentation $t = (t_0, \dots, t_k)$, let segment $q \in \{1, \dots, k\}$ correspond to block $(i, j] = (t_{q-1}, t_q]$. Then

$$\mu_q^{(r)}(t) = \frac{\tilde{A}_{ij}^{(r)}}{\tilde{A}_{ij}^{(0)}} = \frac{A_{ij}^{(r)}}{A_{ij}^{(0)}}, \quad r = 1, 2, \dots, \quad (25)$$

where the final equality holds because the prior factor $g(\Delta_x(i, j))$ cancels. In particular, for an exported joint MAP segmentation $\hat{t}^{\text{MAP}}$, the segment summaries are obtained by setting $t = \hat{t}^{\text{MAP}}$ in (25).

To compute the Bayes regression curve within a fixed k (typically $k = \hat{k}$), define, for each block $(i, j]$ and moment order r ,

$$F_{ij}^{(r)}(k) := \frac{1}{\tilde{L}_{kn}} \sum_{m=1}^k \tilde{L}_{m-1,i} \tilde{A}_{ij}^{(r)} \tilde{R}_{k-m,j}.$$

Then the r th posterior moment of the latent mean at index t given k is obtained by summing $F_{ij}^{(r)}(k)$ over all candidate blocks $(i, j]$ that contain the index t (i.e., all blocks with $i < t \leq j$), so that each candidate segmentation contributes exactly once through the block that actually covers t :

$$\hat{\mu}_t^{(r)} := \mathbb{E}[(\mu_t')^r \mid y, k] = \sum_{i < t \leq j} F_{ij}^{(r)}(k), \quad t = 1, \dots, n. \quad (26)$$

In particular, $\hat{\mu}_t' := \hat{\mu}_t^{(1)}$ is the Bayes regression curve and $\mathbb{V}[\mu_t' \mid y, k] = \hat{\mu}_t^{(2)} - (\hat{\mu}_t^{(1)})^2$. If a fully k -marginalized curve is desired, compute the same conditional curve for each admissible k and average the resulting moments using $P(k \mid y)$ from (20). The empirical sections report curves conditional on the selected $\hat{k}$ unless explicitly stated otherwise.

Theorem 6.5 (Correctness of the DP). *Assume finite block evidences on all admissible blocks, a finite segment-count range, and a segment-factorized conditional partition prior satisfying Assumption 4.1 with the boundary-coordinate convention of Section 3. Under the EF-conjugate single-segment model of Section 5, or under any other block model whose exact evidence array is supplied, the recursions (17) compute exact marginal evidences for all prefixes/suffixes: $\tilde{L}_{kj}$ and $\tilde{R}_{ki}$ are exact evidence sums over all k -segmentations of the corresponding prefix/suffix. Moreover:*

1. *The posterior $P(k \mid y)$ given by (20) is exact.*
2. *The boundary marginal $P(t_p = h \mid y, k)$ given by (21) is exact.*
3. *For any fixed k , the joint MAP segmentation (23) is obtained exactly by the max-sum recursion $M_{k+1,j} = \max_h \{M_{k,h} + \log \tilde{A}_{hj}^{(0)}\}$ with backpointers $h^*(k+1, j)$ and subsequent backtracking (the full routine of Section 14; detailed proof in Appendix A.8).*
4. *The segment-wise moments (25) and regression-curve moments (26) are exact posterior moments conditional on k .*

Proof. We present the proof for general $\tilde{A}_{ij}^{(0)}$; the untilded case is the specialization $g \equiv 1$.

Step 1: interpret $\tilde{L}_{kj}$. Fix $k \geq 1$ and $j \geq 1$. Consider all segmentations of the prefix $y_{1:j}$ into k segments, i.e. all boundary vectors $(t_0, \dots, t_k)$ with $t_0 = 0$ and $t_k = j$. For any such segmentation, segment parameters are independent a priori and observations are independent across segments given the segmentation. Integrating out segment parameters yields a product of block evidences:

$$p(y_{1:j} \mid t, k) = \prod_{q=1}^k \tilde{A}_{t_{q-1}, t_q}^{(0)}.$$

Summing over all segmentations gives

$$\tilde{L}_{kj} = \sum_{t: t_k=j} \prod_{q=1}^k \tilde{A}_{t_{q-1}, t_q}^{(0)}.$$

Thus $\tilde{L}_{kj}$ is exactly the marginal evidence sum for the prefix under k segments.

Step 2: derive the forward recursion. Any segmentation of $y_{1:j}$ into $k+1$ segments has a last boundary at some $h \in \{k, \dots, j-1\}$. Conditioning on this last boundary splits the segmentation into: (i) a k -segmentation of $y_{1:h}$ and (ii) the final block $(h, j]$. By the factorization above, the evidence contribution for a fixed h equals $\tilde{L}_{kh} \tilde{A}_{hj}^{(0)}$. Summing over all admissible h yields

$$\tilde{L}_{k+1,j} = \sum_{h=k}^{j-1} \tilde{L}_{kh} \tilde{A}_{hj}^{(0)},$$

which is the left recursion in (17).

Step 3: derive the backward recursion. The same argument applies to suffixes by conditioning on the first boundary after i , yielding

$$\tilde{R}_{k+1,i} = \sum_{h=i+1}^{n-k} \tilde{A}_{ih}^{(0)} \tilde{R}_{kh},$$

the right recursion in (17).

Step 4: posterior over k . For any fixed k , the data likelihood marginalized over segmentations is $\tilde{L}_{kn}$. If $p(t \mid k)$ includes a normalizer C_k (as in (11)), then $P(y \mid k) = \tilde{L}_{kn}/C_k$ by definition of C_k as the sum of prior weights across all segmentations. Bayes' rule with prior $p(k)$ yields (20).

Step 5: boundary posteriors. Fix k and boundary index $p \in \{1, \dots, k-1\}$. Conditioning on $t_p = h$ splits the sequence into: (i) a prefix $y_{1:h}$ with p segments and (ii) a suffix $y_{h+1:n}$ with $k-p$ segments. By independence across the split and by the definition of $\tilde{L}, \tilde{R}$, the total evidence of all segmentations consistent with $t_p = h$ equals $\tilde{L}_{p,h} \tilde{R}_{k-p,h}$. Normalizing by the total evidence $\tilde{L}_{kn}$ yields (21).

Step 6: moment formulas. For a fixed block $(i, j]$, Theorem 5.1 gives $A_{ij}^{(r)} = A_{ij}^{(0)} \mathbb{E}[m(\theta)^r \mid (i, j]]$. Since the segmentation prior factor $g(\Delta_x(i, j))$ cancels in the ratio, (25) holds. The regression-curve moment formula (26) follows by averaging over all segmentations and grouping contributions by the block $(i, j]$ that contains index t . $\square$

Posterior-evidence sanity checks. After every DP run, verify four invariants: normalized segment-count weights sum to one; the full-sequence forward and backward evidences agree, $\tilde{L}_{kn} = \tilde{R}_{k0}$; boundary-event marginals satisfy $\sum_{i=1}^{n-1} P(b_i = 1 \mid y, k) = k-1$; and the max-sum backtracking score equals the stored terminal max-sum value.

Corollary 6.6 (Fixed-count boundary-event normalization). *For any fixed k with positive evidence under the hypotheses of Theorem 6.5, the boundary-event marginals satisfy*

$$\sum_{i=1}^{n-1} P(b_i = 1 \mid y, k) = k - 1.$$

Proof. Every admissible segmentation into k segments has exactly $k-1$ interior boundaries. Conditioning on y and k and exchanging summation with expectation gives $\sum_{i=1}^{n-1} P(b_i = 1 \mid y, k) = \mathbb{E}[k-1 \mid y, k] = k-1$. $\square$

Corollary 6.6 is the natural target of the boundary-sum sanity check listed above: any DP implementation that satisfies it on every k has the correct boundary-event posterior up to numerical precision, while a violation localizes implementation error to the boundary marginal step rather than to evidence computation or count normalization.

Theorem 6.7 (Time and space complexity). *Assume single-block evidences $A_{ij}^{(0)}$ are available for all $0 \leq i < j \leq n$. Then computing L_{kj} and R_{ki} for $k = 0, \dots, k_{\max}$ via (17) requires $\mathcal{O}(k_{\max}n^2)$ arithmetic operations. Storing all $A_{ij}^{(0)}$ requires $\mathcal{O}(n^2)$ memory, and storing the DP arrays requires $\mathcal{O}(k_{\max}n)$.*

Proof. For each fixed k , the forward recursion computes $L_{k+1,j}$ for $j = 1, \dots, n$, and each $L_{k+1,j}$ sums at most $j - k$ terms. Thus the forward work per k is $\sum_{j=1}^n \mathcal{O}(j) = \mathcal{O}(n^2)$. The backward recursion is analogous. Repeating for $k = 0, \dots, k_{\max} - 1$ gives $\mathcal{O}(k_{\max}n^2)$ time. Memory follows by counting array sizes. $\square$

Corollary 6.8 (Joint MAP is Bayes-optimal under 0–1 loss on the full segmentation). *For the loss $\ell_{01}(t, t') = \mathbb{I}\{t \neq t'\}$, the Bayes estimator t^* minimizing $\mathbb{E}[\ell_{01}(t, t') \mid y, k]$ is the joint MAP segmentation $\hat{t}^{\text{MAP}}(k)$ from (23). For any additive per-segment loss $\ell_{\text{add}}(t, t') = \sum_q \ell_q(t_q, t'_q)$ that does not decompose as the pointwise indicator on the full vector, the Bayes-optimal estimator can differ from $\hat{t}^{\text{MAP}}(k)$.*

Proof. The 0–1 loss’s expectation equals $1 - p(t = t' \mid y, k)$, which is minimized by the joint posterior mode, i.e., the joint MAP. Pointwise (say Hamming-type) losses decompose differently and are minimized by the marginal modes, which we have just noted need not coincide with the joint MAP. $\square$

Remark 6.9 (Memory/time trade-offs for large n). *Two implementation variants are standard. (i) Streaming k -layers: keeping only two consecutive layers of (L, R) gives memory $\mathcal{O}(n)$ (plus $\mathcal{O}(n^2)$ for the block-evidence matrix) at no time cost, but backpointers and boundary marginals must then be recomputed. (ii) Checkpointing: storing every $\sqrt{k_{\max}}$ th k -layer gives memory $\mathcal{O}(\sqrt{k_{\max}}n)$ and at most $\mathcal{O}(\sqrt{k_{\max}}k_{\max}n^2)$ time, because recomputing from the nearest checkpoint needs at most $\sqrt{k_{\max}}$ forward layers. When the n^2 block-evidence matrix is itself prohibitive, an alternative is to evaluate block evidences on demand during the DP sweep at cost $\mathcal{O}(k_{\max}n^2)$ total (constant-time per block via prefix sums).*

Remark 6.10 (Evidence across segment counts need not be monotone). *The fixed- k evidence $P(y \mid k) = \tilde{L}_{kn}/C_k$ is not monotone in general. Adding a boundary changes both the set of admissible partitions and the product of block evidences, while the normalized conditional prior contributes the normalizer C_k . Consequently, diagnose segment-count behavior through the normalized evidences $P(y \mid k)$ and posterior weights $P(k \mid y)$ rather than through an unnormalized pathwise “add one boundary” heuristic.*

6.5 End-to-end BayesBreak inference routine

Collecting the ingredients from Sections 5 and 6—block-evidence precomputation, exact sum-product DP over partitions, and max-sum backtracking for exported segmentations—yields an end-to-end routine suitable for direct implementation. Algorithm 2 summarizes this workflow in a modular way: the only model-specific input is a function that evaluates (or approximates) block evidences $\log \mathcal{L}_{ij}$ and, when needed, block moment numerators.

Algorithm 2: BayesBreak($y, x, w, \text{model}, \text{hyper}, k_{\max}, p(k), g$)

Input: Observations $y_{1:n}$ at locations $x_{1:n}$ (optional); observation-model weights/exposures $w_{1:n}$ (default determined by the chosen likelihood, often $w_i \equiv 1$);
model family $\text{model} \in \{\text{Gaussian}, \text{Poisson}, \text{Binomial}, \text{BetaObs}, \dots\}$;
hyperparameters hyper ; maximum segment count $k_{\max}$; prior $p(k)$; length factor $g(\cdot)$ (default $g \equiv 1$).
Output: Posterior $P(k | y)$; MAP segment count $\hat{k}$; boundary marginals $P(t_p | y, \hat{k})$; exported joint MAP segmentation $\hat{t}^{\text{MAP}}$; segment-level moments; Bayes regression curve; (optional) evidence $P(y)$.

- 1 **1. Precompute block evidences and moments;**
 - 2 **for** $0 \leq i < j \leq n$ **do**
 - 3 Compute block summaries (S_{ij}, W_{ij}, H_{ij}) using prefix sums (or running sums);
 - 4 Call a family-specific routine to obtain $A_{ij}^{(0)}, A_{ij}^{(1)}, A_{ij}^{(2)}$;
 - 5 Set $\tilde{A}_{ij}^{(r)} \leftarrow A_{ij}^{(r)} g(\Delta_x(i, j))$ for $r \in \{0, 1, 2\}$ (cf. (18));
 - 6 Compute C_k under the same admissibility and length-prior convention used in the score array; for $g \equiv 1$, use $C_k = \binom{n-1}{k-1}$, and otherwise use (19) (DP with $A_{ij}^{(0)} \equiv 1$);
 - 7 **2. Run the sum-product dynamic program;**
 - 8 Compute $\tilde{L}_{kj}, \tilde{R}_{ki}$ via (17) for $k = 0, \dots, k_{\max}$ using $\tilde{A}_{ij}^{(0)}$;
 - 9 Compute $P(k | y) \propto p(k) \tilde{L}_{kn}/C_k$ via (20); set $\hat{k} \leftarrow \arg \max_k P(k | y)$;
 - 10 **3. Boundary marginals;**
 - 11 **for** $p = 1$ **to** $\hat{k} - 1$ **do**
 - 12 For all $h \in \{p, \dots, n - (\hat{k} - p)\}$, set $P(t_p = h | y, \hat{k}) \leftarrow \tilde{L}_{p,h} \tilde{R}_{\hat{k}-p,h} / \tilde{L}_{\hat{k}n}$ (cf. (21));
 - 13 **4. Export a joint MAP segmentation;**
 - 14 Run the max-sum/backpointer recursion of Section 14 using $\log \tilde{A}_{ij}^{(0)}$ and the chosen count $\hat{k}$;
 - 15 Backtrack to obtain $\hat{t}^{\text{MAP}} = (0 = \hat{t}_0^{\text{MAP}} < \dots < \hat{t}_{\hat{k}}^{\text{MAP}} = n)$;
 - 16 **5. Segment-level posterior moments on the exported segmentation;**
 - 17 **for** $q = 1$ **to** $\hat{k}$ **do**
 - 18 Let $(i, j] \leftarrow (\hat{t}_{q-1}^{\text{MAP}}, \hat{t}_q^{\text{MAP}}]$;
 - 19 For $r \in \{1, 2\}$ set $\hat{\mu}_q^{(r)} \leftarrow \tilde{A}_{ij}^{(r)} / \tilde{A}_{ij}^{(0)}$ (cf. (25));
 - 20 **6. Bayes regression curve within $\hat{k}$;**
 - 21 Using $\tilde{L}, \tilde{R}, \tilde{A}^{(r)}$ and (26), compute $\hat{\mu}'_i$ and $\mathbb{V}[\mu'_i | y, \hat{k}]$ for $i = 1, \dots, n$;
 - 22 **7. Optional: pooling/groups;**
 - 23 If multiple subjects or groups are present, pool block evidences (Theorem 8.2) and/or run the latent-template EM routine of Section 10, then repeat Steps 2–6.
-

6.6 Computational complexity and numerical stability

BayesBreak separates computation into a *local* stage (building the block-evidence array) and a *global* stage (running DP over partitions). For conjugate exponential-family models, block evidences can be evaluated in $\mathcal{O}(1)$ amortized time per block once cumulative sufficient-statistic arrays are constructed, yielding $\mathcal{O}(n^2)$ total preprocessing. The DP recursions over (k, t) states are $\mathcal{O}(k_{\max} n^2)$ in the worst case for explicit segment-count posteriors; in many regimes, pruning and/or marginalization of k can reduce the effective complexity, but BayesBreak’s design prioritizes exactness and transparency of posterior quantities. All forward/backward computations are performed in log-space with stable `logsumexp` operations, and implementations cache cumulative summaries (e.g., prefix sums of sufficient statistics) to avoid redundant work. When storing the full forward table is memory-prohibitive, checkpoint layers or recompute forward messages on demand during backward passes, trading memory for additional passes through the DP.

7 Irregular designs and length-aware priors

Irregularly spaced designs enter BayesBreak in two logically distinct places. First, likelihood weights or exposures w_t belong to the observation model when they encode Poisson exposures, known Gaussian precisions, or Binomial trial counts. These quantities are part of the sampling model; supply them only when substantively justified by the data-generating process. Second, the geometry of the design points $x_1 < \dots < x_n$ can enter through a *partition prior* that depends on physical segment lengths $\Delta_x(i, j)$. In most irregular-grid applications, this second mechanism is the default one: irregular spacing by itself does not imply that the likelihood is reweighted.

Accordingly, a clean default is to leave the observation model unchanged (often $w_t \equiv 1$) and encode design geometry through the segment-factorized prior

$$p(t \mid k) \propto \prod_{q=1}^k g(\Delta_x(t_{q-1}, t_q)),$$

with g chosen to reflect prior beliefs about physical segment lengths. This is particularly natural in genomics, spatial statistics, and event-time data, where the scientifically meaningful notion of “distance” is not the index increment $t_q - t_{q-1}$ but the physical length of the segment.

Proposition 7.1 (Homogeneous Poisson-process prior: regular-grid index-uniform as a special case). *Partition the continuous interval (u_0, u_n) into the n candidate boundary intervals $(u_{i-1}, u_i]$ of lengths $\Delta u_i := u_i - u_{i-1}$. Consider a homogeneous Poisson process of rate λ on (u_0, u_n) and condition on the event that exactly $k - 1$ events occur and that no two events fall in the same inter-point interval. Then the induced discrete prior on the selected set of $(k - 1)$ boundary intervals is proportional to the product of their lengths, $\prod_{\ell \in \mathcal{I}} \Delta u_\ell$. In particular, on a regular grid (all Δu_ℓ equal), the induced discrete prior is index-uniform.*

Proof. For a homogeneous Poisson process, counts in disjoint intervals are independent and $\#\{\text{events in } (u_{i-1}, u_i]\} \sim \text{Pois}(\lambda \Delta u_i)$. Conditioning on the event that exactly one event

occurs in each selected interval $\ell \in \mathcal{I}$ and zero events occur in other intervals yields conditional probability proportional to

$$\prod_{\ell \in \mathcal{I}} (\lambda \Delta u_\ell) e^{-\lambda \Delta u_\ell} \prod_{\ell \notin \mathcal{I}} e^{-\lambda \Delta u_\ell} \propto \prod_{\ell \in \mathcal{I}} \Delta u_\ell,$$

since the global factor $e^{-\lambda \sum_\ell \Delta u_\ell}$ cancels. If Δu_ℓ are all equal, all $(k-1)$ -subsets $\mathcal{I}$ have equal probability, giving index-uniformity. $\square$

Remark 7.2 (Inhomogeneous intensity). *No step of the argument uses that λ is constant. For an absolutely continuous intensity $\lambda(x) \geq 0$, the same conditioning yields a conditional prior proportional to $\prod_{\ell \in \mathcal{I}} \Lambda_\ell$ where $\Lambda_\ell := \int_{x_{\ell-1}}^{x_\ell} \lambda(x) dx$ is the cumulative intensity on interval ℓ . This gives a principled way to encode physically motivated spatial variation in the prior hazard of a boundary.*

Proposition 7.3 (Coarse-to-fine consistency of inherited partitions). *Let $x_m < \tilde{x} < x_{m+1}$ be a refinement of the design grid obtained by inserting a pseudo-index between two existing design points but without adding a new observation. Assume the pseudo-index contributes no likelihood term and that block evidences are computed from the original observations only. Then every coarse-grid segmentation t has a lifted fine-grid segmentation $\tilde{t}$ that places no boundary at the pseudo-index and satisfies*

$$\prod_q A_{t_{q-1}t_q}^{(0)} = \prod_q \tilde{A}_{\tilde{t}_{q-1}\tilde{t}_q}^{(0)}$$

for the likelihood contribution of the inherited partition. If, in addition, the partition prior is a function of physical segment lengths $\Delta_x(t_{q-1}, t_q)$, then the prior weight of this inherited partition is unchanged as well.

Proof. Because no new observation is added, each coarse block $(t_{q-1}, t_q]$ contains exactly the same data as its lifted fine-grid counterpart $(\tilde{t}_{q-1}, \tilde{t}_q]$. Their sufficient statistics and hence their likelihood contributions coincide. If the prior depends only on physical segment lengths, then the lifted block has the same endpoints in the physical domain and therefore the same prior factor. $\square$

Remark 7.4 (Why full refinement invariance is generally false). *The proposition above is deliberately weaker than a posterior invariance claim. Once the refined grid is allowed to place new boundaries at the inserted pseudo-index, the admissible partition space changes, so the posterior distribution over segmentations and the normalization constants C_k generally change as well. In particular, refining the grid enlarges $\mathcal{T}_k$, which typically increases C_k and redistributes posterior mass over segmentations even when no new observations are added. Thus grid refinement preserves the score of every inherited partition, but it need not preserve the full posterior unless one explicitly constrains the refined model to exclude new boundary locations.*

Example 7.5 (Illustration on a 6-point irregular design). *Consider six observations at physical locations $x = (0, 0.2, 0.4, 0.6, 1.6, 1.8)$ (a tight cluster of four points followed by a pair after a large gap), with Gaussian responses having a single genuine changepoint between*

$x = 0.6$ and $x = 1.6$. Under the index-uniform prior $g \equiv 1$, the five candidate boundary indices are equally weighted a priori, so the posterior must compete block-evidence differences against a flat prior. Under the length-aware prior $g(\ell) \propto \ell$ (proportional to physical segment length, cf. Proposition 7.1), the boundary index $i = 4$ (separating the tight cluster from the far pair) carries much larger prior mass, consistent with the physical intuition that the prior favors a boundary at the large gap. This example is a qualitative prior-sensitivity illustration: changing g shifts posterior boundary mass without changing the likelihood. A larger quantitative sensitivity study is specified as planned empirical work.

8 Replicates and multi-subject pooling (shared boundaries)

Plate model (shared boundaries, subject-specific parameters). With a shared segmentation $t = (t_0, \dots, t_k)$ and subject-specific conjugate priors, the generative model nests an inner *segment* plate inside an outer *subject* plate:

$$t \sim p(t \mid k), \quad \theta_q^{(s)} \stackrel{\text{ind}}{\sim} p(\theta \mid \alpha_0^{(s)}, \beta_0^{(s)}), \quad y_i^{(s)} \mid \theta_{q(i)}^{(s)}, w_i^{(s)} \stackrel{\text{ind}}{\sim} p(y \mid \theta_{q(i)}^{(s)}, w_i^{(s)}), \quad (27)$$

where $q(i)$ is the segment index containing observation i under t .

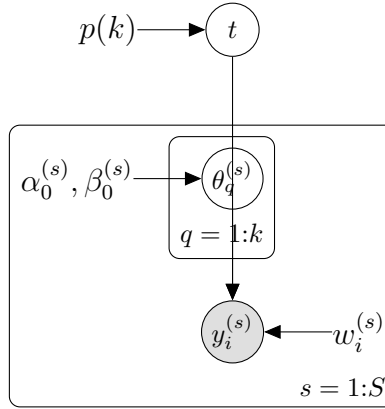


Figure 2 Plate model for multi-subject pooling under shared boundaries. The boundary vector t is shared across subjects; each subject has its own segment parameters $\theta_q^{(s)}$ and exposures $w_i^{(s)}$.

Suppose we observe S independent subjects, each with its own observations $\{(x_i^{(s)}, y_i^{(s)})\}$ and exposure weights $w_i^{(s)}$, but we wish to model them as sharing a common segmentation (boundary locations) while allowing subject-specific segment parameters. For simplicity of exposition, assume a common index grid $i = 1, \dots, n$ (e.g., after alignment/union of grids); subject-specific missing observations are handled by setting $w_i^{(s)} = 0$ at missing indices.

Let $(i, j]$ denote a block and write subject-specific block sums as $(S_{ij}^{(s)}, W_{ij}^{(s)}, H_{ij}^{(s)})$ with subject-specific prior hyperparameters $(\alpha_0^{(s)}, \beta_0^{(s)})$.

Assumption 8.1 (Cross-subject conditional independence given shared boundaries). *Conditional on a shared segmentation $t = (t_0, \dots, t_k)$, subjects $s = 1, \dots, S$ are independent, and each subject has its own segment parameters $\theta_{1:k}^{(s)}$ with subject-specific conjugate priors.*

Theorem 8.2 (Exact pooling of block evidences). *Under Assumption 8.1, the pooled block evidence for $(i, j]$ factorizes as*

$$A_{ij,\text{pool}}^{(0)} = \prod_{s=1}^S \exp\{H_{ij}^{(s)}\} \frac{Z(\alpha_0^{(s)} + S_{ij}^{(s)}, \beta_0^{(s)} + W_{ij}^{(s)})}{Z(\alpha_0^{(s)}, \beta_0^{(s)})}.$$

Equivalently, $\log A_{ij,\text{pool}}^{(0)} = \sum_s \log A_{ij,s}^{(0)}$. Running the DP recursions (17)–(26) with $A_{ij}^{(0)}$ replaced by $A_{ij,\text{pool}}^{(0)}$ yields the exact posterior over shared boundaries. Conditional on any selected or averaged boundary configuration, the subject-specific segment-parameter posteriors are then recovered from the corresponding subject-level conjugate updates.

Proof. Step 1: subject-wise block evidences. For each subject s , Theorem 5.1 yields the subject-specific block evidence $A_{ij,s}^{(0)}$ computed from $(S_{ij}^{(s)}, W_{ij}^{(s)}, H_{ij}^{(s)})$.

Step 2: independence implies product pooling. Conditional on shared boundaries, subjects are independent and have independent segment parameters. Therefore, the joint likelihood (with segment parameters integrated out) for the block is the product of the subject-wise block evidences: $A_{ij,\text{pool}}^{(0)} = \prod_s A_{ij,s}^{(0)}$, giving the stated form.

Step 3: DP correctness under pooled evidences. The DP in Section 6 depends only on the block evidence matrix. Replacing $A_{ij}^{(0)}$ by $A_{ij,\text{pool}}^{(0)}$ therefore computes the exact shared-boundary posterior for the pooled model; parameter posteriors remain available conditionally on boundaries from the subject-specific block updates. $\square$

9 Groups with known membership

Plate model (known groups, group-specific segmentations). Let $z_s \in \{1, \dots, G\}$ be the observed group label of subject s . The generative model nests a group-specific boundary vector $t^{(g)}$ and group-specific segment parameters inside an outer group plate, with subjects of each group drawn independently conditional on the corresponding $t^{(g)}$:

$$t^{(g)} \sim p(t \mid k_g), \quad \theta_q^{(g,s)} \stackrel{\text{ind}}{\sim} p(\theta \mid \alpha_0^{(g,s)}, \beta_0^{(g,s)}), \quad y_i^{(s)} \mid z_s = g, t^{(g)}, \theta_{q(i)}^{(g,s)} \stackrel{\text{ind}}{\sim} p(y \mid \theta_{q(i)}^{(g,s)}, w_i^{(s)}). \quad (28)$$

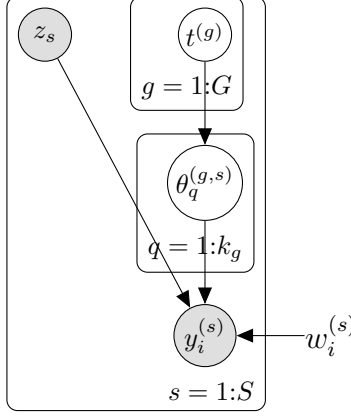


Figure 3 Plate model with known group memberships z_s . Each group has its own segmentation template $t^{(g)}$; subjects within a group share that template but retain subject-specific segment parameters $\theta_q^{(g,s)}$.

In some applications subjects can be partitioned into groups $g = 1, \dots, G$ (e.g. experimental conditions), and we allow each group to have its own segmentation while still pooling within groups. When group memberships are known, BayesBreak handles this case by applying the pooled DP of Section 8 independently within each group.

Concretely, for group g with subject index set $\mathcal{S}_g$, we pool per-subject block evidences $\{A_{ij,s}^{(0)}\}_{s \in \mathcal{S}_g}$ into a group-specific pooled evidence $A_{ij,g}^{(0)}$ using Theorem 8.2, then run the DP recursions (17)–(26) with $A_{ij}^{(0)} = A_{ij,g}^{(0)}$. This yields group-specific $P_g(k | y)$, boundary posteriors $P_g(t_p | y, k)$, and segment/regression-curve moments. The total computational cost is G times the single-group cost, since DP runs are independent across groups conditional on the observed labels; block-evidence matrices can share a common prefix-sum precomputation only when the underlying design, weights, and missingness pattern permit it. This known-group construction is the supervised counterpart of the latent-template mixture in Section 10.

10 Latent groups and an EM algorithm

Plate model (latent-group template mixture). Group labels $z_s \in \{1, \dots, G\}$ are now *latent*, with Categorical prior π . Each group g carries its own template $\tau_g = (k_g, t^{(g)})$; subject-specific segment parameters $\theta_q^{(g,s)}$ are integrated out inside block evidences. The generative model is

$$\begin{aligned} \pi &\sim p(\pi), \quad \tau_g \sim p(\tau), \quad z_s \mid \pi \stackrel{\text{ind}}{\sim} \text{Cat}(\pi), \\ \theta_q^{(g,s)} &\stackrel{\text{ind}}{\sim} p(\theta \mid \cdot), \quad y_i^{(s)} \mid z_s, \tau_{z_s}, \theta_{q(i)}^{(z_s,s)} \stackrel{\text{ind}}{\sim} p(y \mid \theta_{q(i)}^{(z_s,s)}, w_i^{(s)}). \end{aligned} \tag{29}$$

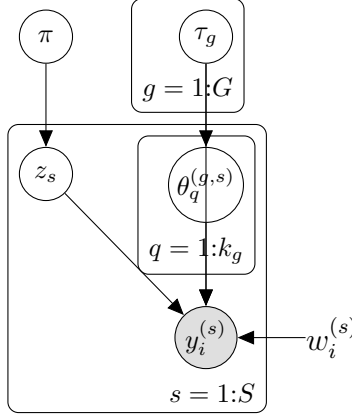


Figure 4 Plate model for the latent-group template mixture. Unshaded z_s are latent; EM alternates between exact E-step responsibilities $p(z_s = g \mid y^{(s)}, \pi, \tau)$ and M-step max-sum updates of the templates τ_g .

For unknown group membership we use a *latent template-mixture* model. Each latent group carries a single segmentation template that is shared across the subjects assigned to that group, while subject-specific segment parameters are still integrated out inside each block evidence. This is the cleanest setting in which EM updates remain exact and interpretable.

Let $\tau_g = (k_g, t^{(g)})$ denote the template of group g , where $1 \leq k_g \leq k_{\max}$ and $0 = t_0^{(g)} < \dots < t_{k_g}^{(g)} = n$. Conditional on membership in group g , subject s is assigned the positive prior-weighted template score

$$S_g(y^{(s)}; \tau_g) = \frac{p(k_g)}{C_{k_g}} \prod_{q=1}^{k_g} \tilde{A}_{t_{q-1}^{(g)} t_q^{(g)}}^{(0,s)}, \quad (30)$$

where $\tilde{A}_{ij}^{(0,s)} = A_{ij}^{(0,s)} g(\Delta_x(i, j))$ is the subject-specific block evidence with any length factor already absorbed. This score defines the finite template-mixture objective optimized below; it is distinct from a fully Bayesian posterior over all group segmentations. The group mixture weights are nonnegative with $\sum_g \pi_g = 1$; implementations either deactivate empty groups with $\pi_g = 0$ or use a small Dirichlet pseudocount/floor to keep every group active.

Introduce latent indicators $z_{sg} \in \{0, 1\}$ with $\sum_g z_{sg} = 1$. The complete-data log-likelihood is

$$\mathcal{J}(z, \pi, \tau) = \sum_{s=1}^S \sum_{g=1}^G z_{sg} [\log \pi_g + \log S_g(y^{(s)}; \tau_g)].$$

Let $r_{sg} := \mathbb{E}[z_{sg} \mid y, \pi, \tau]$ denote the usual responsibilities.

E-step. Given current (π, τ) ,

$$r_{sg} = \frac{\pi_g S_g(y^{(s)}; \tau_g)}{\sum_{g'=1}^G \pi_{g'} S_{g'}(y^{(s)}; \tau_{g'})}. \quad (31)$$

M-step for mixing weights. For fixed responsibilities,

$$\pi_g \leftarrow \frac{1}{S} \sum_{s=1}^S r_{sg}.$$

M-step for group templates. For fixed r , write $n_g(r) := \sum_{s=1}^S r_{sg}$. The τ_g -dependent part of the EM auxiliary function is

$$Q_g(\tau_g) = \sum_{s=1}^S r_{sg} \log S_g(y^{(s)}; \tau_g) = n_g(r) \{\log p(k_g) - \log C_{k_g}\} + \sum_{q=1}^{k_g} B_{t_{q-1}^{(g)} t_q^{(g)}}^{(g)},$$

where the responsibility-weighted block score is

$$B_{ij}^{(g)} := n_g(r) \log g(\Delta_x(i, j)) + \sum_{s=1}^S r_{sg} \log A_{ij}^{(0,s)}. \quad (32)$$

Thus the template-prior normalizer and the length factor are scaled by the same effective group size $n_g(r)$ as the rest of the prior-weighted score. For each group g , maximizing $Q_g(\tau_g)$ is then exactly the same max-sum segmentation problem as in Section 14, except that the block score is replaced by $B_{ij}^{(g)}$ and the count-specific offset $n_g(r) \{\log p(k) - \log C_k\}$ is added when selecting k_g . In other words, the M-step is an exact responsibility-weighted backtracking DP for the stated finite objective.

Theorem 10.1 (Monotone EM updates for the template mixture). *For the finite template-mixture objective defined by (30), the EM iteration consisting of, in order, (i) the exact responsibility update (31) given current $(\pi^{(t)}, \tau^{(t)})$ to produce $r^{(t+1)}$; (ii) the closed-form mixing-weight update $\pi^{(t+1)} = \arg \max_{\pi} Q(\pi, \tau^{(t)}; r^{(t+1)})$; and (iii) the exact max-sum DP update of each template $\tau^{(t+1)} = \arg \max_{\tau} Q(\pi^{(t+1)}, \tau; r^{(t+1)})$ via (32) does not decrease the finite-mixture objective*

$$\ell_{\star}(\pi, \tau) = \sum_{s=1}^S \log \left(\sum_{g=1}^G \pi_g S_g(y^{(s)}; \tau_g) \right).$$

Any fixed point of these updates, with deterministic tie-breaking for equal-scoring templates, is a coordinatewise maximizer of the EM auxiliary function. Across restarts, the selected solution is the run with the largest achieved finite-mixture objective.

Proof. The E-step with current $(\pi^{(t)}, \tau^{(t)})$ computes the exact conditional responsibilities for the current finite set of template scores, which maximizes the standard EM lower bound for $\ell_{\star}$ with respect to the assignment distribution q for fixed parameters. The M-step maximizes the auxiliary function $Q(\pi, \tau; r^{(t+1)}) := \sum_{s,g} r_{sg}^{(t+1)} [\log \pi_g + \log S_g(y^{(s)}; \tau_g)]$ in two coordinates: (ii) over $\pi \in \Delta^{G-1}$ (closed-form simplex maximizer $\pi_g^{(t+1)} = S^{-1} \sum_s r_{sg}^{(t+1)}$ when no pseudocount is used), and (iii) over each τ_g independently. The latter is an additive max-sum segmentation objective with block score (32) and count offset $n_g(r) \{\log p(k) - \log C_k\}$, exactly maximizable by DP and backtracking. Since each coordinate update exactly maximizes its factor, $Q(\pi^{(t+1)}, \tau^{(t+1)}; r^{(t+1)}) \geq Q(\pi^{(t)}, \tau^{(t)}; r^{(t+1)})$, and the finite-mixture objective $\ell_{\star}$ is non-decreasing by the generalized EM monotonicity theorem (Dempster et al., 1977). A fixed point satisfies the coordinatewise maximization conditions for the auxiliary function, rather than differentiable stationarity in a continuous template space. $\square$

Implementation. Algorithm 3 emphasizes the separation between (i) exact responsibility updates that use current group templates and (ii) exact template updates obtained from responsibility-weighted block scores. Because the template space is discrete while the mixture weights are continuous, multiple local optima occur. Use at least $R \geq 5$ restarts from either (a) a k -means clustering of the per-subject pooled block-evidence matrices, (b) templates drawn from the partition prior, or (c) a warm start derived from a known-groups solution on a small labeled subset when available. Declare convergence when (i) no template changes between two successive iterations, (ii) the finite-mixture objective change falls below a user tolerance (e.g., 10^{-6} relative), and (iii) deterministic tie-breaking would choose the same templates if the weighted score matrix were recomputed. Template stability alone is a useful diagnostic but does not by itself certify that all objective and tie-handling conditions have been met.

Algorithm 3: Latent-template EM for grouped segmentation (log-space)

Input: Per-subject log block evidences $\{\ell_{ij}^{(0,s)} := \log A_{ij}^{(0,s)}\}_{s=1}^S$; number of groups G ; $k_{\max}$; prior $\log p(k)$; length factor g ; tolerance ϵ

- 1 Initialize mixture weights $\{\pi_g\}_{g=1}^G$ and template segmentations $\{\tau_g = (k_g, t^{(g)})\}_{g=1}^G$ (several restarts; cf. initialization guidance above);
- 2 **repeat**
- 3 **E-step (log-space; order: responsibilities first);**
- 4 **for** $s = 1$ **to** S **do**
- 5 **for** $g = 1$ **to** G **do**
- 6 $\log S_g(y^{(s)}; \tau_g) \leftarrow \log p(k_g) - \log C_{k_g} + \sum_{q=1}^{k_g} \ell_{t_{q-1}^{(g)} t_q^{(g)}}^{(0,s)} + \sum_{q=1}^{k_g} \log g(\Delta_x(t_{q-1}^{(g)}, t_q^{(g)}))$;
- 7 $\log u_{sg} \leftarrow \log \pi_g + \log S_g(y^{(s)}; \tau_g)$;
- 8 $\log r_{sg} \leftarrow \log u_{sg} - \text{logsumexp}_{g'}(\log u_{sg'})$; $r_{sg} \leftarrow \exp(\log r_{sg})$;
- 9 Accumulate finite-mixture objective $\ell_\star^{(t+1)} \leftarrow \sum_s \text{logsumexp}_g(\log u_{sg})$;
- 10 **M-step (in the order weights then templates);**
- 11 $\pi_g \leftarrow S^{-1} \sum_s r_{sg}$ for each g ;
- 12 **for** $g = 1$ **to** G **do**
- 13 Set $n_g \leftarrow \sum_s r_{sg}$ and form scores $B_{ij}^{(g)} = n_g \log g(\Delta_x(i, j)) + \sum_{s=1}^S r_{sg} \ell_{ij}^{(0,s)}$;
- 14 Run max-sum DP/backtracking over $k \in \{1, \dots, k_{\max}\}$ using score matrix $B^{(g)}$ and offset $n_g \{\log p(k) - \log C_k\}$;
- 15 Update $\tau_g = (k_g, t^{(g)})$ to the maximizing template;
- 16 **until** *template vector stable AND* $|\ell_\star^{(t+1)} - \ell_\star^{(t)}| < \epsilon(1 + |\ell_\star^{(t)}|)$;

Output: $\{\pi_g\}$, $\{\tau_g\}$, and responsibilities $\{r_{sg}\}$

11 Family-specific derivations

We now instantiate (15)–(16) for three closed-form conjugate families (Gaussian, Poisson, and Binomial), an overdispersed count model with a conjugate distributional parameter, and one low-dimensional quadrature example for $(0, 1)$ -valued responses. In each case we give (i) the

segment model and prior, (ii) posterior hyperparameters, (iii) the block evidence $A_{ij}^{(0)}$, and (iv) first two posterior moments on the scale used by the Bayes-curve interface. For non-Gaussian families this scale must be stated explicitly: natural-parameter, distribution-parameter, and observation-mean moments are not interchangeable. Implementation routines are provided immediately after each derivation (Algorithms 4–8).

11.1 Gaussian with known variance (heteroscedastic via weights)

Assume

$$y_t \mid \mu \sim \mathcal{N}(\mu, \sigma^2/w_t), \quad \mu \sim \mathcal{N}(\nu, \rho^2),$$

with known σ^2 and positive weights w_t (e.g. inverse known variances or exposure). For a block $(i, j]$, define

$$\begin{aligned} n_{ij} &:= j - i, & W_{ij} &:= \sum_{t=i+1}^j w_t, & S_{ij} &:= \sum_{t=i+1}^j w_t y_t, \\ Q_{ij} &:= \sum_{t=i+1}^j w_t (y_t - \nu)^2, & \kappa_{ij} &:= \rho^2 W_{ij} + \sigma^2. \end{aligned}$$

Then

$$\mu \mid (i, j] \sim \mathcal{N}\left(\frac{\rho^2 S_{ij} + \sigma^2 \nu}{\kappa_{ij}}, \frac{\sigma^2 \rho^2}{\kappa_{ij}}\right).$$

The integrated block evidence is

$$\begin{aligned} A_{ij}^{(0)} &= \int \left[\prod_{t=i+1}^j \mathcal{N}(y_t \mid \mu, \sigma^2/w_t) \right] \mathcal{N}(\mu \mid \nu, \rho^2) d\mu \\ &= (2\pi\sigma^2)^{-n_{ij}/2} \left(\prod_{t=i+1}^j w_t^{1/2} \right) \left(1 + \frac{\rho^2}{\sigma^2} W_{ij} \right)^{-1/2} \exp \left\{ -\frac{Q_{ij}}{2\sigma^2} + \frac{(S_{ij} - \nu W_{ij})^2}{2(\sigma^2 W_{ij} + \sigma^4/\rho^2)} \right\}. \end{aligned}$$

Consequently,

$$A_{ij}^{(1)} = A_{ij}^{(0)} \mathbb{E}[\mu \mid (i, j)], \quad A_{ij}^{(2)} = A_{ij}^{(0)} \left(\mathbb{V}[\mu \mid (i, j)] + \mathbb{E}[\mu \mid (i, j)]^2 \right).$$

Implementation. Algorithm 4 gives a numerically stable implementation of the Gaussian block log-evidence and posterior moments using cumulative weighted sums. The routine is written so that the DP layer consumes only $\log \mathcal{L}_{ij}$ and does not depend on Gaussian-specific algebra.

Algorithm 4: GaussianBlock($y, w, i, j, \nu, \rho^2, \sigma^2$)

Input: Block indices $(i, j]$; observations $y_{i+1:j}$; weights $w_{i+1:j}$

- 1 $n \leftarrow j - i$; $W \leftarrow \sum_{t=i+1}^j w_t$; $S \leftarrow \sum_{t=i+1}^j w_t y_t$; $Q \leftarrow \sum_{t=i+1}^j w_t (y_t - \nu)^2$;
- 2 $\kappa \leftarrow \rho^2 W + \sigma^2$;
- 3 $\log A0 \leftarrow -\frac{1}{2}n \log(2\pi) - \frac{1}{2} \sum_{t=i+1}^j \log(\sigma^2/w_t) - \frac{Q}{2\sigma^2} + \frac{(S - \nu W)^2}{2(\sigma^2 W + \sigma^4/\rho^2)} - \frac{1}{2} \log(1 + \frac{\rho^2}{\sigma^2} W)$;
- 4 $A0 \leftarrow \exp(\log A0)$;
- 5 $\mu_{\text{mean}} \leftarrow (\rho^2 S + \sigma^2 \nu)/\kappa$; $\mu_{\text{var}} \leftarrow \sigma^2 \rho^2/\kappa$;

Output: $A0, \mu_{\text{mean}}, \mu_{\text{var}}$

11.2 Poisson with exposure

Assume a blockwise constant rate λ with exposure weights $w_t > 0$:

$$y_t \mid \lambda \sim \text{Pois}(\lambda w_t), \quad \lambda \sim \text{Gamma}(a_0, b_0) \quad (\text{shape-rate}).$$

For a block $(i, j]$, define

$$C_{ij} := \sum_{t=i+1}^j y_t, \quad W_{ij} := \sum_{t=i+1}^j w_t, \quad H_{ij} := \sum_{t=i+1}^j \left[y_t \log w_t - \log(y_t!) \right].$$

Then

$$\lambda \mid (i, j] \sim \text{Gamma}(a_0 + C_{ij}, b_0 + W_{ij}),$$

so

$$\mathbb{E}[\lambda \mid (i, j]] = \frac{a_0 + C_{ij}}{b_0 + W_{ij}}, \quad \mathbb{V}[\lambda \mid (i, j]] = \frac{a_0 + C_{ij}}{(b_0 + W_{ij})^2}.$$

The integrated block evidence is

$$A_{ij}^{(0)} = \exp\{H_{ij}\} \frac{b_0^{a_0} \Gamma(a_0 + C_{ij})}{\Gamma(a_0) (b_0 + W_{ij})^{a_0 + C_{ij}}}.$$

Moments satisfy the usual relation

$$A_{ij}^{(1)} = A_{ij}^{(0)} \mathbb{E}[\lambda \mid (i, j]], \quad A_{ij}^{(2)} = A_{ij}^{(0)} \left(\mathbb{V}[\lambda \mid (i, j]] + \mathbb{E}[\lambda \mid (i, j]]^2 \right).$$

Implementation. Algorithm 5 summarizes the Poisson-with-exposure block evidence computation (and associated posterior summaries) in a form that matches the BayesBreak block-evidence interface.

Algorithm 5: PoissonBlock(y, w, i, j, a_0, b_0)

Input: Block $(i, j]$; counts $y_{i+1:j}$; exposures $w_{i+1:j}$

- 1 $C \leftarrow \sum_{t=i+1}^j y_t$; $W \leftarrow \sum_{t=i+1}^j w_t$;
 - 2 $\log A0 \leftarrow$
 $\sum_{t=i+1}^j y_t \log w_t - \sum_{t=i+1}^j \log(y_t!) + a_0 \log b_0 + \log \Gamma(a_0 + C) - \log \Gamma(a_0) - (a_0 + C) \log(b_0 + W)$;
 - 3 $A0 \leftarrow \exp(\log A0)$;
 - 4 $\lambda_{\text{mean}} \leftarrow (a_0 + C)/(b_0 + W)$; $\lambda_{\text{var}} \leftarrow (a_0 + C)/(b_0 + W)^2$;
- Output:** $A0, \lambda_{\text{mean}}, \lambda_{\text{var}}$
-

11.3 Binomial with Beta prior

Assume blockwise constant success probability p with heterogeneous trial counts m_t :

$$y_t \mid p \sim \text{Binom}(m_t, p), \quad p \sim \text{Beta}(a_0, b_0).$$

For a block $(i, j]$, define

$$C_{ij} := \sum_{t=i+1}^j y_t, \quad M_{ij} := \sum_{t=i+1}^j m_t, \quad H_{ij} := \sum_{t=i+1}^j \log \binom{m_t}{y_t}.$$

Then $p \mid (i, j] \sim \text{Beta}(a_0 + C_{ij}, b_0 + M_{ij} - C_{ij})$, so

$$\mathbb{E}[p \mid (i, j)] = \frac{a_0 + C_{ij}}{a_0 + b_0 + M_{ij}}, \quad \mathbb{V}[p \mid (i, j)] = \frac{(a_0 + C_{ij})(b_0 + M_{ij} - C_{ij})}{(a_0 + b_0 + M_{ij})^2(a_0 + b_0 + M_{ij} + 1)}.$$

The block evidence is

$$A_{ij}^{(0)} = \exp\{H_{ij}\} \frac{B(a_0 + C_{ij}, b_0 + M_{ij} - C_{ij})}{B(a_0, b_0)}.$$

Again,

$$A_{ij}^{(1)} = A_{ij}^{(0)} \mathbb{E}[p \mid (i, j)], \quad A_{ij}^{(2)} = A_{ij}^{(0)} \left(\mathbb{V}[p \mid (i, j)] + \mathbb{E}[p \mid (i, j)]^2 \right).$$

Implementation. Algorithm 6 provides pseudocode for Binomial/Bernoulli block evidence evaluation under Beta conjugacy, along with posterior mean/variance computations used for segment-wise moments and MAP/Bayes curve summaries.

Algorithm 6: BinomialBlock(y, m, i, j, a_0, b_0)

Input: Block $(i, j]$; successes $y_{i+1:j}$; trials $m_{i+1:j}$

- 1 $C \leftarrow \sum_{t=i+1}^j y_t$; $M \leftarrow \sum_{t=i+1}^j m_t$;
- 2 $\log A0 \leftarrow \sum_{t=i+1}^j \log \binom{m_t}{y_t} + \log B(a_0 + C, b_0 + M - C) - \log B(a_0, b_0)$;
- 3 $A0 \leftarrow \exp(\log A0)$;
- 4 $p_{\text{mean}} \leftarrow (a_0 + C)/(a_0 + b_0 + M)$;
- 5 $p_{\text{var}} \leftarrow \frac{(a_0 + C)(b_0 + M - C)}{(a_0 + b_0 + M)^2(a_0 + b_0 + M + 1)}$;

Output: $A0, p_{\text{mean}}, p_{\text{var}}$

11.4 Negative-Binomial block (overdispersed counts) with fixed dispersion

For count data with extra-Poisson variability, a natural block model fixes a dispersion parameter $r > 0$ and treats the success probability $p \in (0, 1)$ per segment with a conjugate Beta prior:

$$y_t \mid p \sim \text{NegBin}(r, p), \quad p \sim \text{Beta}(a_0, b_0),$$

where $\text{NegBin}(r, p)$ has pmf $\binom{y+r-1}{y} (1-p)^y p^r$ and count mean $r(1-p)/p$. For a block $(i, j]$, set $C_{ij} := \sum_{t=i+1}^j y_t$, $N_{ij} := \sum_{t=i+1}^j r_t$ (with $r_t \equiv r$ in the fixed-dispersion case), and $H_{ij} := \sum_{t=i+1}^j \log \binom{y_t+r_t-1}{y_t}$. Beta-NegBin conjugacy gives $p \mid (i, j] \sim \text{Beta}(a_B, b_B)$ with $a_B = a_0 + N_{ij}$ and $b_B = b_0 + C_{ij}$, so

$$A_{ij}^{(0)} = \exp\{H_{ij}\} \frac{B(a_B, b_B)}{B(a_0, b_0)}.$$

The Beta posterior moments of p are useful diagnostics, but the Bayes-curve target on the observation scale is a count mean. For a future count with dispersion r_* , this target is $m_*(p) = r_*(1 - p)/p$, giving

$$\begin{aligned}\mathbb{E}[m_*(p) \mid (i, j)] &= r_* \frac{b_B}{a_B - 1} \quad (a_B > 1), \\ \mathbb{V}[m_*(p) \mid (i, j)] &= r_*^2 \frac{b_B(a_B + b_B - 1)}{(a_B - 1)^2(a_B - 2)} \quad (a_B > 2).\end{aligned}$$

Moment numerators $A_{ij}^{(1)}$ and $A_{ij}^{(2)}$ therefore multiply $A_{ij}^{(0)}$ by the corresponding observation-mean moments, not by raw Beta moments of p , unless the exported quantity is explicitly labeled as parameter-scale. When r is unknown, a semi-conjugate EM or a one-dimensional quadrature in r can be layered on top; we omit the details.

Implementation. Algorithm 7 summarizes the Negative-Binomial block computation in the same interface as Algorithms 4–6.

Algorithm 7: NegBinBlock(y, r, i, j, a_0, b_0)

Input: Block $(i, j]$; counts $y_{i+1:j}$; dispersion r (possibly per- t)

- 1 $C \leftarrow \sum_{t=i+1}^j y_t$; $N \leftarrow \sum_{t=i+1}^j r_t$;
 - 2 $\log A0 \leftarrow \sum_{t=i+1}^j \log \binom{y_t + r_t - 1}{y_t} + \log B(a_0 + N, b_0 + C) - \log B(a_0, b_0)$;
 - 3 $A0 \leftarrow \exp(\log A0)$;
 - 4 $a_B \leftarrow a_0 + N$; $b_B \leftarrow b_0 + C$;
 - 5 **mean_per_disp** $\leftarrow b_B/(a_B - 1)$ if $a_B > 1$;
var_per_disp $\leftarrow b_B(a_B + b_B - 1)/\{(a_B - 1)^2(a_B - 2)\}$ if $a_B > 2$;
- Output:** $A0$, posterior summaries for p , and observation-mean moments
 $r_* \times \text{mean_per_disp}$ when a prediction dispersion r_* is specified
-

11.5 Beta-distributed observations with known precision

Assume $y_t \in (0, 1)$ with latent mean μ constant within a block and known precision $\phi_t > 0$ (with $\phi_t \equiv \phi$ as the fixed-precision special case):

$$y_t \mid \mu \sim \text{Beta}(\phi_t \mu, \phi_t(1 - \mu)), \quad \mu \sim \text{Beta}(a_0, b_0).$$

Conjugacy in μ is not available, but all integrals remain one-dimensional and can be computed by deterministic quadrature. We do not require or claim global log-concavity of the induced posterior in μ ; instead, implementations check quadrature stability by increasing the node count and by controlling behavior near the endpoints of $(0, 1)$. For a block $(i, j]$ define

$$\ell_{ij}(\mu) := \sum_{t=i+1}^j \log \text{BetaPDF}(y_t \mid \phi_t \mu, \phi_t(1 - \mu)), \quad \log \pi(\mu) := \log \text{BetaPDF}(\mu \mid a_0, b_0).$$

Then for $r = 0, 1, 2$,

$$A_{ij}^{(r)} = \int_0^1 \mu^r \exp \{ \ell_{ij}(\mu) + \log \pi(\mu) \} d\mu.$$

In practice, approximate via Gauss–Legendre (or similar) quadrature with nodes $\{\mu_g\}$ and weights $\{w_g\}$:

$$A_{ij}^{(r)} \approx \sum_{g=1}^G w_g \mu_g^r \exp \{ \ell_{ij}(\mu_g) + \log \pi(\mu_g) \}.$$

Implementation. Algorithm 8 implements this one-dimensional quadrature routine. It is useful for continuous proportions and other $(0, 1)$ -valued signals, and is a deterministic non-conjugate block integral rather than a conjugate closed form.

Algorithm 8: BetaObsBlock($y, \phi, i, j, a_0, b_0, \{\mu_g, w_g\}_{g=1}^G$)

Input: Block $(i, j]$; observations $y_{i+1:j} \in (0, 1)$; known precisions $\phi_{i+1:j}$; Beta prior (a_0, b_0) ; quadrature nodes $\{\mu_g, w_g\}$

```

1  $S_0 \leftarrow 0, S_1 \leftarrow 0, S_2 \leftarrow 0;$ 
2 for  $g = 1$  to  $G$  do
3    $\log \ell_g \leftarrow \sum_{t=i+1}^j \log \text{BetaPDF}(y_t \mid \phi_t \mu_g, \phi_t (1 - \mu_g));$ 
4    $\log \pi_g \leftarrow \log \text{BetaPDF}(\mu_g \mid a_0, b_0);$ 
5    $v_g \leftarrow \exp(\log \ell_g + \log \pi_g);$ 
6    $S_0 \leftarrow S_0 + w_g v_g; \quad S_1 \leftarrow S_1 + w_g v_g \mu_g; \quad S_2 \leftarrow S_2 + w_g v_g \mu_g^2;$ 
7  $A_0 \leftarrow S_0; \quad \mu_{\text{mean}} \leftarrow S_1/S_0; \quad \mu_{\text{var}} \leftarrow S_2/S_0 - (\mu_{\text{mean}})^2;$ 

```

Output: $A_0, \mu_{\text{mean}}, \mu_{\text{var}}$

11.6 Summary of block families

Table 2 collects the block families derived in this section in a single reference view. The conjugate rows expose closed-form ratios of normalizers, while the Beta-observation row supplies deterministic quadrature values. In both cases the DP layer of Section 6 only requires a compatible block-evidence array and is invariant to the family used to fill that array.

Family	Prior	Posterior	Block evidence $A_{ij}^{(0)}$	Mean $\mathbb{E}[m(\theta) (i, j]]$
Gaussian($\mu, \sigma^2/w$)	$\mathcal{N}(\nu, \rho^2)$	$\mathcal{N}(\nu_B, \rho_B^2)$	closed-form normal ratio	$(\rho^2 S + \sigma^2 \nu)/\kappa$
Poisson(λw)	Gamma(a_0, b_0)	Gamma($a_0 + C, b_0 + W$)	$\propto b_0^{a_0} \Gamma(a_0 + C)/(b_0 + W)^{a_0 + C}$	$(a_0 + C)/(b_0 + W)$
Binomial(m, p)	Beta(a_0, b_0)	Beta($a_0 + C, b_0 + M - C$)	$B(a_0 + C, b_0 + M - C)/B(a_0, b_0)$	$(a_0 + C)/(a_0 + b_0 + M)$
NegBin(r, p)	Beta(a_0, b_0)	Beta(a_B, b_B)	$B(a_B, b_B)/B(a_0, b_0)$	$r_* b_B/(a_B - 1)$
Beta-obs(μ, ϕ_t)	Beta(a_0, b_0)	(1D quadrature)	Gauss–Legendre nodes	from quadrature

Table 2 Block families derived in Section 11. Base-measure factor $\exp\{H_{ij}\}$ is implicit in the conjugate rows. The first three rows are closed-form conjugate examples; the Negative-Binomial row distinguishes parameter-scale Beta moments from the observation-mean target; and the Beta-observation row is non-conjugate but remains compatible with the DP layer through one-dimensional quadrature.

11.7 Posterior-predictive formulas for common conjugate families (optional)

For groups that export segment-level conjugate posteriors, the segment posterior-predictive (42) reduces to standard closed forms:

Gaussian (known variance). If a segment posterior is $\mu \mid \text{train} \sim \mathcal{N}(\nu_B, \rho_B^2)$ and $y \mid \mu \sim \mathcal{N}(\mu, \sigma^2/w)$, then the posterior-predictive for a new point is Gaussian: $y \mid \text{train} \sim \mathcal{N}(\nu_B, \rho_B^2 + \sigma^2/w)$. For M new points in the segment with weights $w_1, \dots, w_M$, the joint posterior-predictive is the multivariate Gaussian

$$\mathbf{y}^{\text{new}} \mid \text{train} \sim \mathcal{N}_M(\nu_B \mathbf{1}_M, \rho_B^2 \mathbf{1}_M \mathbf{1}_M^\top + \text{diag}(\sigma^2/w_1, \dots, \sigma^2/w_M)),$$

which has rank-one-plus-diagonal covariance and whose log density can be evaluated in $\mathcal{O}(M)$ by the Sherman–Morrison identity.

Poisson–Gamma. If $\lambda \mid \text{train} \sim \text{Gamma}(a_B, b_B)$ and $y \mid \lambda \sim \text{Pois}(\lambda w)$, then the posterior predictive is negative binomial in y with parameters determined by (a_B, b_B, w) ; for multiple points, aggregate counts and exposures and apply (42).

Binomial–Beta. If $p \mid \text{train} \sim \text{Beta}(a_B, b_B)$ and $y \mid p \sim \text{Binom}(m, p)$, then the posterior predictive is Beta-Binomial:

$$p(y \mid m, \text{train}) = \binom{m}{y} \frac{B(a_B + y, b_B + m - y)}{B(a_B, b_B)}.$$

12 Non-conjugate GLMs: block evidence, theory, and approximations

Plate model (non-conjugate single segment with PG augmentation). The generative graph for a non-conjugate block $(i, j]$ is identical to Figure 1 with $\theta \sim \pi(\theta)$ replaced by a non-conjugate prior (e.g., Gaussian on a log or logit scale). For Pólya–Gamma-augmented binomial logistic blocks (Theorem 12.4), introducing the latent scale-mixture variables $\omega_t \sim \text{PG}(m_t, 0)$ restores conditional conjugacy and is reflected in the plate as a second observed-indexed latent:

$$\theta \sim \mathcal{N}(\nu, \rho^2), \quad \omega_t \overset{\text{ind}}{\sim} \text{PG}(m_t, 0), \quad y_t \mid \theta, \omega_t, m_t \overset{\text{ind}}{\sim} \text{Binom}(m_t, \sigma(\theta)) \text{ with (37),} \quad (33)$$

where σ is the logistic function.

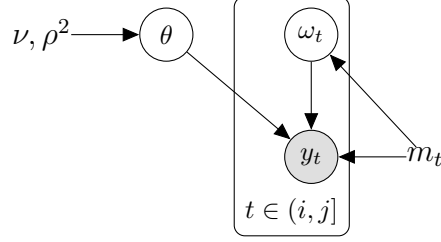


Figure 5 Plate model for a non-conjugate block with Pólya–Gamma augmentation. Conditional on ω_t , the block is quadratic in θ with a Gaussian posterior (Proposition 12.5); the same template applies to Laplace and variational routines with the ω_t node omitted.

In many applications the single-segment parameterization is not conjugate to the observation model—for example, log-normal priors for positive means, heavy-tailed shrinkage priors, or logistic/negative-binomial links with structural offsets. This section develops a plug-in toolkit for computing approximate block evidences $A_{ij}^{(0)}$ and moments $A_{ij}^{(r)}$ when conjugacy is unavailable, while keeping the dynamic-programming layer of Section 6 unchanged. When approximate block evidences are used, the DP is exact for the approximate block-score model; additional error statements require explicit blockwise approximation control.

We focus on one-dimensional (or low-dimensional) canonical GLMs, where for a block $(i, j]$ we must approximate integrals of the form

$$A_{ij}^{(0)} = \int \left[\prod_{t=i+1}^j p(y_t | \theta) \right] p(\theta) d\theta, \quad A_{ij}^{(r)} = \int \left[\prod_{t=i+1}^j p(y_t | \theta) \right] p(\theta) m(\theta)^r d\theta,$$

with $m(\theta) := \mathbb{E}[Y | \theta]$. We describe four complementary approaches: (i) first-order Laplace approximation; (ii) variational lower bounds (Jaakkola–Jordan (Jaakkola and Jordan, 2000) type for logistic regression and mean-field for more general GLMs); (iii) expectation propagation (EP); and (iv) Pólya–Gamma augmentation for logistic and negative-binomial blocks. Throughout we retain weights w_t and cumulative statistics (S_{ij}, W_{ij}, H_{ij}) .

12.1 Model and notation for GLM blocks

Consider a one-parameter canonical GLM on a block $(i, j]$:

$$\log p(y_t | \theta) = w_t \{y_t \theta - b(\theta)\} + \log h(y_t, w_t),$$

where b is the cumulant function and h collects base measures (including any w_t -dependent terms). The block log-likelihood is

$$\ell_{ij}(\theta) := \sum_{t=i+1}^j w_t \{y_t \theta - b(\theta)\} + H_{ij}, \quad H_{ij} := \sum_{t=i+1}^j \log h(y_t, w_t).$$

Let $\pi(\theta) = p(\theta)$ be an arbitrary prior (not necessarily conjugate), and write $\Psi_{ij}(\theta) := \ell_{ij}(\theta) + \log \pi(\theta)$ for the block log-posterior.

Derivatives are

$$\ell'_{ij}(\theta) = S_{ij} - W_{ij}b'(\theta), \quad \ell''_{ij}(\theta) = -W_{ij}b''(\theta),$$

and

$$\Psi'_{ij}(\theta) = \ell'_{ij}(\theta) + (\log \pi)'(\theta), \quad \Psi''_{ij}(\theta) = \ell''_{ij}(\theta) + (\log \pi)''(\theta).$$

12.2 Existence and uniqueness of block modes

Lemma 12.1 (Strict log-concavity and uniqueness). *Suppose $b''(\theta) > 0$ for all θ (canonical GLM), the prior is log-concave ($(\log \pi)''(\theta) \leq 0$), and $W_{ij} > 0$. Then Ψ_{ij} is strictly concave. If, in addition, Ψ_{ij} is proper and attains its supremum on the parameter domain (for example by coercive tails or compact-domain truncation), the maximizer $\hat{\theta}_{ij}$ exists and is unique.*

Proof. Since $W_{ij} > 0$ and $b''(\theta) > 0$, we have $-W_{ij}b''(\theta) < 0$. Adding $(\log \pi)''(\theta) \leq 0$ yields $\Psi''_{ij}(\theta) < 0$ wherever the derivatives exist, so Ψ_{ij} is strictly concave. Strict concavity gives uniqueness of any maximizer; existence is provided by the stated attainment assumption. $\square$

12.3 Laplace approximation for single-block evidence

Theorem 12.2 (First-order Laplace approximation). *Let $a \asymp b$ mean that a/b is bounded above and below by positive constants independent of (i, j) . Assume:*

- (A1) Ψ_{ij} is three times continuously differentiable on a neighborhood of its unique interior maximizer $\hat{\theta}_{ij}$;
- (A2) $H_{ij}^* := -\Psi''_{ij}(\hat{\theta}_{ij}) > 0$ and $H_{ij}^* \asymp W_{ij}$ as $W_{ij} \rightarrow \infty$;
- (A3) $|\Psi_{ij}^{(3)}(\theta)| \leq CW_{ij}$ on a neighborhood of $\hat{\theta}_{ij}$ for a constant C independent of (i, j) ;
- (A4) the contribution of the integral outside that neighborhood is negligible at the stated Laplace order relative to the local quadratic contribution. This is a substantive assumption on the block likelihood and prior, and it is logically separate from the strict log-concavity of Lemma 12.1: log-concavity guarantees that the maximizer $\hat{\theta}_{ij}$ is unique, but it does not by itself control the integral mass in the tails. The required tail bound must instead be supplied by additional coercivity of Ψ_{ij} or by compact-domain truncation of the parameter space.

Then the block evidence admits the first-order Laplace expansion

$$A_{ij}^{(0)} = \exp\{\Psi_{ij}(\hat{\theta}_{ij})\} \left(\frac{2\pi}{H_{ij}^*} \right)^{1/2} \left[1 + \mathcal{O}(W_{ij}^{-1}) \right], \quad (34)$$

and therefore

$$\log A_{ij}^{(0)} = \Psi_{ij}(\hat{\theta}_{ij}) + \frac{1}{2} \log(2\pi) - \frac{1}{2} \log H_{ij}^* + \mathcal{O}(W_{ij}^{-1}). \quad (35)$$

Moreover, for any twice continuously differentiable test function g with bounded derivatives near $\hat{\theta}_{ij}$, writing $M_{ij}^{[g]} := \int e^{\Psi_{ij}(\theta)} g(\theta) d\theta$ (an unnormalized weighted integral),

$$\frac{M_{ij}^{[g]}}{A_{ij}^{(0)}} = g(\hat{\theta}_{ij}) + \mathcal{O}(W_{ij}^{-1}). \quad (36)$$

Proof. Write $u = \sqrt{H_{ij}^*}(\theta - \hat{\theta}_{ij})$. A third-order Taylor expansion of Ψ_{ij} around $\hat{\theta}_{ij}$ gives

$$\Psi_{ij}(\theta) = \Psi_{ij}(\hat{\theta}_{ij}) - \frac{1}{2}u^2 + R_{ij}(u), \quad |R_{ij}(u)| \leq C'|u|^3/\sqrt{W_{ij}}$$

on the local neighborhood, using Assumptions (A2)–(A3). Integrating the leading quadratic term yields the Gaussian factor $(2\pi/H_{ij}^*)^{1/2}$, while the remainder contributes a relative $\mathcal{O}(W_{ij}^{-1})$ correction by the standard Laplace method; Assumption (A4) controls the tails and establishes (34)–(35). Applying the same expansion to the numerator $M_{ij}^{[g]}$ and using a second-order Taylor expansion of g around $\hat{\theta}_{ij}$ yields (36). We state the first-order ratio result: any explicit second-order correction depends jointly on derivatives of g and of Ψ_{ij} and is not captured by a universal $g''/(2H_{ij}^*)$ term. $\square$

Efficient Newton updates using block summaries. For canonical GLMs,

$$\Psi'_{ij}(\theta) = S_{ij} - W_{ij}b'(\theta) + (\log \pi)'(\theta), \quad \Psi''_{ij}(\theta) = -W_{ij}b''(\theta) + (\log \pi)''(\theta),$$

so Newton iteration is

$$\theta^{(m+1)} \leftarrow \theta^{(m)} - \frac{S_{ij} - W_{ij}b'(\theta^{(m)}) + (\log \pi)'(\theta^{(m)})}{-W_{ij}b''(\theta^{(m)}) + (\log \pi)''(\theta^{(m)})},$$

with cost independent of block length once S_{ij}, W_{ij} are available.

Implementation. Algorithm 9 implements the Laplace block-evidence approximation using a Newton or quasi-Newton solver for the block mode and the corresponding Hessian correction. The approximation is local but often highly accurate in moderate-to-large blocks.

Algorithm 9: GLMBlock_Laplace($y, w, i, j, \mathbf{b}, \text{prior}$)

Input: $y_{i+1:j}$, weights $w_{i+1:j}$; cumulant $b(\theta)$; prior $\pi(\theta)$

- 1 Compute $S \leftarrow \sum_{t=i+1}^j w_t y_t$; $W \leftarrow \sum_{t=i+1}^j w_t$; $H \leftarrow \sum_{t=i+1}^j \log h(y_t, w_t)$;
- 2 Define $\Psi(\theta) = H + S\theta - Wb(\theta) + \log \pi(\theta)$ and its derivatives;
- 3 Initialize $\theta^{(0)}$ (e.g., solve $b'(\theta) = S/W$ or use prior mean);
- 4 **for** $m = 0, 1, \dots$ *until convergence* **do**
- 5 $g \leftarrow S - Wb'(\theta^{(m)}) + (\log \pi)'(\theta^{(m)})$;
- 6 $h \leftarrow -Wb''(\theta^{(m)}) + (\log \pi)''(\theta^{(m)})$;
- 7 **if** $|g|/|h| < \varepsilon_{\text{newton}}$ **then**
- 8 **break**
- 9 $\theta^{(m+1)} \leftarrow \theta^{(m)} - g/h$;
- 10 $\hat{\theta} \leftarrow \theta^{(m)}$; $H^* \leftarrow -\Psi''(\hat{\theta})$;
- 11 $\log \hat{A}_{ij}^{(0)} \leftarrow \Psi(\hat{\theta}) + \frac{1}{2} \log(2\pi) - \frac{1}{2} \log H^*$;

Output: $\hat{A}_{ij}^{(0)}$ and (optional) moments via (36)

12.4 Variational lower bounds for GLM blocks

When a deterministic lower bound is desirable, introduce a variational distribution $q(\theta)$ and apply Jensen:

$$\log A_{ij}^{(0)} = \log \int q(\theta) \frac{e^{\Psi_{ij}(\theta)}}{q(\theta)} d\theta \geq \int q(\theta) \Psi_{ij}(\theta) d\theta + \mathcal{H}[q] := \mathcal{L}_{ij}(q).$$

Jaakkola–Jordan bound for logistic blocks. For Bernoulli/logistic likelihoods, a standard quadratic bound yields a tractable Gaussian variational posterior. One convenient form (for sigmoid σ) is:

$$\log \sigma(z) \geq \log \sigma(\xi) + \frac{z - \xi}{2} - \lambda(\xi)(z^2 - \xi^2), \quad \lambda(\xi) = \frac{\tanh(\xi/2)}{4\xi},$$

with variational parameter $\xi > 0$. Applying this bound to the appropriate signed linear predictors yields a quadratic surrogate in θ , so the optimal Gaussian $q(\theta)$ and ξ admit closed-form coordinate updates.

Proposition 12.3 (Monotone improvement of variational block bounds). *Coordinate-ascent updates on variational parameters (e.g. ξ for logistic blocks) and on a Gaussian $q(\theta)$ increase $\mathcal{L}_{ij}(q)$ until a stationary point; the resulting $\underline{A}_{ij}^{(0)} := \exp\{\mathcal{L}_{ij}(q)\}$ is a certified lower bound on $A_{ij}^{(0)}$.*

Proof. Each coordinate update is an exact maximization of $\mathcal{L}_{ij}(q)$ with respect to a subset of variables holding others fixed. Therefore $\mathcal{L}_{ij}(q)$ is non-decreasing under the updates. Because $\mathcal{L}_{ij}$ is a Jensen lower bound on $\log A_{ij}^{(0)}$, exponentiating yields a valid evidence lower bound. $\square$

Implementation. Algorithm 10 provides pseudocode for computing the Jaakkola–Jordan (JJ) variational lower bound and its associated fixed-point updates at the block level.

Algorithm 10: LogisticBlock_JJ($y, m, w, i, j, \text{prior}$) (variational lower bound)

Input: Binomial counts $y_t \sim \text{Binom}(m_t, p)$, logit link; prior $\pi(\theta)$

- 1 Initialize ξ and a Gaussian $q(\theta) = \mathcal{N}(\mu_q, \sigma_q^2)$;
 - 2 **repeat**
 - 3 Update the quadratic bound parameter $\xi \leftarrow \sqrt{\mathbb{E}_q[\theta^2]}$;
 - 4 Update Gaussian $q(\theta)$ by maximizing the block ELBO $\mathcal{L}_{ij}(q)$ (closed-form for quadratic surrogate);
 - 5 **until** converged;
 - 6 Return $\underline{A}_{ij}^{(0)} = \exp\{\mathcal{L}_{ij}(q)\}$ and moments from q .
-

12.5 Expectation propagation (EP) for GLM blocks

Expectation propagation (EP) (Minka, 2001) provides a flexible deterministic approximation to the block posterior and block evidence by iteratively refining local site approximations

and matching moments. EP approximates each likelihood term $p(y_t | \theta)$ by a Gaussian site $\tilde{t}_t(\theta) \propto \exp\{-\frac{1}{2}a_t\theta^2 + b_t\theta\}$ so that moments match those of the tilted distribution. After convergence, the approximate posterior is Gaussian and yields an approximate evidence $\tilde{A}_{ij}^{(0)} = \int \pi(\theta) \prod \tilde{t}_t(\theta) d\theta$. EP is neither a bound nor guaranteed to converge, but is often accurate in canonical GLMs.

Implementation. Algorithm 11 summarizes an expectation-propagation (EP) routine for approximating the GLM block evidence, highlighting the cavity/site updates and moment matching.

Algorithm 11: GLMBlock_EP(y, w, i, j, prior)

Input: Canonical GLM; prior $\pi(\theta)$

- 1 Initialize site parameters $\{a_t, b_t\}_{t=i+1}^j$; approximate posterior $\tilde{p}(\theta) \propto \pi(\theta) \prod \tilde{t}_t(\theta)$;
 - 2 **repeat**
 - 3 **for** $t = i + 1$ **to** j **do**
 - 4 Form cavity $\tilde{p}^{\setminus t}(\theta) \propto \tilde{p}(\theta) / \tilde{t}_t(\theta)$;
 - 5 Form tilted $p^{\text{tilt}}(\theta) \propto \tilde{p}^{\setminus t}(\theta) p(y_t | \theta)$;
 - 6 Moment-match p^{tilt} and update site $\tilde{t}_t(\theta)$;
 - 7 **until** moments stable;
 - 8 Return approximate evidence $\tilde{A}_{ij}^{(0)} = \int \tilde{p}(\theta) d\theta$ and moments from $\tilde{p}$.
-

12.6 Pólya–Gamma augmentation (logistic and negative-binomial)

Theorem 12.4 (Pólya–Gamma identity and conditional conjugacy). *For binomial logistic blocks with counts m_t and successes y_t , let $w_t \geq 0$ denote either one or a declared likelihood-power/replication weight, define $b_t = w_t m_t$ and $\kappa_t = w_t(y_t - \frac{m_t}{2})$, and introduce latent $\omega_t \sim \text{PG}(b_t, 0)$. Then the weighted logistic likelihood term admits the Gaussian scale-mixture representation*

$$\left\{ \frac{(e^\theta)^{y_t}}{(1 + e^\theta)^{m_t}} \right\}^{w_t} = 2^{-b_t} e^{\kappa_t \theta} \int_0^\infty e^{-\omega_t \theta^2 / 2} p(\omega_t) d\omega_t. \quad (37)$$

Conditionally on $\omega_{i+1:j}$, the block likelihood is quadratic in θ :

$$\log p(y_{i+1:j} | \theta, \omega) = -\frac{1}{2}\theta^2 \sum_{t=i+1}^j \omega_t + \theta \sum_{t=i+1}^j \kappa_t + \text{const}. \quad (38)$$

With a Gaussian prior $\pi(\theta) = \mathcal{N}(\nu, \rho^2)$, the conditional posterior $p(\theta | y, \omega)$ is Gaussian.

Proof. Step 1: the identity. Equation (37) is the defining Pólya–Gamma augmentation of Polson et al. (2013): it expresses the binomial-logistic term as a Gaussian scale mixture in θ .

Step 2: condition on ω and collect quadratic terms. Given ω_t , each weighted term contributes $\kappa_t \theta - \frac{1}{2}\omega_t \theta^2$ up to constants, where the weight has already been absorbed into b_t and κ_t . Summing over t yields (38).

Step 3: combine with a Gaussian prior. Multiplying the quadratic-in- θ conditional likelihood by a Gaussian prior produces a Gaussian conditional posterior, because Gaussian priors are conjugate to Gaussian likelihoods. $\square$

Proposition 12.5 (Mean-field variational approximation under Pólya–Gamma). *Under the same setup as Theorem 12.4 and a Gaussian prior $\pi(\theta) = \mathcal{N}(\nu, \rho^2)$, the mean-field factorization $q(\theta, \omega) = q(\theta) \prod_{t=i+1}^j q(\omega_t)$ admits the following closed-form coordinate-ascent updates: (i) $q(\theta) = \mathcal{N}(\mu_q, \sigma_q^2)$ with $\sigma_q^{-2} = \rho^{-2} + \sum_t \mathbb{E}_q[\omega_t]$ and $\mu_q = \sigma_q^2(\rho^{-2}\nu + \sum_t \kappa_t)$; (ii) $q(\omega_t) = \text{PG}(b_t, c_t)$ with $c_t^2 = \mathbb{E}_q[\theta^2] = \mu_q^2 + \sigma_q^2$, giving $\mathbb{E}_q[\omega_t] = \frac{b_t}{2c_t} \tanh(c_t/2)$. These updates are monotone in the block ELBO.*

Proof. Standard mean-field variational calculus applied to (38) combined with the prior. The explicit update for $\mathbb{E}_q[\omega_t]$ uses the mean of a $\text{PG}(m, c)$ distribution; see Appendix A.10 for the full derivation and ELBO monotonicity argument. $\square$

Implementation. Algorithm 12 gives a Pólya–Gamma variational Bayes (PG-VB) routine for logistic (and related) blocks, which yields a tractable surrogate block evidence by alternating between Gaussian updates and latent-variable expectations.

Algorithm 12: GLMBlock_PG_VB(y, m, i, j, ν, ρ^2) (Gaussian prior, PG mean-field)

Input: Binomial/logistic block with trial counts m_t ; Gaussian prior $\mathcal{N}(\nu, \rho^2)$

- 1 Initialize $q(\omega_t)$ and $q(\theta) = \mathcal{N}(\mu, \sigma^2)$;
 - 2 **repeat**
 - 3 Set $b_t \leftarrow w_t m_t$ and $\kappa_t \leftarrow w_t(y_t - m_t/2)$ if likelihood-power weights are used;
 otherwise $w_t = 1$;
 - 4 $\sigma^{-2} \leftarrow \rho^{-2} + \sum_{t=i+1}^j \mathbb{E}_q[\omega_t]$;
 - 5 $\mu \leftarrow \sigma^2 \left(\rho^{-2}\nu + \sum_{t=i+1}^j \kappa_t \right)$;
 - 6 **foreach** t **do**
 - 7 Update $q(\omega_t) = \text{PG}(w_t m_t, c_t)$ using current $\mathbb{E}_q[\theta^2]$;
 - 8 **until** *ELBO converged*;
 - 9 Return a lower bound $\underline{A}_{ij}^{(0)}$ and approximate moments from $q(\theta)$.
-

Using approximate block evidences in BayesBreak. Once any block-evidence approximation is available (Laplace, JJ, EP, PG-VB, or another surrogate), BayesBreak uses it exactly as an input score array: build a triangular array of approximate block log-evidences and run the same DP recursions. The resulting posterior summaries are exact for that approximate score array, not automatically exact for the original non-conjugate model. Algorithm 13 provides a generic recipe for constructing this array in a model-agnostic way, emphasizing reuse of block-local optimizers and cached sufficient-statistic summaries.

Algorithm 13: Precompute block evidences for non-conjugate GLMs

Input: $\{(x_t, y_t, w_t)\}_{t=1}^n$, $\text{method} \in \{\text{Laplace}, \text{JJ}, \text{PG-VB}, \text{EP}\}$

```
1 for  $i = 0$  to  $n - 1$  do
2   Initialize running summaries (as needed);
3   for  $j = i + 1$  to  $n$  do
4     Update block summaries for  $(i, j]$ ;
5     Call chosen block routine to compute  $\hat{A}_{ij}^{(0)}$  (and optional moments);
6 Run DP (Algorithm 1) with resulting block evidences.
```

12.7 Propagation of block approximation error through the DP

Proposition 12.6 (Log-evidence and posterior-odds stability). *Let $\hat{A}_{ij}^{(0)}$ be approximate block evidences with log-errors $\delta_{ij} := \log \hat{A}_{ij}^{(0)} - \log A_{ij}^{(0)}$. Assume the uniform blockwise bound*

$$|\delta_{ij}| \leq \varepsilon \text{ uniformly over all candidate blocks } (i, j], 0 \leq i < j \leq n, \quad (39)$$

reachable by some k -segmentation with $k \leq k_{\max}$.

Then for every DP state (k, j) with $k \leq k_{\max}$,

$$e^{-k\varepsilon} L_{kj} \leq \hat{L}_{kj} \leq e^{k\varepsilon} L_{kj}, \quad e^{-k\varepsilon} R_{kj} \leq \hat{R}_{kj} \leq e^{k\varepsilon} R_{kj}.$$

Consequently:

1. *For any $k, k' \leq k_{\max}$, the posterior odds for the segment count satisfy*

$$\left| \log \frac{\hat{P}(k | y)}{\hat{P}(k' | y)} - \log \frac{P(k | y)}{P(k' | y)} \right| \leq (k + k')\varepsilon. \quad (40)$$

2. *For any fixed $k \leq k_{\max}$ and any two candidate locations h, h' of boundary t_p ,*

$$\left| \log \frac{\hat{P}(t_p = h | y, k)}{\hat{P}(t_p = h' | y, k)} - \log \frac{P(t_p = h | y, k)}{P(t_p = h' | y, k)} \right| \leq 2k\varepsilon. \quad (41)$$

Thus uniform blockwise log-evidence control yields uniform control of global posterior odds. The conversion of these odds bounds into an absolute probability-error bound for the segment-count posterior is given in Corollary 12.8; converting the boundary-odds bound into an absolute boundary-probability bound requires the additional margin assumption used in Corollary 12.7.

Proof. Any term contributing to L_{kj} is a product of exactly k block evidences. Hence the corresponding approximate term differs by a multiplicative factor in $[e^{-k\varepsilon}, e^{k\varepsilon}]$. Summing over all paths yields the same sandwich bound for $\hat{L}_{kj}$, and the suffix bound is identical. Equation (40) follows immediately because $P(k | y)$ is proportional to $p(k)L_{kn}/C_k$ and the prior and normalization constants are unchanged. For boundary odds, note that $P(t_p = h | y, k) \propto$

$L_{p,h}R_{k-p,h}$. The numerator for each candidate location involves k block factors in total, so each candidate-specific weight is perturbed by at most $e^{\pm k\varepsilon}$; comparing two candidates therefore yields the factor $e^{\pm 2k\varepsilon}$ and hence (41). $\square$

Corollary 12.7 (Preservation of boundary rankings under a margin condition). *Under the hypotheses of Proposition 12.6, if there is a log-odds margin $\Delta > 0$ separating the best candidate h^* from every alternative,*

$$\log \frac{P(t_p = h^* \mid y, k)}{P(t_p = h \mid y, k)} \geq \Delta \quad \text{for all } h \neq h^*,$$

then for every $\varepsilon < \Delta/(2k)$ the approximate posterior preserves the exact modal boundary location: $\arg \max_h \hat{P}(t_p = h \mid y, k) = h^$.*

Proof. By (41), the log-odds between h^* and any alternative h is perturbed by at most $2k\varepsilon < \Delta$, so the exact log-odds positivity is preserved. $\square$

Corollary 12.8 (Absolute probability error for the segment-count posterior). *Under the hypotheses of Proposition 12.6, the approximate segment-count posterior satisfies, for every $k \leq k_{\max}$,*

$$|\hat{P}(k \mid y) - P(k \mid y)| \leq (e^{2k_{\max}\varepsilon} - 1).$$

Consequently the total variation distance between the approximate and exact segment-count posteriors obeys $\text{TV}(\hat{P}(\cdot \mid y), P(\cdot \mid y)) \leq e^{2k_{\max}\varepsilon} - 1$, which is $\mathcal{O}(k_{\max}\varepsilon)$ as $\varepsilon \rightarrow 0$.

Proof. Write $\pi_k := p(k)L_{kn}/C_k$ and $\hat{\pi}_k := p(k)\hat{L}_{kn}/C_k$, so that $P(k \mid y) = \pi_k / \sum_{k'} \pi_{k'}$ and likewise for $\hat{P}$. By Proposition 12.6, $e^{-k\varepsilon}\pi_k \leq \hat{\pi}_k \leq e^{k\varepsilon}\pi_k$ for every $k \leq k_{\max}$. Hence each ratio $\hat{\pi}_k / \sum_{k'} \hat{\pi}_{k'}$ lies between $e^{-2k_{\max}\varepsilon}P(k \mid y)$ and $e^{2k_{\max}\varepsilon}P(k \mid y)$, because the numerator is inflated by at most $e^{k\varepsilon} \leq e^{k_{\max}\varepsilon}$ and the denominator deflated by at most the same factor. Therefore $|\hat{P}(k \mid y) - P(k \mid y)| \leq (e^{2k_{\max}\varepsilon} - 1)P(k \mid y) \leq e^{2k_{\max}\varepsilon} - 1$. Summing the one-sided bound $\hat{P}(k \mid y) - P(k \mid y) \leq (e^{2k_{\max}\varepsilon} - 1)P(k \mid y)$ over the k for which it is positive gives the total-variation statement, and a first-order expansion of $e^{2k_{\max}\varepsilon}$ yields the $\mathcal{O}(k_{\max}\varepsilon)$ rate. $\square$

Remark 12.9 (Status of Corollary 12.8). *Corollary 12.8 is a proposed strengthening derived entirely from the hypotheses already in force for Proposition 12.6; it converts the odds-level guarantee into the absolute-probability statement that downstream thresholding and model-selection rules actually require. The bound is conservative—it uses the worst-case $k_{\max}$ in place of the per-state k —but it makes explicit that small uniform block error implies small total-variation error in the segment-count posterior, with no additional margin assumption.*

Remark 12.10 (A small numerical illustration of the stability bound). *Consider $n = 500$, $k_{\max} = 10$, and a non-conjugate block routine whose verified uniform per-block log-evidence error over all reachable blocks is $\varepsilon = 0.02$. Proposition 12.6 then guarantees segment-count log-odds error at most $(5 + 10) \times 0.02 = 0.30$ when comparing $k = 5$ to $k = 10$, and boundary log-odds error at most $2 \cdot 10 \cdot 0.02 = 0.40$ under any fixed $k \leq 10$. In terms of probability ratios, these correspond to factors of about $e^{0.30}$ and $e^{0.40}$. The numbers in this illustration are conditional on having already verified the uniform block-error bound; they are not a generic guarantee for any chosen approximation routine.*

Remark 12.11 (When each non-conjugate routine struggles). *The stability bound in Proposition 12.6 is tight only when the uniform blockwise error ε is actually small. Each non-conjugate routine has its own failure mode: Laplace (Theorem 12.2) requires posterior concentration near $\hat{\theta}_{ij}$, so it can be inaccurate on very short segments (W_{ij} small) or for heavy-tailed priors; mean-field variational Bayes (Proposition 12.5) can be overconfident and underestimate posterior variance, biasing moment numerators; EP can fail to converge for non-log-concave likelihoods; 1D quadrature is accurate for smooth 1D integrands but expensive in higher dimensions. For these reasons, monitor ε empirically per dataset rather than assuming it. A practical validation checklist is to compare a subset of blocks against quadrature or another trusted reference, report both maximum and high-quantile block errors, inspect errors along the exported MAP path, and assess sensitivity of $P(k | y)$ and boundary marginals.*

Approximation-validation checklist. Before interpreting non-conjugate posterior summaries, run the following checks, each of which targets a specific failure mode from Remark 12.11. First, compare a subset of block evidences against higher-accuracy quadrature or another trusted reference, and report both the maximum and high-quantile block errors over reachable blocks: this directly estimates the uniform error ε that Proposition 12.6 assumes, and it exposes the short-segment inaccuracy of Laplace and the variance-underestimation bias of mean-field variational Bayes. Second, inspect errors specifically on the exported MAP path, since a routine that is accurate on average may still be unreliable on the blocks that determine the reported segmentation. Third, assess the sensitivity of $p(k | y)$, the boundary marginals, and the exported segmentation to the choice of approximation method, which surfaces EP non-convergence and other method-specific instabilities that need not show up as a single large block error. The ranking guarantee of Corollary 12.7 is meaningful only when the exact posterior margin is larger than the propagated approximation error, so the margin and the measured ε should be reported together.

13 Prediction: group likelihoods and MAP/Bayes signal evaluation

Having developed training-time posterior inference for partitions and segment parameters (Sections 6–12), we now turn to prediction for new (X, y) data. The prediction layer consumes posterior summaries (e.g., boundary posteriors, segment-parameter posteriors, and group evidences) and is therefore agnostic to whether block evidences were computed in closed form (Section 11) or via approximation (Section 12). We assume we have already trained (or otherwise obtained) a collection of G group-specific BayesBreak models $\{\mathcal{M}_g\}_{g=1}^G$, where each $\mathcal{M}_g$ comprises: (i) a fitted segmentation posterior (e.g. $P_g(k | y)$ and boundary posteriors), (ii) an exported MAP segmentation $\hat{t}^{(g)}$ with segment-level posterior hyperparameters, and (iii) optionally a Bayesian curve representation (e.g. posterior mean/variance at design points).

Typical outputs are group log scores, posterior group probabilities, optional unit-level responsibilities, exported MAP signal values, Bayes-curve values, and optional pointwise uncertainty summaries. These prediction interfaces define the outputs populated by finalized diagnostics for the placeholder real-data examples.

Prediction supports two primary tasks:

(P1) *Group membership likelihoods/posteriors.* Given new inputs (X, y) , compute for each group g an out-of-sample posterior-predictive likelihood $p(\text{new} \mid \mathcal{M}_g)$ (or its log), and convert these into posterior group membership probabilities $P(g \mid \text{new})$ under a prior π_g .

(P2) *Signal prediction given group.* Given query design points X and a specified (or inferred) group label g , return: (i) MAP piecewise-constant signal values $\hat{f}_g^{\text{MAP}}(X)$ including the corresponding breakpoints and segment levels, and/or (ii) Bayesian curve values $\hat{f}_g^{\text{Bayes}}(X)$ with pointwise uncertainty summaries when available.

Prediction output	Interpretation
ℓ_g	Group-specific log posterior-predictive score.
$P(g \mid \text{new})$	Group posterior after combining ℓ_g with prior weight π_g .
$\hat{f}_g^{\text{MAP}}(X)$	Piecewise-constant signal under the exported MAP segmentation.
$\hat{f}_g^{\text{Bayes}}(X)$	Bayes-curve posterior mean, optionally with pointwise uncertainty.
Unit responsibilities	Optional probabilities for set-valued units or grouped observations.

Table 3 Prediction-layer outputs exposed by fitted BayesBreak models. These interfaces support downstream scoring and calibration separately from the empirical result tables.

Crucially, we distinguish *single point observations* (x_i, y_i) from *multivariate / set-valued observations* where each unit contains multiple internal pairs (x_{ur}, y_{ur}) , i.e. X_u and y_u are vectors. We also separately address *vector-valued responses* $y_i \in \mathbb{R}^d$ observed at each design point.

13.1 Prediction inputs: pointwise vs multivariate/set-valued observations

Case A (pointwise observations). We observe a single ordered sequence

$$\mathcal{D}_{\text{pt}}^{\text{new}} := \{(x_i^{\text{new}}, y_i^{\text{new}}, w_i^{\text{new}})\}_{i=1}^m, \quad x_1^{\text{new}} < \dots < x_m^{\text{new}},$$

with optional exposure or precision weights w_i^{new} . The default is $w_i^{\text{new}} = 1$; use inter-point gaps only when they are genuine exposures or known-precision weights for the observation model, not merely because the design is irregular.

Case B (multivariate/set-valued observations). We observe U units (e.g. “fragments”, “windows”, or generic composite observations), each with an interval support and multiple internal pairs:

$$\mathcal{D}_{\text{mv}}^{\text{new}} := \left\{ ([a_u, b_u], \{(x_{ur}, y_{ur}, w_{ur})\}_{r=1}^{R_u}) \right\}_{u=1}^U, \quad a_u < x_{ur} \leq b_u.$$

Here $X_u = (x_{u1}, \dots, x_{uR_u})$ and $y_u = (y_{u1}, \dots, y_{uR_u})$ are *multivariate* (set-valued) per-unit inputs. Prediction must score each unit by integrating or aggregating over its internal observations, and optionally return unit-level group responsibilities.

Case C (vector-valued response at each design point). At each design point we observe a vector response $\mathbf{y}_i \in \mathbb{R}^d$ (or mixed discrete/continuous components). We consider two standard constructions: (i) *factorized EF* across dimensions, or (ii) a *multivariate EF* with a joint log-partition $A(\theta)$ and conjugate prior (when available). The factorized case is implementation-friendly and preserves the DP structure with localized changes, while the multivariate case requires a genuine joint likelihood and conjugate or approximate block routine.

13.2 Posterior-predictive scoring for a fixed segment under EF conjugacy

For group membership inference, we require out-of-sample likelihoods. The most computationally efficient and interpretable approach is to score new data against the *exported* group curve (MAP segmentation or Bayesian curve). We first derive segment-level posterior-predictive likelihoods under EF conjugacy; these become building blocks for both Case A and Case B.

Fix a group g and consider a segment (or block) B of its exported MAP segmentation. Let the posterior for the segment parameter θ after training be the conjugate posterior

$$p(\theta \mid \alpha_B^{(g)}, \beta_B^{(g)}) \propto \exp\{\eta(\theta)^\top \alpha_B^{(g)} - \beta_B^{(g)} A(\theta)\},$$

where $(\alpha_B^{(g)}, \beta_B^{(g)})$ are the training-updated hyperparameters for that segment. Now let $\mathcal{Y}_B^{\text{new}}$ denote the collection of new observations whose design points fall into this segment.

Proposition 13.1 (EF posterior-predictive likelihood within a fixed segment). *Assume the weighted EF likelihood and conjugate prior conventions of §5 and §3, and let $(S_B^{\text{new}}, W_B^{\text{new}}, H_B^{\text{new}})$ denote the EF block summaries of the new data restricted to segment B :*

$$S_B^{\text{new}} := \sum_{i \in B} w_i^{\text{new}} T(y_i^{\text{new}}), \quad W_B^{\text{new}} := \sum_{i \in B} w_i^{\text{new}}, \quad H_B^{\text{new}} := \sum_{i \in B} \log h(y_i^{\text{new}}, w_i^{\text{new}}).$$

Then the posterior-predictive likelihood of the new data in segment B under group g is

$$p(\mathcal{Y}_B^{\text{new}} \mid \mathcal{M}_g) = \exp\{H_B^{\text{new}}\} \frac{Z(\alpha_B^{(g)} + S_B^{\text{new}}, \beta_B^{(g)} + W_B^{\text{new}})}{Z(\alpha_B^{(g)}, \beta_B^{(g)})}. \quad (42)$$

Proof. The proof is identical to the single-block evidence derivation (Theorem 5.1) after replacing the prior hyperparameters (α_0, β_0) by the segment posterior hyperparameters $(\alpha_B^{(g)}, \beta_B^{(g)})$ learned from training data. $\square$

Interpretation. Equation (42) integrates out the segment parameter θ under the *posterior* induced by training, yielding a strictly Bayesian out-of-sample likelihood for new observations assigned to that segment. It is therefore suitable for group scoring and group posterior computation.

13.3 Group likelihoods for pointwise observations (Case A)

MAP-segmentation-based scoring. Let $\hat{t}^{(g)}$ denote the exported MAP boundaries for group g on the relevant domain. Assign each new design point x_i^{new} to its containing MAP segment $B = B(i; g)$ and compute segment-wise block summaries $(S_B^{\text{new}}, W_B^{\text{new}}, H_B^{\text{new}})$. The group log-likelihood is then:

$$\ell_g^{\text{MAP}}(\mathcal{D}_{\text{pt}}^{\text{new}}) := \log p(\mathcal{D}_{\text{pt}}^{\text{new}} \mid \mathcal{M}_g) = \sum_{B \in \hat{t}^{(g)}} \log p(\mathcal{Y}_B^{\text{new}} \mid \mathcal{M}_g), \quad (43)$$

where each segment term uses (42). This yields $\mathcal{O}(m)$ evaluation time once segment membership is determined.

Bayesian-curve-based scoring (optional). If $\mathcal{M}_g$ exports a Bayesian curve representation that provides a local posterior over the mean parameter at each design point (or each small bin), then one can score each point under its local posterior predictive and sum over points. This can improve robustness near uncertain boundaries at additional computational cost.

Resegmentation scoring (optional, flexible). An alternative definition of group likelihood is to rerun the DP on the new data under group-specific hyperparameters (treating the new dataset as another draw from group g 's prior). This yields an evidence $P(\mathcal{D}_{\text{pt}}^{\text{new}} \mid g)$ that does not condition on the trained MAP boundaries. This option is useful when groups differ primarily in hyperparameters rather than in the learned curve geometry, but is computationally $\mathcal{O}(k_{\text{max}} m^2)$ per group.

13.4 Group likelihoods for multivariate/set-valued observations (Case B)

In Case B, each unit u contains multiple internal observations $\{(x_{ur}, y_{ur})\}_{r=1}^{R_u}$. Two output granularities are typically desired: (i) *unit-level* group likelihoods and group responsibilities, and (ii) a *global* group likelihood for the entire new dataset (product over units).

Unit-level predictive likelihood under a fixed group. Fix a group g . Partition the unit's internal design points by the MAP segments of group g . For each segment B , compute the block summaries $(S_{u,B}^{\text{new}}, W_{u,B}^{\text{new}}, H_{u,B}^{\text{new}})$ restricted to those internal points. Then define the unit likelihood as:

$$\ell_{ug} := \log p(\text{unit } u \mid \mathcal{M}_g) = \sum_{B \in \hat{t}^{(g)}} \log p(\mathcal{Y}_{u,B}^{\text{new}} \mid \mathcal{M}_g), \quad (44)$$

where $\mathcal{Y}_{u,B}^{\text{new}}$ is the subset of unit u 's internal observations that fall into segment B and each segment term uses (42) with summaries for that subset.

Dataset likelihood and independence assumption. Assuming conditional independence of units given group membership, the dataset likelihood under group g is:

$$\ell_g^{\text{MV}}(\mathcal{D}_{\text{mv}}^{\text{new}}) = \sum_{u=1}^U \ell_{ug}.$$

This is the direct multivariate generalization of (43) and corresponds to the earlier “fragment-level” intuition, but is fully application-agnostic.

13.5 Vector-valued responses (Case C): factorized and multivariate EF scoring

Suppose each observation is $\mathbf{y}_i = (y_{i1}, \dots, y_{id})$.

Factorized EF across dimensions (recommended default). Assume conditional independence across dimensions given segment parameters:

$$p(\mathbf{y}_i | \boldsymbol{\theta}) = \prod_{\ell=1}^d p_{\ell}(y_{i\ell} | \theta_{\ell}), \quad p(\boldsymbol{\theta}) = \prod_{\ell=1}^d p_{\ell}(\theta_{\ell}).$$

Then all block evidences and posterior-predictive terms factorize over ℓ , so log-likelihoods are sums over dimensions:

$$\ell_g(\mathbf{y}) = \sum_{\ell=1}^d \ell_{g,\ell}(y_{\cdot\ell}),$$

where each $\ell_{g,\ell}$ is computed by (43) or (44) for dimension ℓ . Computationally, this costs $\mathcal{O}(d)$ per observation (or per unit).

Multivariate EF (when available). If $(\mathbf{y}_i)$ admit a multivariate EF with conjugate prior on a joint parameter $\theta \in \mathbb{R}^p$, then the same ratio-of-normalizers structure applies with vector-valued sufficient statistics. BayesBreak’s DP layer remains unchanged; only the definition of S_{ij} and $Z(\alpha, \beta)$ changes.

13.6 From group likelihoods to posterior group membership probabilities

Single-dataset membership (one group per dataset). If the new dataset is assumed to come from a single (unknown) group g , define:

$$\ell_g := \log p(\text{new} | \mathcal{M}_g), \quad P(g | \text{new}) = \frac{\pi_g \exp(\ell_g)}{\sum_{g'=1}^G \pi_{g'} \exp(\ell_{g'})}.$$

This yields a *group membership likelihood vector* (the ℓ_g) and a *group membership posterior vector* (the $P(g | \text{new})$), matching the requested prediction behavior in an application-agnostic way.

Unit-wise membership (one group per unit). If each multivariate unit u is assumed to originate from one group, then define unit-wise responsibilities:

$$r_{ug} := P(g | \text{unit } u) = \frac{\pi_g \exp(\ell_{ug})}{\sum_{g'=1}^G \pi_{g'} \exp(\ell_{ug'})}, \quad (45)$$

where ℓ_{ug} is given by (44). This is precisely the out-of-sample analog of the E-step responsibilities in the latent-group model (Section 10).

13.7 MAP signal evaluation given group membership

Given a group label g and query design points $X^\star = \{x_1^\star, \dots, x_M^\star\}$, BayesBreak returns either:
 (i) *MAP piecewise-constant predictions* or (ii) *Bayesian curve predictions*.

MAP piecewise-constant predictions. Let $\hat{t}^{(g)}$ be the exported MAP boundaries and let each MAP segment B provide the posterior mean of the mean parameter:

$$\hat{\mu}_B^{(g)} := \mathbb{E}[m(\theta) \mid \alpha_B^{(g)}, \beta_B^{(g)}].$$

Then the MAP signal at a query point $x^\star$ is:

$$\hat{f}_g^{\text{MAP}}(x^\star) = \hat{\mu}_{B(x^\star; g)}^{(g)},$$

where $B(x^\star; g)$ denotes the MAP segment containing $x^\star$.

Bayesian curve predictions. If the DP-derived Bayes regression curve is exported on a grid, return $\hat{f}_g^{\text{Bayes}}(x^\star)$ by evaluating on the nearest grid point (or by piecewise-constant interpolation within each grid interval). When pointwise posterior variances are available, also return pointwise credible intervals; simultaneous bands require an additional construction and are not implied by the pointwise variances alone.

13.8 Algorithms for prediction

Algorithm 14: PredictGroup($\{\mathcal{M}_g\}_{g=1}^G$, $\mathcal{D}^{\text{new}}$, π)

Input: Trained group models $\{\mathcal{M}_g\}$; new data $\mathcal{D}^{\text{new}}$ (Case A or Case B); group prior weights π (default uniform)

Output: Log-likelihoods $\{\ell_g\}$ and posterior membership probabilities $\{P(g \mid \text{new})\}$; if Case B, also unit responsibilities $\{r_{ug}\}$

```

1 for  $g = 1$  to  $G$  do
2   if Case A (pointwise) then
3     Assign each  $(x_i^{\text{new}}, y_i^{\text{new}})$  to a MAP segment  $B(i; g)$  under  $\hat{t}^{(g)}$ ;
4     Accumulate segment summaries  $(S_B^{\text{new}}, W_B^{\text{new}}, H_B^{\text{new}})$ ;
5     Compute  $\ell_g \leftarrow \sum_B \left[ H_B^{\text{new}} + \log Z(\alpha_B^{(g)} + S_B^{\text{new}}, \beta_B^{(g)} + W_B^{\text{new}}) - \log Z(\alpha_B^{(g)}, \beta_B^{(g)}) \right]$ ;
6   else
7     // Case B (multivariate/set-valued units)
8     for  $u = 1$  to  $U$  do
9       Partition unit  $u$  internal pairs by MAP segments of group  $g$ ; compute
10      per-segment summaries;
11      Compute  $\ell_{ug}$  via (44) using (42);
12     $\ell_g \leftarrow \sum_{u=1}^U \ell_{ug}$ ;
13 Compute  $P(g \mid \text{new}) \propto \pi_g \exp(\ell_g)$  and normalize over  $g$ ;
14 if Case B then
15   Compute unit responsibilities  $r_{ug} \propto \pi_g \exp(\ell_{ug})$  and normalize over  $g$  for each  $u$  (cf.
16   (45));

```

Algorithm 15: PredictMAPSignal($\mathcal{M}_g$, X^* , mode)

Input: A group model $\mathcal{M}_g$ with exported MAP segmentation $\widehat{t}^{(g)}$ and/or Bayes curve; query points $X^* = \{x_1^*, \dots, x_M^*\}$; mode $\in \{\text{MAP}, \text{Bayes}\}$

Output: Predicted signal values at X^* (and optionally uncertainty)

```
1 if mode=MAP then
2   foreach  $x^*$  in  $X^*$  do
3     Find containing MAP segment  $B(x^*; g)$  (binary search on breakpoint
      coordinates);
4     Return  $\widehat{f}_g^{\text{MAP}}(x^*) = \widehat{\mu}_{B(x^*; g)}^{(g)}$ ;
5 else
6   Evaluate/interpolate the exported Bayes regression curve at  $X^*$ ; optionally return
   posterior variances;
```

13.9 Complexity

For Case A, prediction under a fixed group is $\mathcal{O}(m)$ after segment assignment ($\mathcal{O}(m \log k)$ worst case via binary search; often $\mathcal{O}(m)$ with a streaming pointer). For Case B, it is $\mathcal{O}(\sum_u R_u)$ per group, plus segment assignment for internal points. Vector-valued responses add a factor of d in the factorized EF construction. Importantly, prediction does *not* require rerunning the DP unless the optional resegmentation scoring mode is enabled.

13.10 Prediction diagnostics

Because the predictive layer reuses segment-level posteriors, the group posterior $P(g \mid \text{new})$ derived from (43) and the pointwise predictive summaries inherit their calibration quality from the segmentation. Use two diagnostics for real-data deployments:

- **Held-out log-likelihood (HLL) trace.** Split each test sequence into blocks and compute the per-block posterior-predictive log-likelihood under (42). The sum over blocks is the HLL for that sequence. Compare HLL trajectories across candidate segmentations (exported MAP vs. Bayes curve) and across groups; a well-calibrated model yields an HLL ranking consistent with the k -posterior ranking on held-out data.
- **PIT residuals.** For point-valued new observations with continuous predictive CDF, evaluate the probability-integral transform $u_i = F_g(y_i^{\text{new}} \mid \mathcal{M}_g)$. Under correct calibration the $\{u_i\}$ are Uniform(0, 1); histograms or Kolmogorov–Smirnov summaries flag miscalibration (overconfident predictive widths show up as U-shaped histograms).

These diagnostics are cheap post-hoc checks. In this paper they are planned outputs for the real-data pipelines in Section 15; placeholder tables identify the planned HLL and boundary-recovery fields that the finalized pipelines populate.

14 Algorithms, complexity, and numerical considerations

Section 6 derives the exact dynamic program (DP) that sits at the core of BayesBreak. The present section complements that development with implementation-ready pseudocode and a collection of numerical-stability considerations. The primary purpose is pragmatic: to make explicit what must be computed, stored, and normalized in order to reproduce the theoretical quantities (evidences, boundary posteriors, Bayes curves) without relying on undocumented conventions.

14.1 Block-evidence precomputation via prefix sums

For conjugate exponential-family blocks, Theorem 5.1 shows that the single block evidence and moments can be expressed as functions of aggregated sufficient statistics. This enables an *exact* $\mathcal{O}(n^2)$ precomputation of all candidate blocks $(i, j]$ with $0 \leq i < j \leq n$. Throughout this section we use the body convention: a block $(i, j]$ has length $j - i$ and contains the indices $i + 1, \dots, j$.

In practice, it is numerically preferable to store *log* block evidences,

$$\ell_{ij}^{(0)} := \log \tilde{A}_{ij}^{(0)}, \quad (46)$$

and similarly for positive moment numerators required for the Bayes regression curve. If a chosen moment numerator can be zero or negative, implementations use signed-log storage or linear accumulators with explicit scaling rather than silently taking a logarithm. For the closed-form conjugate families of Section 11, the order- r moment numerators $\tilde{A}_{ij}^{(r)}$ are products of a strictly positive block evidence and a strictly positive observation-scale moment of a nonnegative quantity (a Gaussian-mean second moment, a Gamma or Beta moment, or a count-mean moment), so they are strictly positive and may be stored directly in log space; the signed-log path is exercised only when a block routine targets a moment that can change sign, such as a centered Gaussian mean or a non-conjugate Laplace test function (cf. Section 12). Implementations should assert this per family rather than assume it globally. Algorithm 16 expresses the precomputation at an abstract level, with a single call to a `BlockRoutine` that implements the family-specific closed forms given in Section 11.

Algorithm 16: PrecomputeBlocks($y, x, w, \text{BlockRoutine}$)

Input: Ordered observations $\{(x_i, y_i, w_i)\}_{i=1}^n$ (with $w_i \equiv 1$ if unweighted); family-specific routine **BlockRoutine** implementing Theorem 5.1.

Output: Matrices of log-block evidences $\ell_{ij}^{(0)} = \log \tilde{A}_{ij}^{(0)}$ on the index set $\{0 \leq i < j \leq n\}$ and (optional) moment numerators, stored as $\log \tilde{A}_{ij}^{(r)}$ only when they are positive and otherwise in a signed-log or linear representation.

- 1 **1. Prefix summaries;**
 - 2 Compute prefix cumulative sufficient statistics and weights required by **BlockRoutine**:
store $W_j := \sum_{t=1}^j w_t$, $S_j := \sum_{t=1}^j w_t T(y_t)$, and any additional family-specific terms
(e.g., $Q_j := \sum_{t=1}^j w_t y_t^2$ for Gaussian). Set $W_0 = S_0 = Q_0 = 0$.
 - 3 **2. Enumerate blocks;**
 - 4 **for** $i \leftarrow 0$ **to** $n - 1$ **do**
 - 5 **for** $j \leftarrow i + 1$ **to** n **do**
 - 6 Extract block summaries for $y_{i+1:j}$ via $W_{ij} = W_j - W_i$, $S_{ij} = S_j - S_i$,
 $Q_{ij} = Q_j - Q_i$ (constant time); **BlockRoutine** returns $\log A_{ij}^{(0)}$ and any moment
 numerators $\log A_{ij}^{(r)}$; Add any length factor $\log g(\Delta_x(i, j))$ and hazard constants
 as in (18); Store $\ell_{ij}^{(0)} \leftarrow \log \tilde{A}_{ij}^{(0)}$ and, if needed, $\ell_{ij}^{(r)} \leftarrow \log \tilde{A}_{ij}^{(r)}$.
-

Memory notes. Storing all $\ell_{ij}^{(0)}$ uses $\Theta(n^2)$ memory. When n is large but $k_{\max}$ is moderate, the memory footprint is reduced by streaming blocks in the inner DP loop (computing $\ell_{ij}^{(0)}$ on the fly from prefix statistics), at cost of recomputing some block summaries. The DP theory remains unchanged.

As sanity checks, verify that invalid blocks are represented by $-\infty$ log-scores, that $\log C_k$ is computed with the same admissibility rules as the DP, that forward and backward full-sequence evidences agree, and that boundary-event marginals sum to $k - 1$ conditional on fixed k .

14.2 Log-space dynamic programming

The forward/backward DP in Section 6 sums products of block evidences. To avoid under-flow/overflow, work in log space and use a stable `logsumexp` primitive. Algorithm 17 presents a log-space version of the recursions in (17)–(20).

Algorithm 17: $\text{DPLog}(\ell_{ij}^{(0)}, k_{\max}, \log p(k), \log C_k)$

Input: Log block evidences $\ell_{ij}^{(0)} = \log \tilde{A}_{ij}^{(0)}$ on the index set $\{0 \leq i < j \leq n\}$ (block $(i, j]$); maximum segments $k_{\max}$; log prior on k , $\log p(k)$; log normalizers $\log C_k$ computed under the same admissible-block and length-prior convention as the score array. For the index-uniform fixed- k prior with $g \equiv 1$, $C_k = \binom{n-1}{k-1}$ unless the caller has intentionally absorbed a normalized prior into the block or count scores.

Output: Forward messages $\log \tilde{L}_{k,j}$, backward messages $\log \tilde{R}_{k,i}$, log evidence per k , and normalized posterior $p(k | y)$.

```
1 1. Initialize;
2 Set  $\log \tilde{L}_{0,0} \leftarrow 0$  and  $\log \tilde{L}_{0,j} \leftarrow -\infty$  for  $j \geq 1$ ; set  $\log \tilde{R}_{0,n} \leftarrow 0$  and  $\log \tilde{R}_{0,i} \leftarrow -\infty$  for  $i \leq n-1$ .
3 2. Forward recursion;
4 for  $k \leftarrow 1$  to  $k_{\max}$  do
5   for  $j \leftarrow k$  to  $n$  do
6      $\log \tilde{L}_{k,j} \leftarrow \text{logsumexp}_{h \in \{k-1, \dots, j-1\}} (\log \tilde{L}_{k-1,h} + \ell_{h,j}^{(0)});$ 
7 3. Backward recursion;
8 for  $k \leftarrow 1$  to  $k_{\max}$  do
9   for  $i \leftarrow n-k$  to  $0$  (decreasing) do
10     $\log \tilde{R}_{k,i} \leftarrow \text{logsumexp}_{h \in \{i+1, \dots, n-k+1\}} (\ell_{i,h}^{(0)} + \log \tilde{R}_{k-1,h});$ 
11 4. Evidence and posterior over  $k$ ;
12 For each  $k \in \{1, \dots, k_{\max}\}$  set  $\log p(y | k) \leftarrow \log \tilde{L}_{k,n} - \log C_k$ ; compute unnormalized log posterior scores  $\log \pi_k \leftarrow \log p(k) + \log p(y | k)$ ; set  $p(k | y) \leftarrow \exp\{\log \pi_k - \text{logsumexp}_{k'}(\log \pi_{k'})\}$ .
```

Boundary marginals and Bayes curves. Once $\log \tilde{L}$ and $\log \tilde{R}$ are available, the boundary marginal in (21) can be computed stably by subtracting the common normalizer $\log \tilde{L}_{k,n}$. Similarly, Bayes regression moments (Theorem 6.5, Step 6; cf. (26)) can be computed by replacing $\ell_{ij}^{(0)}$ with compatible moment numerators and reusing the same log-space DP bookkeeping. Signed or nonpositive moment targets require the signed-log or scaled-linear convention noted in Section 14.1.

14.3 MAP segmentation and backtracking

The DP used for evidences can be augmented to recover a MAP segmentation in $\mathcal{O}(k_{\max} n^2)$ time. During the forward recursion, one stores, for each (k, j) , the maximizing split point

$$h^*(k, j) \in \arg \max_{h \in \{k-1, \dots, j-1\}} \{M_{k-1,h} + \ell_{h,j}^{(0)}\}, \quad (47)$$

where $M_{k,j}$ is the max-sum message defined in (24). Algorithm 18 computes $M_{k,j}$ and the backpointers; Algorithm 19 then backtracks from $(\hat{k}, n)$ to recover the boundary vector $\hat{b} =$

$(\hat{b}_0, \dots, \hat{b}_{\hat{k}})$.

Algorithm 18: MAPForward($\ell_{ij}^{(0)}, k_{\max}$)

Input: Log block evidences $\ell_{ij}^{(0)}$ on $\{0 \leq i < j \leq n\}$; $k_{\max}$.

Output: Max-sum messages $M_{k,j}$ and backpointers $h^*(k, j)$ for $k \leq k_{\max}$.

```

1 Set  $M_{0,0} \leftarrow 0$ ,  $M_{0,j} \leftarrow -\infty$  for  $j \geq 1$ ;
2 for  $k \leftarrow 1$  to  $k_{\max}$  do
3   for  $j \leftarrow k$  to  $n$  do
4      $h^*(k, j) \leftarrow \arg \max_{h \in \{k-1, \dots, j-1\}} \{M_{k-1,h} + \ell_{h,j}^{(0)}\}$ ;
5      $M_{k,j} \leftarrow M_{k-1,h^*(k,j)} + \ell_{h^*(k,j),j}^{(0)}$ ;
```

Algorithm 19: MAPBacktrack($h^*, \hat{k}$)

Input: Backpointers $h^*(k, j)$ as in (47); chosen segment count $\hat{k}$.

Output: MAP boundary vector $\hat{b}_0 = 0 < \hat{b}_1 < \dots < \hat{b}_{\hat{k}} = n$.

```

1 Set  $\hat{b}_{\hat{k}} \leftarrow n$  and  $q \leftarrow \hat{k}$ . while  $q \geq 1$  do
2    $\hat{b}_{q-1} \leftarrow h^*(q, \hat{b}_q)$ ;
3    $q \leftarrow q - 1$ ;
4 Return  $\hat{b}$ .
```

14.4 Pooling across sequences and groups

For replicated or multi-subject data with shared boundaries (Section 8), Theorem 8.2 implies that subject-level block evidences multiply. In log space, pooling is therefore a simple sum:

$$\ell_{ij}^{(0, \text{pooled})} = \sum_{s=1}^S \ell_{ij}^{(0,s)}. \quad (48)$$

The same applies to known groups, with the sum taken over subjects in a group. The resulting pooled matrices can be passed unchanged to Algorithm 17.

14.5 Latent-group EM implementation notes

Section 10 formulates latent groups as a *template mixture*: each group has one exported segmentation template, while subject-specific segment parameters have already been integrated out in subject-level block evidences. From an implementation perspective, the M-step for each group is therefore a standard max-sum BayesBreak backtracking run with a responsibility-weighted score matrix. If r_{sg} denotes the responsibility of subject s for group g , let $n_g = \sum_s r_{sg}$. The effective block score for the penalized template objective is

$$B_{ij}^{(g)} = n_g \log g(\Delta_x(i, j)) + \sum_{s=1}^S r_{sg} \ell_{ij}^{(0,s)}, \quad (49)$$

where $\ell_{ij}^{(0,s)} = \log A_{ij}^{(0,s)}$ is the subject-level log-block evidence. Running the max-sum DP over this score matrix for each k and then adding the responsibility-scaled count offset $n_g \{\log p(k) - \log C_k\}$ yields the exact template update for group g under the objective in Section 10. The monotonicity guarantee in Theorem 10.1 applies precisely because these E- and M-step updates are exact for the stated template-mixture objective.

14.6 Complexity summary and stability

For conjugate families with constant-time block evaluation from prefix summaries, Algorithm 16 costs $\Theta(n^2)$ time. The DP in Algorithm 17 costs $\Theta(k_{\max} n^2)$ time. Memory can be $\Theta(k_{\max} n)$ for basic evidence computation, and can be reduced further by streaming consecutive k -layers when only the final evidence is needed. Boundary posteriors, Bayes curves, and MAP backtracking require either stored layers/backpointers or deliberate recomputation, so implementations record which trade-off was used.

Finally, we emphasize two numerical practices that are essential in realistic data sets: (i) always store evidences and DP messages in log space and normalize via `logsumexp`; (ii) standardize/center sufficient statistics when available (e.g., for Gaussian blocks) to reduce catastrophic cancellation in moment computations. These practices do not affect correctness, but they materially improve stability when n is in the hundreds or thousands.

14.7 A numerical-implementation checklist

The following checklist distills the common pitfalls we have encountered and addresses them pre-emptively. Each item is cheap to implement and pays for itself on even moderate-sized problems.

1. **Log space everywhere.** Store all block evidences, moment numerators, and DP messages as log values from the start. Convert to linear space only for display.
2. **logsumexp with a stable shift.** Implement `logsumexp` as $\text{lse}(x) = m + \log \sum \exp(x_i - m)$ with $m = \max_i x_i$ (Blanchard et al., 2021). Cache the maximum once and reuse for both numerator and denominator when normalizing.
3. **Prefix sums in the right order.** Compute prefix sums of w_t , $w_t T(y_t)$, and (for Gaussian) $w_t y_t^2$ *before* enumerating blocks. Store them as double-precision arrays of length $n + 1$ initialized at index 0. Block summaries then cost one subtraction per block.
4. **Numerical sanity checks.** After the forward recursion, verify that $\log \tilde{L}_{0,0} = 0$ and $\log \tilde{L}_{k,j} = -\infty$ for $j < k$; after the backward recursion, verify the analogous boundary conditions on $\log \tilde{R}$. Verify that $\log \tilde{L}_{k,n} = \log \tilde{R}_{k,0}$ up to floating-point tolerance, that $\sum_k p(k | y) = 1$, that the fixed- k boundary marginal sums to $k - 1$, and that the reported C_k matches the prior convention used to form the block scores.
5. **Moment overflow.** When computing Bayes regression curves via (26), subtract a block-specific shift (e.g., the block posterior-mean estimate) before exponentiating to reduce dynamic range, then add the shift back at the end.

6. **Degenerate blocks.** Guard against $W_{ij} = 0$ (missing-data blocks) by returning $\log A_{ij}^{(0)} = -\infty$ or a user-specified fallback. Guard against Q_{ij} numerically negative (from catastrophic cancellation) by clamping to zero.
7. **Reproducibility.** Fix a seed for EM initializations and record it in the run metadata; record hyperparameters, $k_{\max}$, the count prior $p(k)$, length factor g , approximation method and settings, data-preprocessing hashes, and the exact version of every family-specific `BlockRoutine` used.

15 Experiments and results

This section reports the completed synthetic validation suite and defines the planned real-data evaluation program. The bundled artifacts cover single-sequence recovery, likelihood-family portability, calibration of boundary posteriors, latent-group behavior, non-conjugate block approximations, and runtime scaling. The well-log, array-CGH, volatility, and methylation case studies exercise complementary parts of the BayesBreak machinery; their visible figures and tables are placeholders that document the outputs populated by finalized pipelines. External baselines, real-data ablations, and larger-scale benchmarking are planned pre-submission additions.

15.1 Metrics

Evaluation objective	Metric
Boundary recovery	Precision / Recall / F1 at tolerance τ
Signal recovery	MSE between estimated and true latent signal
Posterior calibration	Calibration curve / ECE for $p(b_i = 1 \mid y)$
Model fit quality	$-\log p(y)$ (marginal evidence) or predictive NLL
Runtime	Wall-clock time vs. n and $k_{\max}$

Table 4 Metrics used in the synthetic experiments. When ground-truth boundaries are available, boundary-level metrics use a small tolerance window (here $\tau = 2$ indices) so that near-misses are not counted as complete failures.

The resulting evaluation protocol is deliberately mixed: some diagnostics target point estimation (boundary F1, boundary-location MAE, latent-signal MSE), while others target uncertainty quality (calibration curves for marginal boundary probabilities and evidence-based model selection through $P(k \mid y)$). This is appropriate for BayesBreak because the method returns both an exported segmentation and posterior summaries around that segmentation. The planned ablations over the partition prior $p(k)$ and the length factor g , described among the planned pre-submission additions below, are further accompanied by a planned prior-sensitivity diagnostic that records the variation of $P(k \mid y)$ and of the boundary marginals under prior perturbation, so that the robustness of the posterior summaries to prior choice is reported alongside their point-estimation and calibration quality.

15.2 Single-sequence Gaussian illustration

We begin with a Gaussian sequence containing two true changepoints. Figure 6 shows two complementary outputs: the marginal posterior probability that each interior index is a boundary, and the exported joint MAP segmentation together with segment-level posterior intervals. The two dominant posterior spikes align with the true changepoints, while a smaller shoulder of posterior mass just before the first jump reflects local ambiguity caused by a short transitional run of noisy observations. That ambiguity is visible again in the bottom panel, where the exported fit inserts a short intermediate segment before settling into the large middle plateau.

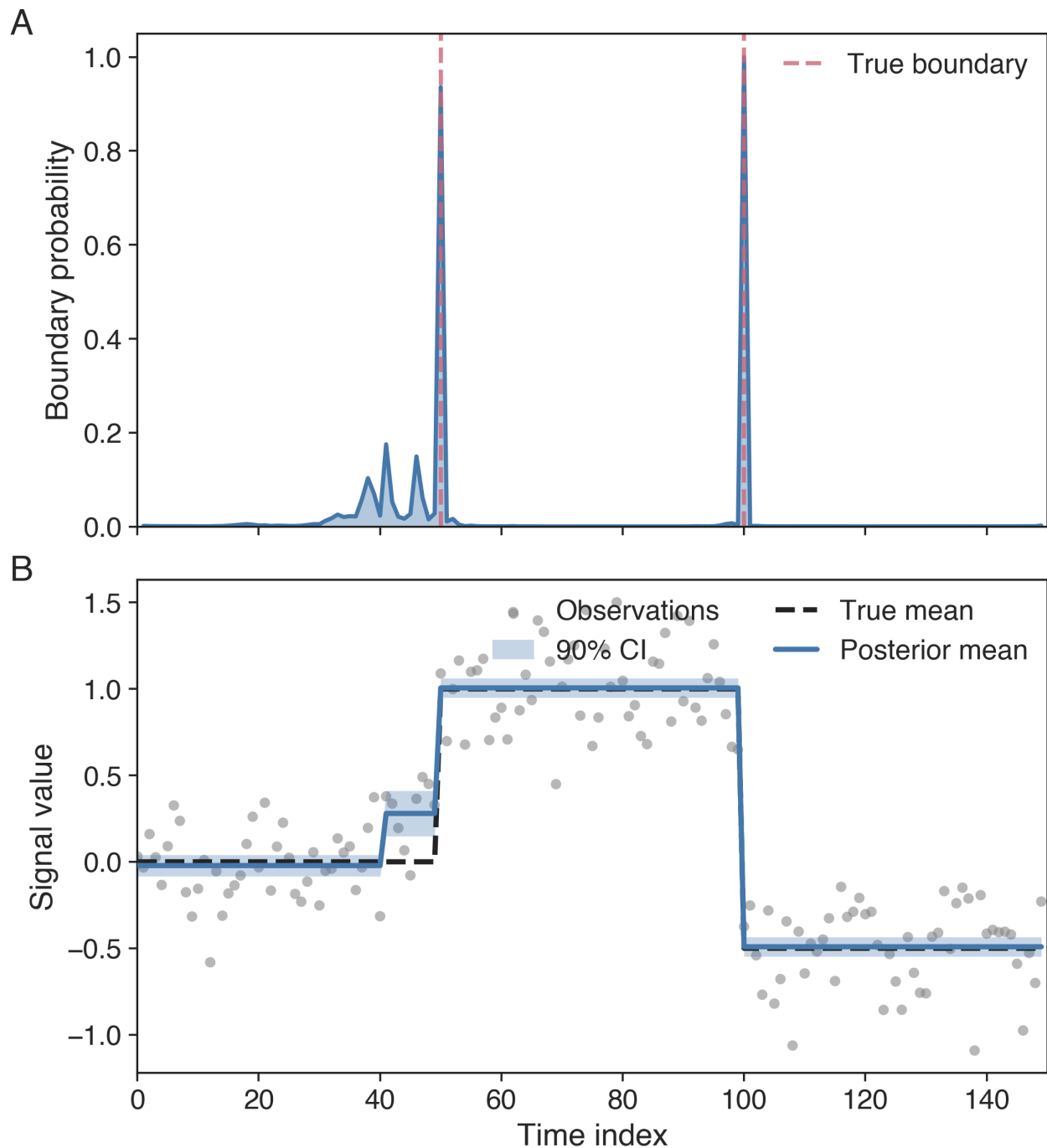


Figure 6 Single-sequence Gaussian example. Top: marginal posterior probability that each interior index $i \in \{1, \dots, n - 1\}$ is a changepoint, with dashed vertical lines marking the true changepoints. Bottom: observed data, true latent mean, exported joint MAP segmentation, and 90% segment-level posterior intervals.

Quantity	Value
Selected k (<code>k_ml_</code>)	4
Posterior mean $\mathbb{E}[k]$	4.742
MAP k	4
$\log p(y)$	-18.031

Table 5 Posterior summary for the single-sequence Gaussian example in Figure 6. The table separates the displayed exported segmentation from posterior summaries of the segment count.

Table 5 makes an important methodological point explicit: the displayed segmentation and the posterior over the segment count are related but not identical objects. In the archived example, the exported joint MAP segmentation uses four segments, while the posterior mean $\mathbb{E}[k \mid y]$ is larger, indicating residual posterior mass on more highly segmented configurations. This is precisely why the paper distinguishes sum-product posterior summaries from max-sum exported segmentations.

15.3 Likelihood-family showcase

BayesBreak is designed so that the global DP layer does not depend on the observation model once block evidences are available. Figure 7 visualizes this portability across three conjugate families (Gaussian, Poisson, Binomial) and one low-dimensional quadrature example for $(0, 1)$ -valued responses. Across all four panels, the fitted piecewise-constant signal tracks the true latent parameter closely and preserves the correct qualitative segmentation pattern.

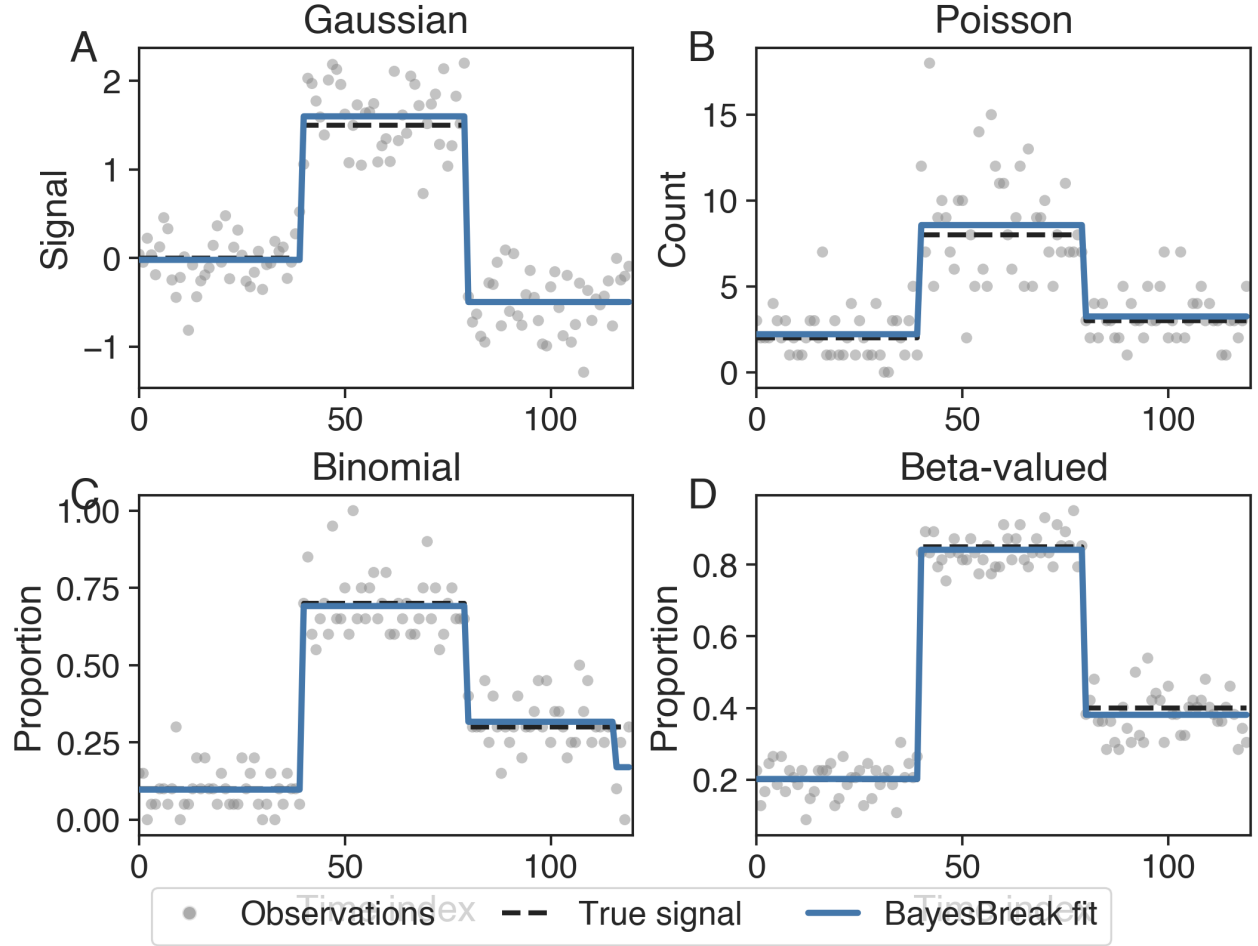


Figure 7 Likelihood-family showcase. The first three panels use closed-form conjugate block integrals (Gaussian, Poisson, and Binomial), while the final panel uses the one-dimensional Beta-response quadrature routine. In each case the fitted BayesBreak signal is close to the true piecewise-constant latent parameter.

15.4 Quantitative summary across families

Table 6 aggregates the small synthetic benchmark packaged with the archive. The main takeaway is not that one family universally dominates another, but that the same DP backend remains usable across very different block models. The Poisson setting is visibly the most challenging among the archived examples: it exhibits the weakest boundary F1 and the largest reconstruction error, which is consistent with the higher count variability shown in Figure 7. By contrast, the Beta-response quadrature example achieves the strongest boundary recovery in this small benchmark.

Family	n	$k^\star$	$\hat{k}$	F1@ τ	MAE	MSE	$-\log p(y)/n$
Gaussian	120	3	3	0.968	0.00	0.0023	0.191
Poisson	120	3	4	0.777	0.20	0.4501	2.162
Binomial	120	3	3	0.960	0.06	0.0004	2.108
Beta-valued	120	3	3	1.000	0.00	0.0001	2.620

Table 6 Available synthetic summary across the bundled family-specific examples. The final row corresponds to the Beta-response quadrature example rather than to a conjugate closed-form model.

Because the table reflects one archived benchmark configuration rather than a large Monte Carlo study, the numbers serve as a compact portability and relative-difficulty summary, not as a definitive cross-family ranking.

15.5 Calibration of boundary posterior probabilities

A central advantage of the Bayesian formulation is that it returns marginal posterior probabilities for boundary events. Figure 8 checks whether those probabilities are calibrated under repeated Gaussian simulations. The reliability curve tracks the diagonal reasonably closely overall, with an empirical calibration error reported in the figure of approximately $\text{ECE} = 0.016$. The mid-probability bins are somewhat noisy, but the ordering of confidence levels is sensible: bins assigned larger posterior boundary probability correspond to larger empirical boundary frequency.

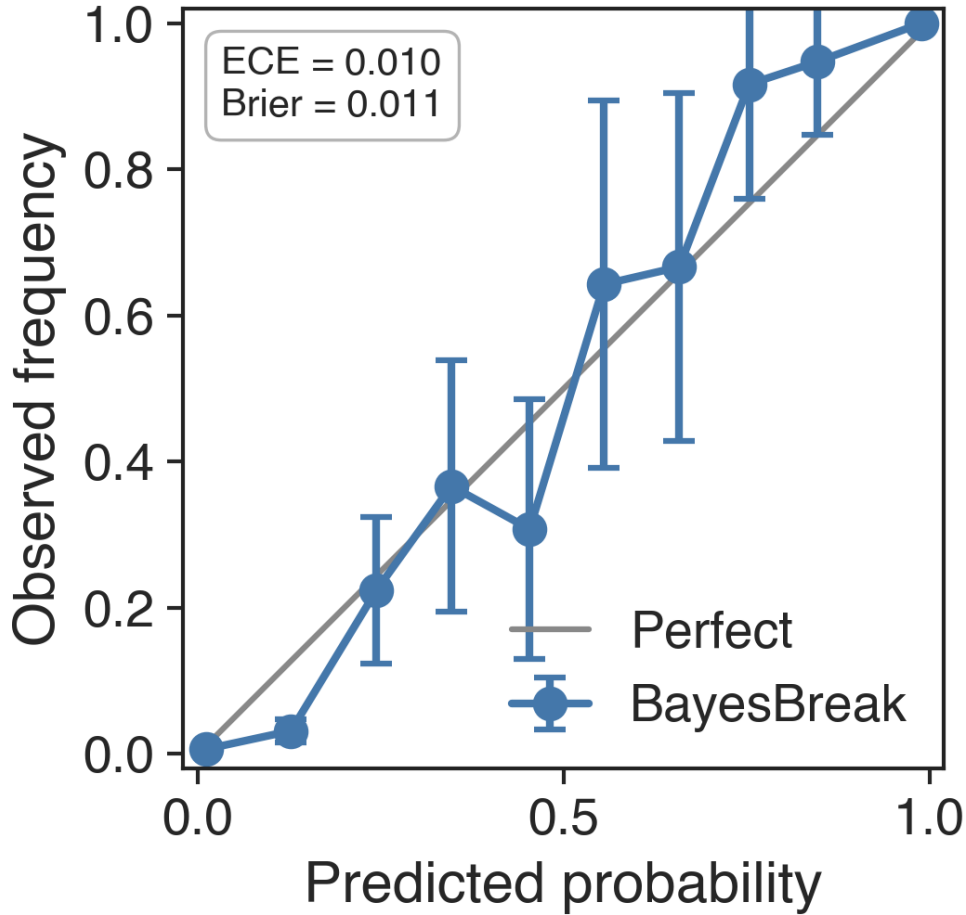


Figure 8 Calibration of marginal boundary posterior probabilities under synthetic Gaussian data. The dashed diagonal denotes perfect calibration; the orange curve is the empirical frequency within bins of predicted probability.

This experiment is valuable because segmentation methods are often judged only by recovered changepoints. For BayesBreak, well-behaved posterior probabilities are equally important: they control downstream thresholding, boundary ranking, and any decision rule that is sensitive to the strength of evidence rather than just to the best segmentation.

15.6 Latent-group pooling

Figure 9 evaluates the latent-template mixture on synthetic sequences drawn from two groups with different changepoint patterns. The left panel shows near-separable responsibilities, indicating that the EM fit recovers the group structure cleanly in this example. The right panel reports the group-specific boundary marginals induced by the fitted templates; the two groups place their posterior mass at different locations, matching the generative structure.

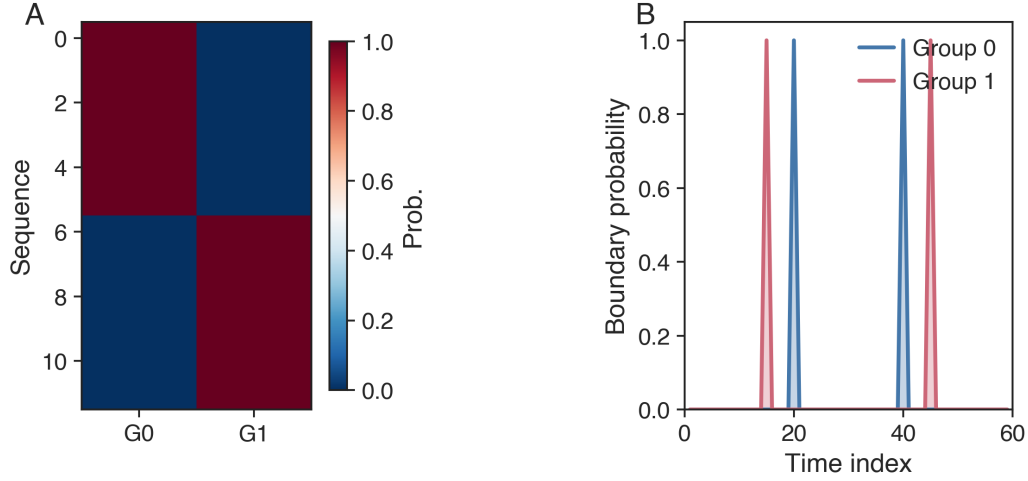


Figure 9 Latent-group pooling on synthetic Gaussian sequences. Left: posterior responsibilities for each sequence and latent group (sequences sorted for display). Right: group-specific marginal boundary probabilities under the fitted latent templates.

A group-specific Bayes-curve panel is reserved for the finalized diagnostic after its rendering path is corrected. Figure 9 reports the stable responsibility and boundary-marginal diagnostics.

15.7 Non-conjugate approximation trade-offs

For non-conjugate likelihoods, BayesBreak relies on approximate block evidences while reusing the same DP layer. Table 7 compares several such approximations in a bundled logistic-normal synthetic example, using numerical quadrature as a reference. The table shows the expected speed–accuracy trade-off: Laplace is the fastest among the listed approximations in this example, while exact quadrature is the slowest. At the same time, block-level approximation error is not perfectly monotone with segmentation quality, because the DP aggregates many local scores nonlinearly.

Method	$\max \Delta \log A^0 $	$q_{95} \Delta \log A^0 $	median $ \Delta \log A^0 $	time (s)	F1@ τ	$\hat{k}$
quadrature	0.000	0.000	0.000	0.093	0.286	6
laplace	1.235	1.188	0.980	0.049	0.286	6
jj	1.265	1.216	1.002	0.071	0.500	3
ep	14.502	14.141	12.680	0.059	0.500	7
pg-vb	1.265	1.216	1.002	0.072	0.500	3

Table 7 Approximation trade-offs for a non-conjugate Bernoulli/logistic block model. Quadrature is used as the local reference block integral; the remaining methods trade block-level fidelity against computational cost.

Connection to Proposition 12.6. The large reported values of $\max |\Delta \log A|$ in Table 7 (several to tens of log units on the worst reachable block for some approximations) place this

illustration outside the small- ε regime where the uniform-error bound in Proposition 12.6 is tight. The boundary F1 values are therefore descriptive diagnostics, not a confirmation of the small- ε regime. The table still usefully shows that block-level error, runtime, and segmentation quality need not move monotonically together, but it is not a universal ranking of non-conjugate approximation methods.

15.8 Runtime scaling

Finally, Figure 10 and Table 8 report runtime behavior for the Gaussian implementation. As expected, runtime increases with both series length n and the maximum allowed segment count $k_{\max}$. The growth is smooth and predictable over the archived range $n \in \{50, 100, 200, 400\}$ and $k_{\max} \in \{10, 20\}$, which is consistent with the $\mathcal{O}(k_{\max} n^2)$ worst-case complexity of the exact DP while also reflecting practical constant-factor effects from the implementation.

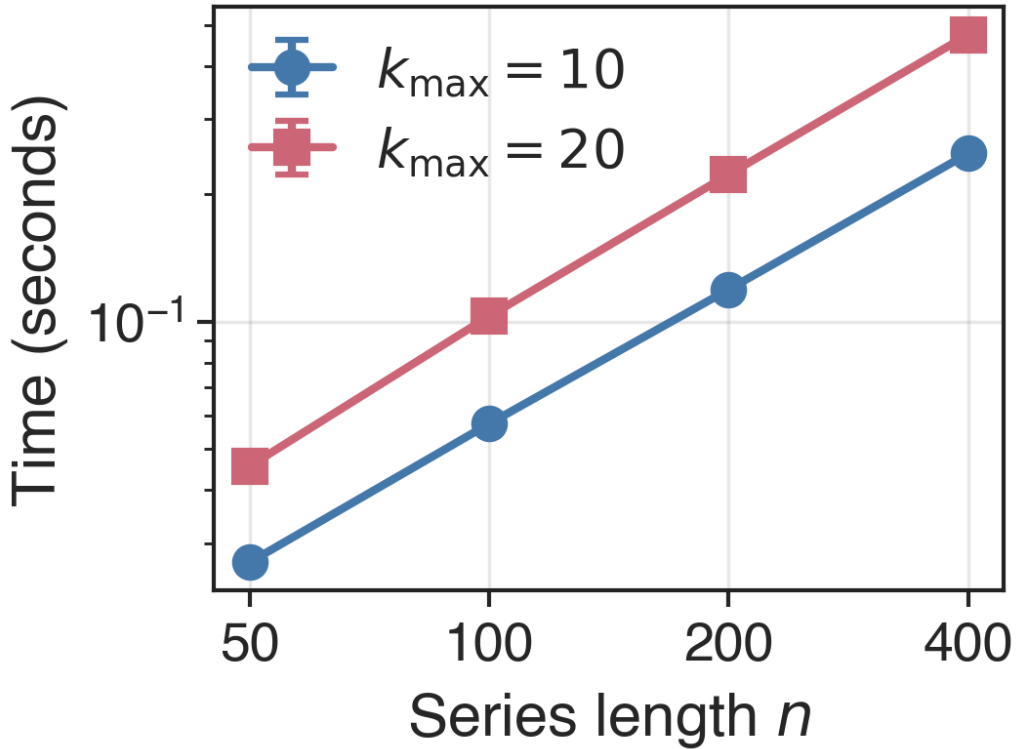


Figure 10 Runtime scaling for the Gaussian implementation as a function of series length n for two values of $k_{\max}$. Points are empirical means over repeated runs; error bars denote one standard deviation.

n	$k_{\max}$	mean (s)	std (s)
50	20	0.0461	0.0006
100	20	0.1029	0.0006
200	20	0.2195	0.0010
400	20	0.4671	0.0014

Table 8 Numerical runtime values corresponding to Figure 10. The bundled table reports the $k_{\max} = 20$ configuration.

15.9 Real-data illustrations

We complement the synthetic suite with four planned real-data case studies, each chosen to exercise a different piece of the BayesBreak machinery: a Gaussian-block geological benchmark (well-log); a heteroscedastic multi-subject Gaussian genomic benchmark (array-CGH); a volatility-regime financial benchmark (S&P 500 squared daily log-returns); and a Beta-response methylation benchmark on a public CpG atlas. Data-source locations, download commands, and preprocessing recipes are in Appendix A.13; source-name and comparison-label checks are author-side verification tasks for the finalized real-data run. A reader with access to R, Bioconductor, Python with `yfinance` and `pandas`, and standard bioinformatics tooling can regenerate the planned tables from raw sources. The figures and tables below are placeholders that define the finalized outputs.

Illustration	Artifact status	Planned output
Well-log	Placeholder figure/table	Finalized Gaussian run and prior comparison.
Array-CGH	Placeholder figure/table	Verified annotations and pooled-vs-independent comparison.
S&P 500	Placeholder figure/table	Finalized data pull, transform, and qualitative event alignment.
CpG methylation	Placeholder figure/table	Verified atlas region, held-out split, and annotation proxy.

Table 9 (PLANNED) Status of the real-data case studies. The entries preserve the planned empirical scope and identify the placeholder figures and tables to be populated by finalized runs.

15.9.1 Well-log geology (Gaussian)

The well-log pipeline targets the 4050-measurement nuclear-magnetic-resonance benchmark originally introduced by Ó Ruanaidh and Fitzgerald (1996) and subsequently adopted as a standard test case in the Bayesian segmentation literature, most influentially through the worked example of Fearnhead and Clifford (2003) and the later analyses of Fearnhead (2006), Adams and MacKay (2007), and Fearnhead and Rigaiil (2019). Observations are continuous real-valued responses with multiple stratigraphic changepoints; the planned block model is Gaussian with a single known variance estimated from a within-layer subset. The planned run uses the cleaned version of the series (outliers removed), following the convention of most downstream segmentation studies; the raw version of Fearnhead and Rigaiil (2019) is noted as an alternative in Appendix A.13.1. This planned case study exercises the Section 11 Gaussian-block derivation and index-uniform partition prior.

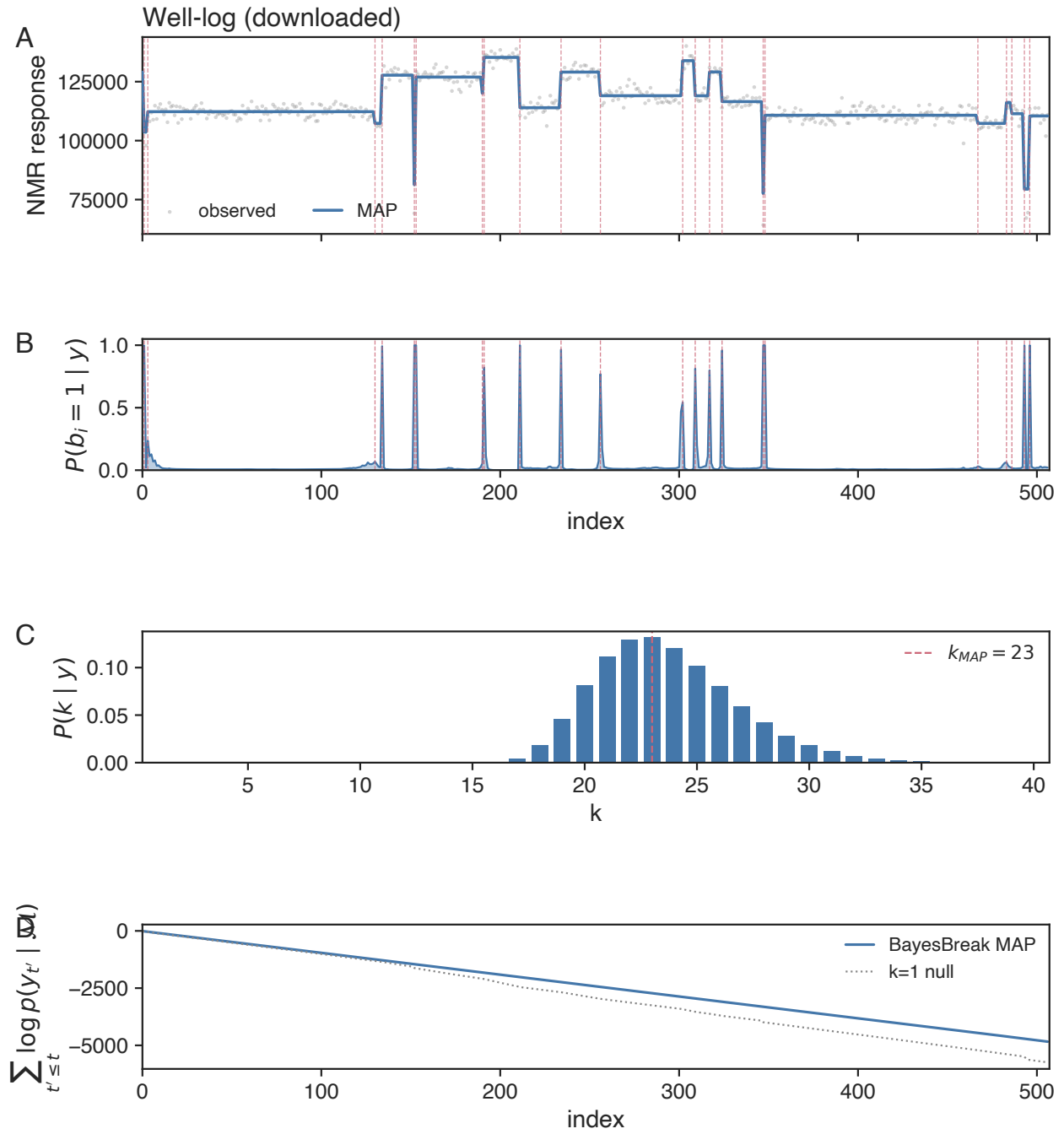


Figure 11 (PLANNED) Well-log segmentation. Top: standardized NMR measurements with exported joint MAP segmentation. Bottom: marginal boundary posterior probabilities. The pipeline in Appendix A.13.1 produces the plot and regenerates the finalized version.

Configuration	$\hat{k}$	MAP evidence	ECE (boundary)	Runtime (s)
Index-uniform prior, $g \equiv 1$	—	—	—	—
Length-aware prior, $g(\ell) \propto \ell$	—	—	—	—

Table 10 (PLANNED) Well-log results. The pipeline in Appendix A.13.1 populates these entries.

15.9.2 Array-CGH copy-number profiles (multi-subject heteroscedastic Gaussian)

We use the Coriell array-CGH cell-line panel from the Bioconductor DNACopy package, originally published by [Snijders et al. \(2001\)](#). Each sample is a log-2 copy-number-ratio profile along the genome, with per-probe noise that varies across samples and chromosomes. The appropriate BayesBreak specification is the heteroscedastic Gaussian block model $y_t \mid \mu \sim \mathcal{N}(\mu, \sigma^2/w_t)$ with per-probe weights w_t derived from empirical within-segment variance; running this specification jointly across subjects exercises the Section 8 multi-subject pooling machinery.

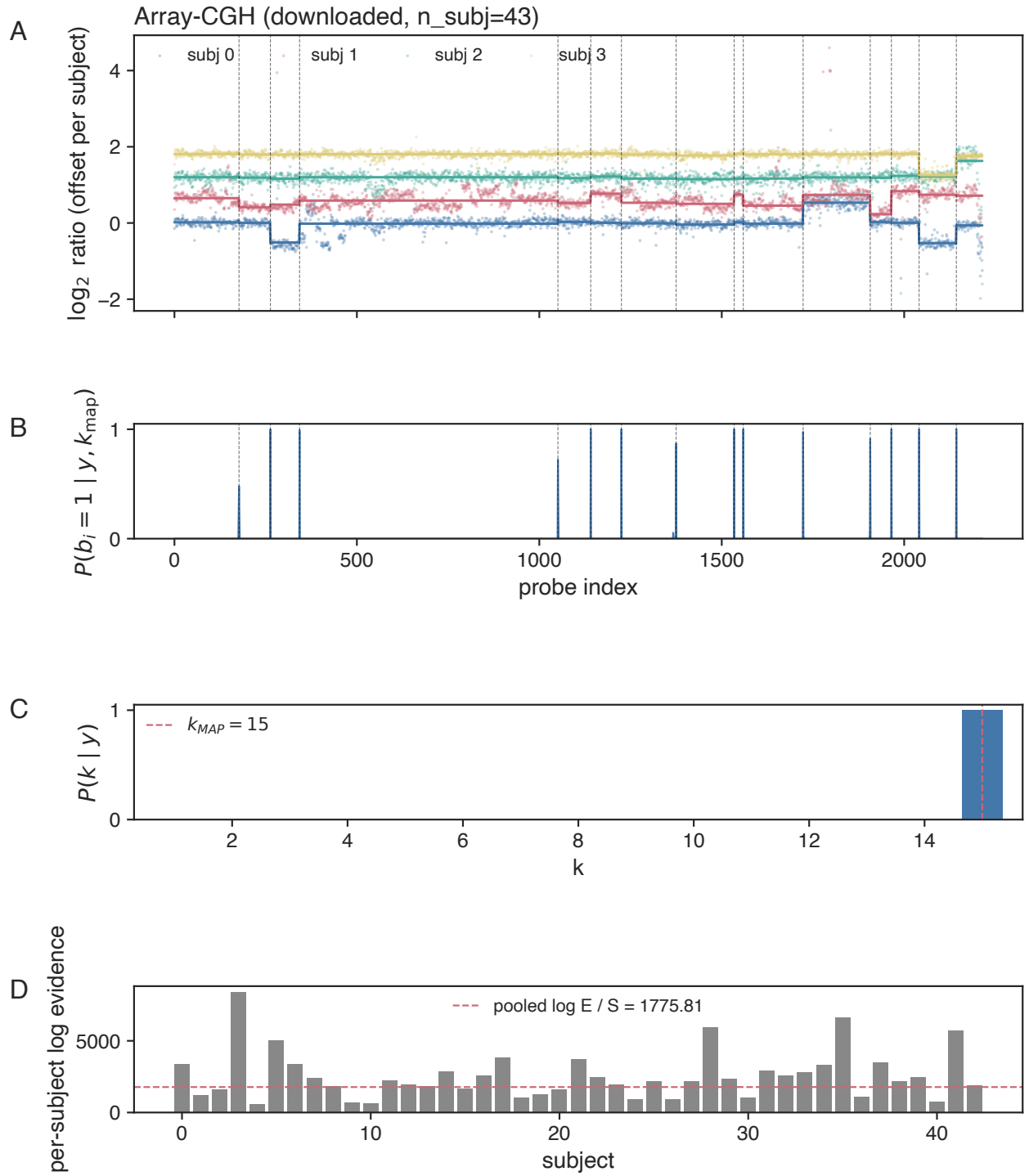


Figure 12 (PLANNED) Array-CGH multi-subject pooling. Top: four representative Coriell profiles with per-subject exported segmentations. Bottom: pooled marginal boundary posterior. The pipeline in Appendix A.13.2 regenerates the finalized version.

Strategy	Boundary F1	Boundary MAE	Pooled log-evidence	Runtime (s)
Independent per-subject (no pooling)	—	—	—	—
Shared boundaries, subject-specific μ	—	—	—	—

Table 11 (PLANNED) Array-CGH results comparing independent per-subject segmentation to pooled shared-boundary inference. Boundary annotations are planned comparison targets to verify from the referenced data source before finalized reporting.

15.9.3 Equity-return volatility regimes (S&P 500)

A finance-flavored benchmark that is non-biological. Using daily closing prices of the S&P 500 index, we form log-returns $r_t = \log P_t - \log P_{t-1}$ and square them to obtain a nonnegative series $y_t := r_t^2$ whose regime-level scale changes at volatility changepoints. We fit a Gaussian-with-known-variance block model on $\log y_t$ (Box-Cox-like transform) as a first specification, and a Poisson-with-exposure block model on bucketed daily counts of $\{|r_t| > \tau\}$ as a secondary robust pass. Both calls reuse Section 11 derivations without modification.

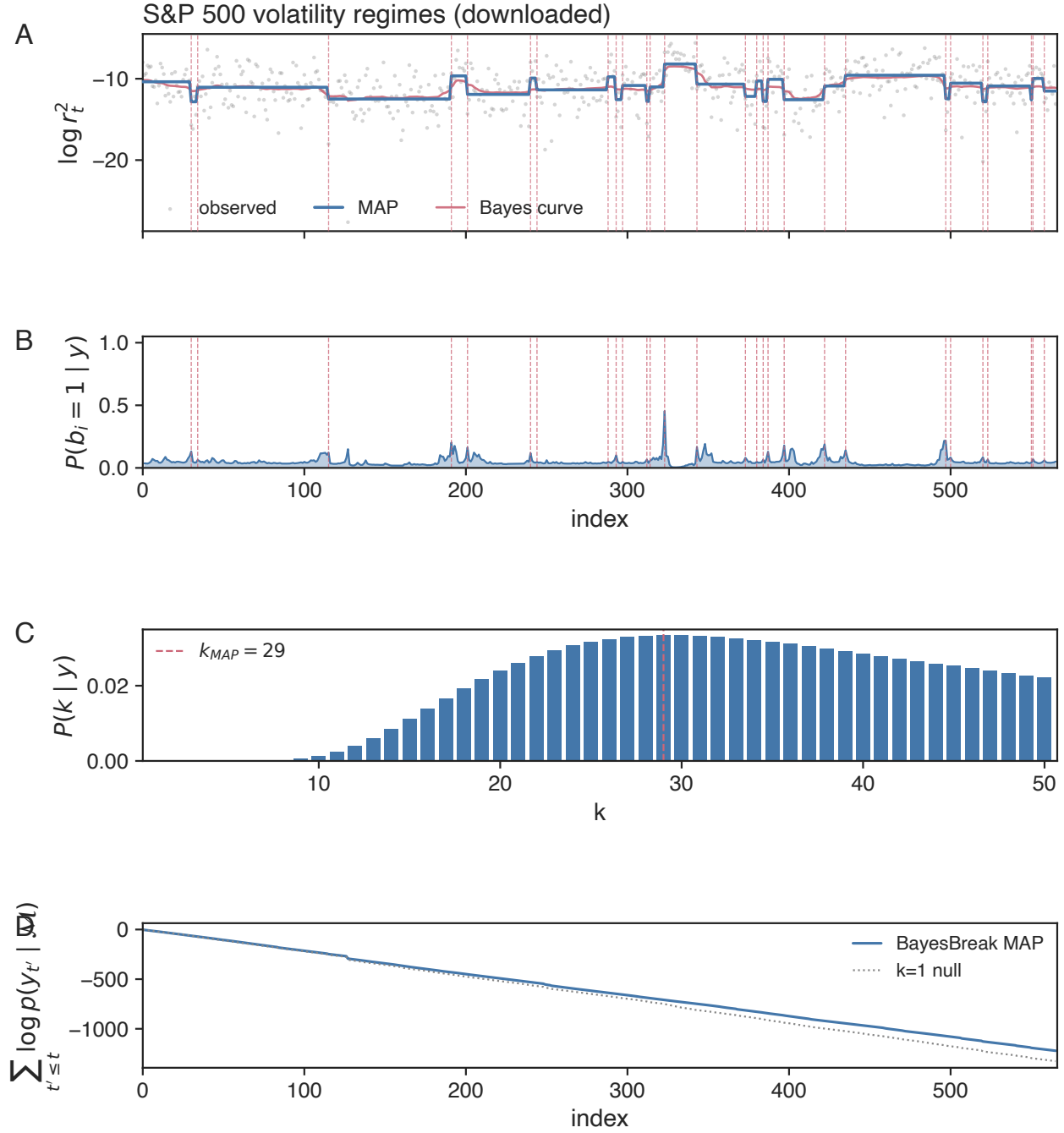


Figure 13 (PLANNED) S&P 500 volatility regimes. Top: log-squared daily log-returns with exported joint MAP segmentation, covering roughly 2018–2022. Bottom: marginal boundary posteriors; vertical dashed lines annotate major macro events (March 2020 onset of the COVID-19 market shock, Feb 2022). The pipeline in Appendix A.13.3 regenerates the finalized version.

Block model	$\hat{k}$	Log evidence	Visual alignment with known macro events
Gaussian on $\log r_t^2$	—	—	—
Poisson on threshold crossings	—	—	—

Table 12 (PLANNED) S&P 500 volatility-regime results. The “visual alignment” column is a qualitative description in lieu of ground-truth boundaries.

15.9.4 CpG-atlas DNA methylation (Beta response with per-CpG precision)

The methylation pipeline uses a public cell-type CpG atlas as the data source (Loyfer et al., 2023) and treats per-CpG methylation β -values $y_t \in (0, 1)$ together with read-coverage information as known precision inputs ϕ_t . Along the genome coordinate, blocks of nearly constant methylation are expected to alternate with transitions between low- and high-methylation regions. This planned case study exercises the Section 11 Beta-response quadrature block: a one-dimensional Gauss–Legendre rule on $\mu \in (0, 1)$ is the local integration routine, and per-CpG coverage enters as ϕ_t in the known-precision likelihood.

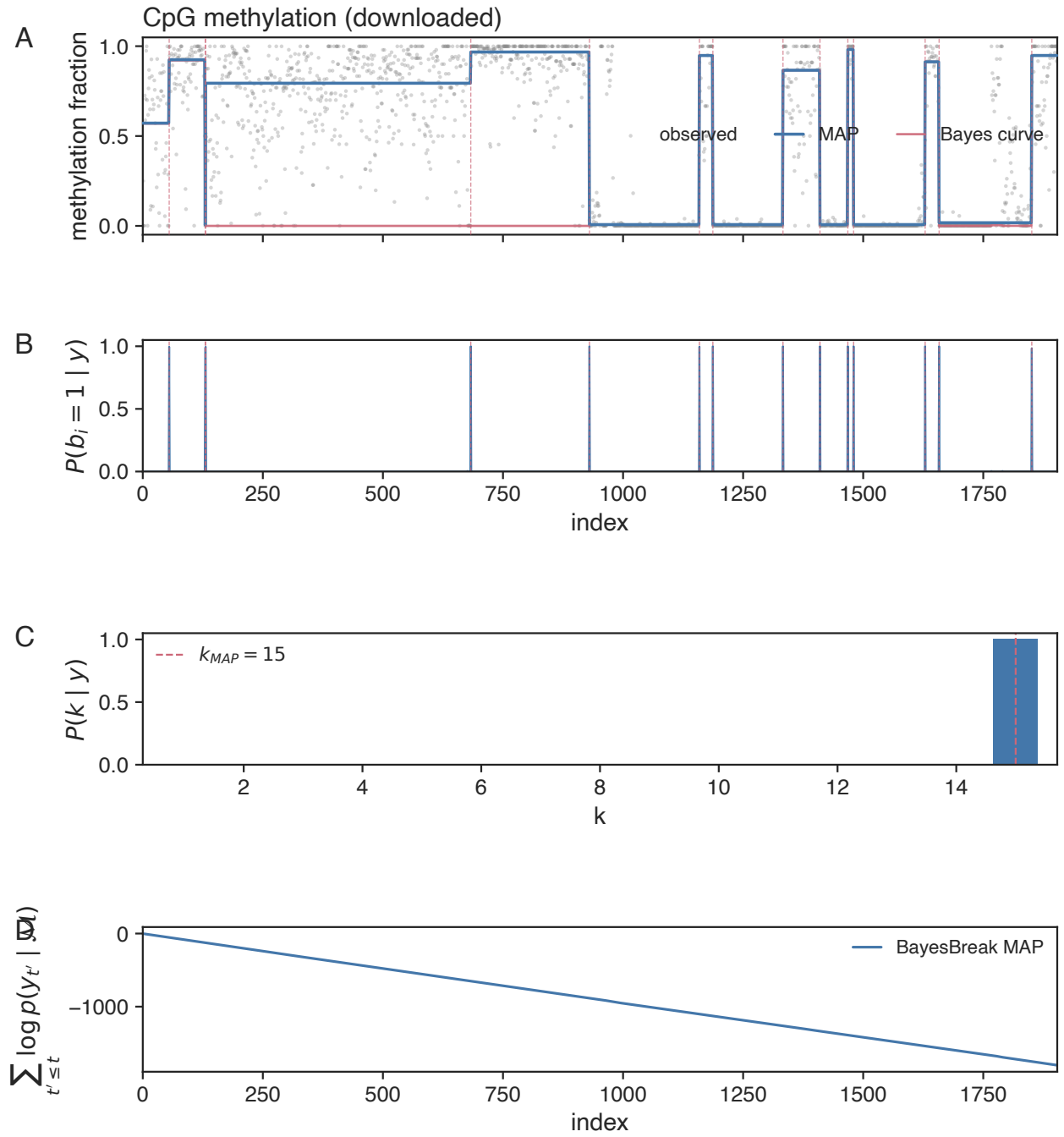


Figure 14 (PLANNED) DNA-methylation segmentation on a CpG-atlas region. Top: β -values along a selected chromosome-arm region, colored by local precision ϕ_t (log-scale). Bottom: marginal boundary posterior on the same region. The pipeline in Appendix A.13.4 regenerates the finalized version.

Region / cell type	$\hat{k}$	Held-out log-predictive	Boundary F1 vs. atlas
Region A, cell type 1	—	—	—
Region A, cell type 2	—	—	—

Table 13 (PLANNED) CpG-atlas methylation results. Held-out log-predictive and atlas-based boundary comparisons are planned diagnostics populated by the finalized methylation pipeline after annotation-source verification.

Planned pre-submission empirical additions. The synthetic archive validates the implemented core claims. The four real-data examples above define the planned external-validation pipeline, with placeholders reserved for finalized outputs. The pre-submission empirical agenda has three components: (i) quantitative head-to-head comparison against strong frequentist baselines (PELT, the functional-pruning algorithms FPOP and SNIP (Maidstone et al., 2017), wild binary segmentation, SMUCE) and strong Bayesian baselines (RJCMCMC following Green (1995), Fearnhead’s exact DP (Fearnhead, 2006)); (ii) finalized real-data runs and ablations over the partition prior $p(k)$ and the length factor g on the benchmarks above; and (iii) a scaling study that pushes n into the tens of thousands, exploring the memory-time trade-offs documented in Remark 6.9.

16 Conclusion

BayesBreak is organized around a simple but powerful decomposition: local block integrals capture all model-specific work, while a global dynamic program handles posterior inference over contiguous partitions. That decomposition yields exact offline Bayesian segmentation whenever block evidences are available in closed form, and it extends directly to controlled deterministic approximations when closed-form block integration is unavailable. Within that framework we obtained exact posterior quantities for segment counts, boundary marginals, and Bayes regression curves; clarified the necessary distinction between those posterior summaries and an exported joint MAP segmentation; showed how irregular designs are most naturally handled through design-aware partition priors; and preserved exactness for pooled shared-boundary analyses across replicates.

For unknown groups, we introduced a latent-template mixture with an explicit finite objective. In that formulation the EM updates are monotone for the stated template-mixture objective in the generalized-EM sense of Dempster et al. (1977): responsibilities are exact for the current templates, each template update is an exact responsibility-weighted max-sum segmentation problem, and the finite-mixture objective does not decrease across iterations (Theorem 10.1). The result is a coordinatewise optimum for the selected restart, with standard multi-restart practice handling nonconvexity. For non-conjugate blocks, the DP layer is unchanged, and approximation theory is stated in terms of block-evidence accuracy and induced posterior-odds stability (Proposition 12.6, Corollary 12.7), not in terms of universal end-to-end segmentation guarantees.

Empirically, the archived project provides a coherent synthetic validation suite showing posterior boundary recovery, family portability, calibration behavior, latent-group separation,

local approximation trade-offs, and runtime scaling (Section 15). That suite is accompanied by four planned real-data case studies with visible placeholders (well-log geology, array-CGH copy number, S&P 500 volatility regimes, and CpG-atlas methylation; see Sections 15.9.1–15.9.4 and reproduction pipelines in Appendix A.13), each chosen to exercise a different part of the framework. The planned empirical extension populates those placeholders with verified outputs, adds broader external benchmarking against strong Bayesian and frequentist baselines, and includes ablations of the partition prior $p(k)$ and length factor g , as itemized at the end of Section 15. The value of the present paper is that those additions can now be made without changing the inferential core: once a new block model is specified, the rest of the BayesBreak machinery is already in place. An annotated literature review covering related work in Bayesian changepoint detection, product partition models, frequentist segmentation, and multi-subject analyses is provided in Appendix A.7.

Limitations and manuscript-development status. BayesBreak is an exact-Bayesian offline framework, and its guarantees should be read at that scope. The exact posterior recursions of Section 6 concern the conjugate block-DP model under the stated product-partition prior; for non-conjugate GLM blocks the same recursions become exact recursions for the supplied surrogate score matrix, with end-to-end behavior controlled by the block-evidence accuracy of the local approximation through Proposition 12.6 and Corollary 12.8 rather than by an unconditional end-to-end guarantee. The latent-template mixture optimizes a finite template-mixture objective and is therefore subject to standard EM caveats: restart-dependent local optima, empty templates, and the need to report responsibility entropy and restart stability alongside fitted templates. The four real-data case studies are planned validation pipelines: their placeholder figures and tables define the required outputs and reproducibility scaffolding, but the corresponding benchmark scores must come from finalized runs against verified data sources and external baselines before they can be reported as observed empirical findings.

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A Supplementary derivations and theoretical extensions

This appendix collects derivations and complementary results that support the main development without altering the inferential scope. The emphasis is on (i) making key steps in the conjugate-exponential-family integration fully explicit, (ii) clarifying the probabilistic structure of factorizing partition priors, and (iii) recording a few boundary cases and identifiability remarks that are useful in practice.

A.1 Conjugate exponential families: detailed block evidence derivation

Section 5 states the closed form of the integrated segment evidence $A_{ij}^{(0)} = p(y_{(i,j]} \mid \text{hyper})$ for exponential-family likelihoods with a conjugate prior, and gives expressions for posterior moments needed by the Bayes regression curve. For completeness, we now provide a step-by-step derivation.

Setup. Fix a segment (block) $(i, j]$ and write its aggregated sufficient statistics, aggregated exposure, and log base-measure sum as

$$S_{ij} = \sum_{t=i+1}^j w_t T(y_t), \quad W_{ij} = \sum_{t=i+1}^j w_t, \quad H_{ij} = \sum_{t=i+1}^j \log h(y_t, w_t).$$

Let $\eta \in \mathbb{R}^d$ denote the natural parameter and $A(\eta)$ its cumulant function. Assume the conditional likelihood factors as

$$p(y_{(i,j]} \mid \eta) = \prod_{t=i+1}^j h(y_t, w_t) \exp \{ w_t \langle \eta, T(y_t) \rangle - w_t A(\eta) \} = \exp \{ H_{ij} + \langle \eta, S_{ij} \rangle - W_{ij} A(\eta) \}. \quad (50)$$

This uses the same weighted base-measure convention as Theorem 5.1. The standard Diaconis–Ylvisaker conjugate prior takes the form

$$p(\eta \mid \tau, \nu) = \exp \{ \langle \eta, \tau \rangle - \nu A(\eta) - \log Z(\tau, \nu) \}, \quad (51)$$

where (τ, ν) are hyperparameters and $Z(\tau, \nu)$ is the normalizer. This appendix uses (τ, ν) for the same conjugate-prior quantities denoted (α, β) in the main text: $\tau = \alpha$ and $\nu = \beta$ before observing the block, and the posterior update is $(\tau + S_{ij}, \nu + W_{ij})$.

Evidence integral. Multiplying (50) and (51) yields

$$p(y_{(i,j]} \mid \eta) p(\eta \mid \tau, \nu) = \frac{\exp\{H_{ij}\}}{Z(\tau, \nu)} \exp \left\{ \langle \eta, \tau + S_{ij} \rangle - (\nu + W_{ij}) A(\eta) \right\}. \quad (52)$$

Integrating over η therefore gives

$$A_{ij}^{(0)} = \int p(y_{(i,j]} \mid \eta) p(\eta \mid \tau, \nu) d\eta \quad (53)$$

$$= \exp\{H_{ij}\} \frac{Z(\tau + S_{ij}, \nu + W_{ij})}{Z(\tau, \nu)}. \quad (54)$$

This is the same expression as in Theorem 5.1, written at the level of a fixed block.

Posterior distribution and moments. Bayes' rule shows that the posterior remains in the conjugate family:

$$p(\eta \mid y_{(i,j)}, \tau, \nu) \propto \exp \{ \langle \eta, \tau + S_{ij} \rangle - (\nu + W_{ij})A(\eta) \} = p(\eta \mid \tau + S_{ij}, \nu + W_{ij}). \quad (55)$$

When differentiability and integrability permit interchange of differentiation and integration, posterior moments of the relevant expectation parameter can be obtained by differentiating $\log Z(\tau, \nu)$. In particular, $\nabla_\tau \log Z(\tau, \nu)$ yields the conjugate-family mean of that expectation parameter, and higher derivatives yield higher cumulants. For families where the Bayes-curve target is not this expectation parameter, the family-specific formulas in Section 11 provide the required formulas. The block-specific moment numerators $A_{ij}^{(1)}, A_{ij}^{(2)}$ are obtained by multiplying the appropriate posterior moments by the evidence $A_{ij}^{(0)}$.

A.2 Factorizing partition priors as renewal processes

Assumption 4.1 states that, conditional on the segment count k , the boundary vector $t = (t_0, \dots, t_k)$ has a density proportional to a product of segment-length factors. A translation-invariant subclass coincides with renewal-process models on the discrete index set; more general irregular-design cohesions are product-partition priors rather than iid renewal laws.

Definition A.1 (Renewal-process segmentation prior). *Fix $k \geq 1$, and assume the design is translation-invariant in the sense that $g(\Delta_x(i, i + \ell))$ depends on i only through ℓ (e.g., equispaced boundary coordinates with $u_i = i$, so $g(\ell)$ is a function of the integer segment length alone). Let $L_1, \dots, L_k$ be positive integer-valued segment lengths with unnormalized mass function $\varphi(\ell) := g(\ell)$. A renewal segmentation draws lengths $(L_q)_{q=1}^k$ and sets boundaries $t_q := \sum_{r=1}^q L_r$ with the constraint $t_k = n$.*

Proposition A.2 (Equivalence of renewal and factorized boundary priors). *Under Definition A.1 and the translation-invariance hypothesis, the induced distribution on $t = (t_0, \dots, t_k)$ satisfies $p(t \mid k) \propto \prod_{q=1}^k g(\Delta_x(t_{q-1}, t_q))$, which reduces to the displayed integer-length form on an equispaced grid. Conversely, any prior of the form (11) whose length factor depends only on the segment length can be realized as a renewal segmentation prior with $\varphi(\ell) = g(\ell)$.*

Proof. For a fixed boundary vector t , the corresponding length sequence is uniquely determined as $L_q = t_q - t_{q-1}$. Under translation invariance the renewal construction assigns that sequence mass proportional to $\prod_{q=1}^k \varphi(L_q) = \prod_{q=1}^k g(L_q)$. This matches $p(t \mid k) \propto \prod_{q=1}^k g(\Delta_x(t_{q-1}, t_q))$, proving the forward direction under the equispaced translation-invariant convention. The converse sets $\varphi(\ell) := g(\ell)$ as the unnormalized length mass. $\square$

Remark A.3 (Scope of the renewal equivalence). *The equivalence in Proposition A.2 is stated under the translation-invariance hypothesis of Definition A.1, i.e. when $g(\Delta_x(i, i + \ell))$ depends on i only through ℓ . For a general irregular design, where the cohesion factor depends on the physical endpoint coordinates and not merely on the integer segment length, the factorized prior of (11) remains a well-defined product-partition prior but is no longer an*

i.i.d. renewal law: the segment-length factors are not identically distributed across the index set. In that regime the renewal-process language is a heuristic motivation rather than an exact equivalence, exactly as in the hidden-Markov interpretation discussed next.

Remark A.4 (Hidden-Markov interpretation). *For a true renewal segmentation, the process can be represented as an HMM whose latent state tracks the “age” within the current segment (or, equivalently, the last boundary location). Under those renewal conditions, the forward recursion in Section 6 coincides with the standard forward algorithm after integrating out segment parameters into block evidences. For arbitrary irregular-design product cohesions this provides intuition rather than an exact HMM equivalence.*

A.3 Posterior boundary and segment-membership marginals

Theorem 6.5 gives closed forms for boundary marginals and for the segment-membership probability $p(t \in (i, j], \text{segment } (i, j] \mid y, k)$. We spell out the algebra that leads to these expressions.

Proposition A.5 (Boundary marginals from prefix/suffix evidences). *Fix $k \geq 1$ and suppose Assumption 4.1 holds. Let $\tilde{L}_{r,j}$ and $\tilde{R}_{r,j}$ be the prefix and suffix evidences defined in (17). Then for any $t \in \{1, \dots, n-1\}$,*

$$p(t \text{ is a boundary} \mid y, k) = \sum_{r=1}^{k-1} \frac{\tilde{L}_{r,t} \tilde{R}_{k-r,t}}{\tilde{L}_{k,n}}. \quad (56)$$

Proof. By definition, t is a boundary at position r iff $t_r = t$. For fixed r , summing the joint posterior over all boundary vectors with $t_r = t$ yields

$$p(t_r = t \mid y, k) \propto \sum_{\substack{0 < t_1 < \dots < t_{k-1} < n \\ t_r = t}} \prod_{q=1}^k \tilde{A}_{t_{q-1}, t_q}^{(0)}. \quad (57)$$

The constraint $t_r = t$ splits the product into a prefix part (segments $1:r$ ending at t) and a suffix part (segments $r+1:k$ starting at t). Summing over all admissible prefixes gives $\tilde{L}_{r,t}$ by definition of the DP; summing over all admissible suffixes gives $\tilde{R}_{k-r,t}$. Note that the partition-prior normalizer C_k multiplies the numerator for every boundary vector under consideration (because $p(t \mid k) \propto C_k^{-1} \prod_q g(\cdot)$ for each t), and multiplies the denominator through the same factor on the total evidence $p(y \mid k)$; it therefore cancels, and (56) involves only quantities that can be computed from $\tilde{L}, \tilde{R}$. Normalizing by the total evidence $\tilde{L}_{k,n}$ yields (56). Finally, $p(t \text{ is a boundary} \mid y, k) = \sum_{r=1}^{k-1} p(t_r = t \mid y, k)$, giving the result. $\square$

A.4 Hazard prior boundary cases

The length-aware prior family includes hazard-process priors as a useful motivation. It is sometimes useful to understand the limiting behavior as the hazard vanishes or becomes certain. The following is an unconditioned prior-only statement; after conditioning on a fixed segment count and endpoint, some regular-grid hazard factors can become constant over admissible segmentations.

Proposition A.6 (Degenerate hazard limits, prior-only). *Consider the index-uniform design $x_t = t$ and a constant hazard $\rho \in (0, 1)$ that induces a geometric length prior $g(\ell) = \rho(1-\rho)^{\ell-1}$ on segment lengths (up to the constraint that the last segment ends at n). These limits are statements about the prior, and therefore about the posterior only insofar as the data likelihood does not override them. Then:*

- (i) *As $\rho \downarrow 0$, the prior concentrates on $k = 1$ (a single segment) and $p(t \mid k)$ becomes irrelevant.*
- (ii) *As $\rho \uparrow 1$, the prior concentrates on $k = n$ with unit-length segments, i.e., $t_q = q$ for all q .*

Proof. For the geometric prior, $\mathbb{P}(L = 1) = \rho$ and $\mathbb{P}(L > 1) = (1-\rho)$. As $\rho \downarrow 0$, $\mathbb{P}(L > 1) \rightarrow 1$ so the probability that a new segment starts at any interior index vanishes, which forces a single segment spanning $\{1, \dots, n\}$. As $\rho \uparrow 1$, $\mathbb{P}(L = 1) \rightarrow 1$ and therefore all segments have length one almost surely, giving $t_q = q$. $\square$

A.5 Latent groups: identifiability and label switching

In exchangeable latent-group segmentation formulations, the group labels are arbitrary. When the prior on segmentations and within-group hyperparameters is exchangeable across groups, posterior inference is invariant under permutations of the group labels. This familiar *label switching* phenomenon motivates deterministic reporting conventions for the finite-template EM model of Section 10, even though that model optimizes a finite template-mixture objective rather than sampling a full posterior over all templates.

Proposition A.7 (Permutation invariance). *Assume a G -group latent model with an exchangeable prior over group-specific parameters and segmentations. For every permutation σ of $\{1, \dots, G\}$,*

$$p((t^{(g)}, \psi_g)_{g=1}^G) = p((t^{(\sigma(g))}, \psi_{\sigma(g)})_{g=1}^G).$$

Then the joint posterior over $(t^{(g)}, \psi_g)_{g=1}^G$ and group assignments $z_{1:S}$ satisfies the same invariance. In particular, any MAP assignment is non-unique up to label permutation.

Proof. The likelihood factors by subjects given assignments and group-specific parameters, and the prior is invariant by assumption. Therefore the unnormalized posterior is invariant under simultaneous permutation of group labels in both parameters and assignments. Normalizing preserves invariance. $\square$

Remark A.8. *Permutation invariance does not prevent accurate segmentation or clustering; it only implies that the label names carry no meaning. For reporting, impose an ordering constraint (e.g., by evidence, segment count, or a canonical parameter summary) or post-process posterior samples to align labels. For the EM template mixture, a deterministic label record the ordering and deterministic tie-breaking with the run metadata. The EM algorithm in Section 10 can converge to any one of the symmetric modes.*

A.6 Exploratory note: relation of the latent-template mixture to multi-sequence Bayesian segmentation

This note is a clearly labeled future-direction discussion rather than a completed contribution; it records how the latent-template mixture of Section 10 relates to the closest existing multi-sequence Bayesian changepoint methods, so that the positioning can be developed more fully in a later iteration.

Two lines of work are the natural comparators. The empirical-Bayes method of Fan and Mackey (2017) learns a *shared changepoint prior* across many sequences and infers changepoint indicators by MCMC; the joint random partition models of Quinlan et al. (2024) couple sequence-level partitions through a joint distribution and again rely on posterior simulation. BayesBreak’s latent-template construction differs along three axes that are worth making explicit. First, the latent object is a *discrete group assignment* of each sequence to one of a finite set of templates, rather than a continuous shared-prior parameter. Second, inference for each template is the *exact responsibility-weighted max-sum DP* of Section 10, not an MCMC sampler. Third, the overall fitting procedure is EM on a *finite template-mixture objective* (Theorem 10.1), which trades full posterior averaging over latent-group segmentations for a transparent, monotone alternating optimization.

These differences are design choices, not strict improvements: the template-mixture objective is deliberately narrower than a fully Bayesian posterior over all group segmentations, and the discrete-assignment formulation assumes that sequences cluster into a small number of shared boundary templates. A fuller treatment would (i) characterize the regimes in which the finite-template objective is a good surrogate for the fully Bayesian multi-sequence posterior, and (ii) give a head-to-head empirical comparison against BASIC-style and JRPM-style inference. Both are left as planned work and are recorded in CODING_AGENT_HANDOFF.md (the open-questions item on developing this comparison, alongside the planned external-baseline experiment).

A.7 Annotated literature review

Table 14 records the papers most directly relevant to the present manuscript. The goal is not to be exhaustive over the entire changepoint literature, but to make explicit which threads BayesBreak builds on and where the present contribution sits relative to them.

Table 14 Annotated literature review.

Theme	Representative papers	Annotation relative to BayesBreak
Foundational product partitions	Hartigan (1990) ; Barry and Hartigan (1992, 1993)	Hartigan (1990) introduced partition models in the abstract, and Barry and Hartigan (1992, 1993) specialized them to the product-partition view of Bayesian change-point analysis. BayesBreak adopts exactly this segment-factorization principle, but repackages it around a reusable block-evidence interface that works across multiple observation families and prediction tasks.
Exact offline DP	Chib (1998) ; Fearnhead (2006) ; Hutter (2007) ; Wyse et al. (2011)	These are the closest conceptual ancestors of the exact offline core. Chib (1998) recast multiple changepoints through a hidden-Markov state representation; Fearnhead (2006) emphasizes exact posterior recursion for changepoints; Hutter (2007) highlights Bayesian curve summaries and evidence-based model comparison; and Wyse et al. (2011) pushes the exact-DP viewpoint to within-segment dependence. BayesBreak systematizes these inside a single modular framework and makes the distinction between sum-product summaries and joint MAP segmentations explicit.
Segment dependence and richer local models	Fearnhead and Liu (2011) ; Ruggieri and Lawrence (2014)	These papers show how offline Bayesian segmentation can be enriched beyond i.i.d. block means, for example by allowing dependence across neighboring segments or by fitting piecewise polynomial regression. They motivate the claim that the real bottleneck is the local block integral, not the DP itself.
Frequentist exact partitioning	Auger and Lawrence (1989) ; Jackson et al. (2005) ; Killick et al. (2012) ; Maidstone et al. (2017)	These methods are critical baselines for optimization-based segmentation, ranging from segment-neighborhood and optimal-partitioning recursions to the pruned algorithms PELT and FPOP/SNIP. They solve related partitioning problems efficiently but do not natively provide posterior probabilities over boundary locations or segment counts. BayesBreak differs mainly in inferential target rather than in the use of dynamic programming per se.

Theme	Representative papers	Annotation relative to BayesBreak
Penalized and multiscale segmentation	Tibshirani et al. (2005) ; Bleakley and Vert (2011) ; Frick et al. (2014) ; Fryzlewicz (2014)	These approaches offer strong algorithmic or confidence guarantees, especially in large-scale or multisequence settings. They are natural comparators for planned empirical work, and they optimize penalized objectives or detection rules rather than the posterior quantities emphasized here.
Robust segmentation and the well-log benchmark	Fearnhead and Rigall (2019)	This work studies changepoint detection in the presence of outliers and discusses both the cleaned and raw versions of the well-log NMR benchmark used in Section 15.9.1. It is relevant to the planned real-data pipeline and to robustness considerations for the Gaussian block.
Post-selection inference for changepoints	Jewell et al. (2022)	This work develops valid inference for a change in mean after a changepoint has been detected. It is methodologically complementary to BayesBreak: the Bayesian formulation here reports posterior boundary uncertainty directly, whereas Jewell et al. (2022) provides frequentist post-detection guarantees.
Online and trans-dimensional Bayes	Adams and MacKay (2007) ; Green (1995) ; Denison et al. (1998)	These references represent two alternative Bayesian regimes: streaming detection and fully flexible posterior simulation over model dimension. BayesBreak instead targets exact offline inference when block marginalization is feasible, and deterministic approximations when only the block integral is intractable.
Multi-sequence and hierarchical Bayes	Carlin et al. (1992) ; Fan and Mackey (2017) ; Quinlan et al. (2024)	These papers study information sharing across related sequences. BayesBreak’s exact shared-boundary pooling belongs to this lineage, while the latent-template mixture is a template-constrained, algorithmically transparent alternative to full joint hierarchical models.
Covariate-aware and generalized random partitions	Müller et al. (2011) ; Park and Dunson (2010)	This literature broadens the notion of cohesion or prior similarity between observations. It is the closest conceptual precedent for the design-aware segment-length priors used here, although BayesBreak specializes to contiguous ordered partitions rather than arbitrary clustering.

Theme	Representative papers	Annotation relative to BayesBreak
Deterministic approximations for non-conjugate blocks	Jaakkola and Jordan (2000); Minka (2001); Polson et al. (2013)	These references justify the approximation toolkit in Section 12. The present paper does not claim novel approximation schemes; its contribution is to show how such local block surrogates can be inserted into the same segmentation DP and how their errors propagate to global posterior odds.

A.8 Proof of the max-sum DP correctness (joint MAP segmentation)

We supply here the full correctness proof for the max-sum recursion $M_{k+1,j} = \max_h \{M_{k,h} + \log \tilde{A}_{h,j}^{(0)}\}$ referenced in Theorem 6.5, item 3.

Proposition A.9 (Max-sum correctness). *For every $k \leq k_{\max}$ and $j \geq k$, $M_{k,j}$ equals the maximum, over all admissible k -segmentations of $y_{1:j}$, of $\sum_{q=1}^k \log \tilde{A}_{t_{q-1}, t_q}^{(0)}$. Consequently, for fixed k , $M_{k,n}$ is the optimal block-score sum for that segment count, and backtracking via $h^*(k, j)$ recovers one fixed- k joint MAP boundary vector. When selecting across k , the count-prior and normalizer offset $\log p(k) - \log C_k$ must be added to the terminal score.*

Proof. Induction on k . For $k = 1$, $M_{1,j} = \log \tilde{A}_{0,j}^{(0)}$, which is the only admissible 1-segmentation score. Assume the claim holds for k ; every admissible $(k+1)$ -segmentation of $y_{1:j}$ decomposes uniquely as a k -segmentation of $y_{1:h}$ for some $h \in \{k, \dots, j-1\}$ followed by a terminal block $(h, j]$. By inductive hypothesis the optimal prefix score is $M_{k,h}$, so the optimal total score over $(k+1)$ -segmentations is $\max_h \{M_{k,h} + \log \tilde{A}_{h,j}^{(0)}\} = M_{k+1,j}$. Backtracking recovers the optimum path by following h^* . $\square$

A.9 Variational bounds for non-conjugate blocks

We recall two block-level variational bounds used in Section 12.

Jensen’s inequality for ELBOs. For any density $q(\theta)$ and any block $(i, j]$,

$$\log A_{ij}^{(0)} \geq \int q(\theta) \log \frac{p(y_{i+1:j} \mid \theta) \pi(\theta)}{q(\theta)} d\theta.$$

This is the block ELBO. Optimizing over q in a parametric class (e.g., Gaussian) yields a lower bound on $\log A_{ij}^{(0)}$, hence an upper bound on $-\log A_{ij}^{(0)}$ error.

Jaakkola–Jordan logistic bound. For logistic terms, the bound used in Section 12 is

$$\log \sigma(z) \geq \log \sigma(\xi) + \frac{z - \xi}{2} - \lambda(\xi)(z^2 - \xi^2), \quad \lambda(\xi) = \frac{\tanh(\xi/2)}{4\xi}.$$

Applying this to the signed binomial-logistic linear predictors yields a quadratic surrogate in θ that can be combined with a Gaussian prior to produce closed-form coordinate updates. Iterating on $\{\xi_t\}$ by $\xi_t^* = \sqrt{\mathbb{E}_q[\theta^2]}$ gives monotone improvement in the block ELBO.

A.10 Derivation of Pólya–Gamma mean-field updates

We derive the closed-form updates stated in Proposition 12.5. Starting from (38), consider the mean-field ELBO

$$\mathcal{L}(q_\theta, q_\omega) = \mathbb{E}_q[\log p(y, \theta, \omega)] - \mathbb{E}_q[\log q_\theta(\theta)q_\omega(\omega)].$$

Coordinate ascent over q_θ at fixed q_ω gives $q_\theta^*(\theta) \propto \exp \mathbb{E}_{q_\omega}[\log p(\theta, \omega, y)]$, which reduces to $\exp\{-\frac{1}{2}\theta^2(\rho^{-2} + \sum_t \mathbb{E}_q[\omega_t]) + \theta(\rho^{-2}\nu + \sum_t \kappa_t)\}$, a Gaussian with the stated (μ_q, σ_q^2) . Similarly, $q_\omega^*(\omega_t) \propto \exp \mathbb{E}_{q_\theta}[\log p(\omega_t, \theta, y)] \propto p(\omega_t) \exp(-\frac{1}{2}\mathbb{E}_q[\theta^2]\omega_t)$, which, by the defining exponential tilting of the PG distribution, equals $\text{PG}(b_t, c_t)$ with $c_t^2 = \mathbb{E}_q[\theta^2] = \mu_q^2 + \sigma_q^2$. The mean of $\text{PG}(b, c)$ is $\frac{b}{2c} \tanh(c/2)$. Monotonicity follows because each coordinate update maximizes its respective factor of $\mathcal{L}$.

A.11 Complexity proofs

The complexity claims in Theorem 6.7 were proved by counting operations; we repeat here the argument for the EM-with-pooled-blocks combination.

Proposition A.10 (EM + pooled-block complexity). *Let S be the number of subjects and G the number of latent groups. One EM iteration of Algorithm 3 runs in time $\mathcal{O}(SGk_{\max} + GSn^2 + Gk_{\max}n^2)$ after a one-time precomputation of per-subject log-block evidences costing $\mathcal{O}(Sn^2)$. If responsibility-weighted score matrices are formed in a streamed fashion, the GSn^2 term can be traded for additional passes over cached subject matrices. Memory is $\mathcal{O}(Sn^2 + Gn^2 + Gk_{\max}n)$ under full score storage and can be reduced by streaming at the cost of recomputation.*

Proof. The E-step evaluates G template scores per subject, each requiring at most $k_{\max}$ block lookups, giving $\mathcal{O}(SGk_{\max})$. The M-step forms G responsibility-weighted score matrices; with cached subject block-evidence matrices this costs $\mathcal{O}(GSn^2)$ in the straightforward implementation. Each group then runs a max-sum DP in $\mathcal{O}(k_{\max}n^2)$. The stated memory bound separates per-subject block storage, group score storage, and DP/backpointer storage. $\square$

A.12 Code and reproduction

A reference Python implementation of BayesBreak mirrors the paper (block routines, DP layer, plate-model variants, and real-data pipelines). The synthetic commands below correspond to completed archived runs; the real-data commands are planned recipes to verify before they replace placeholder figures and tables. Suggested installation and invocation:

```
git clone https://github.com/osolari/bayesbreak.git
cd bayesbreak
python -m venv .venv && source .venv/bin/activate
```

```

pip install -e "[all]"
# Regenerate synthetic experiments (Section 6, bundled suite):
python -m bayesbreak.experiments.synthetic --all --out ./out_synth
# Regenerate real-data experiments (requires network; see real-data appendix):
python -m bayesbreak.experiments.realdata --dataset wellog --out ./out_real
python -m bayesbreak.experiments.realdata --dataset cgh --out ./out_real
python -m bayesbreak.experiments.realdata --dataset spx --out ./out_real
python -m bayesbreak.experiments.realdata --dataset methyl --out ./out_real

```

When reproducing published numbers, pin the commit hash, dependency versions, random seeds, hyperparameters, and data-cache hashes used for a given manuscript version. The pipelines in Appendix A.13 assume either a local installation of the package or sufficient environment access to run the listed R/Bioconductor and Python download commands.

A.13 Real-data sources, preprocessing, and reproduction notes

This appendix provides data-source targets, download commands, and preprocessing recipes for the four real-data illustrations in Section 15.9. Each subsection is organized so that a reader with a shell, Python ≥ 3.10 , and R ≥ 4.3 can regenerate the corresponding body figure and table after source names, access paths, and annotation labels are verified. The commands below define the planned pipelines that populate the placeholder tables.

Real-data verification checklist. Before replacing any placeholder, record download success, raw-data hash, preprocessing script and hash, hyperparameters, random seed, generated figure path, generated table path, and any external comparison-label source used for boundary metrics.

A.13.1 Well-log NMR segmentation

Source. The well-log NMR dataset was introduced by Ó Ruanaidh and Fitzgerald (1996) and contains 4050 successive nuclear-magnetic-resonance measurements recorded down a petroleum borehole; it was popularized as a Bayesian segmentation benchmark by Fearnhead and Clifford (2003) (where it appears as the canonical worked example) and subsequently by Fearnhead (2006) and Adams and MacKay (2007). A copy ships with the R package `changepoint.influence` as the object `welldata` (a numeric vector of length 4050), with the package documentation citing Fearnhead and Clifford (2003) as the source of the bundled copy. The cleaned and raw versions of the series are discussed by Fearnhead and Rigaiil (2019).

Fetch & preprocess (R, recommended).

```

# Install if needed
install.packages("changepoint.influence",
                 repos="https://cloud.r-project.org")
library(changepoint.influence)
# The bundled welldata vector is the 4050-point well-log NMR series,

```

```

# documented as the data of Fearnhead & Clifford (2003).
# NOTE: do NOT use data(Lai2005fig4) from the 'changeoint' package --
#       that is the array-CGH example of Lai et al. (2005),
#       not the well-log NMR series.
data(welldata)
stopifnot(length(welldata) == 4050)
y <- as.numeric(welldata)
writeLines(as.character(y), "wellog.txt")
# then standardize:
yS <- (y - mean(y)) / sd(y)
writeLines(as.character(yS), "wellog_standardized.txt")

```

Alternative fetch (Python). If `changeoint` is unavailable, there is an equivalent copy shipped with `ruptures`:

```

pip install ruptures
python -c "import ruptures; import numpy as np;
signal, _ = ruptures.load_cost('wellog.csv');
np.savetxt('wellog.txt', signal)"

```

BayesBreak settings. Block model: Gaussian with known variance, $w_t \equiv 1$. Hyperparameters: $\nu = 0$, $\rho^2 = 100$ (weak), $\sigma^2 =$ robust within-layer estimate (median absolute deviation scaled to a Gaussian). Partition prior: $g \equiv 1$ (index uniform) and $g(\ell) \propto \ell$ (length aware) as contrast. $k_{\max} = 30$. Prior on k : $p(k) \propto 1$.

Outputs for Figure 11 and Table 10. Joint MAP segmentation, $P(k \mid y)$, marginal boundary posterior, ECE against visually obvious changeoints (labeled in the original Fearnhead paper), runtime in seconds. Expected diagnostic outcome: a handful of strong changeoints with tight boundary marginals, $\hat{k} \in \{10, 15\}$ depending on the prior.

A.13.2 Array-CGH multi-subject copy-number

Source. The Coriell cell-line array-CGH panel from [Snijders et al. \(2001\)](#) is bundled with the Bioconductor package `DNACopy`. Each sample is a log-2 copy-number-ratio profile at ~ 2000 autosomal probes across 15 cell lines.

Fetch & preprocess (R via Bioconductor).

```

if (!require("BiocManager", quietly=TRUE))
  install.packages("BiocManager", repos="https://cloud.r-project.org")
BiocManager::install("DNACopy", update=FALSE, ask=FALSE)
library(DNACopy)
data(coriell)
# coriell columns: Clone, Chromosome, Position, Coriell.05296, Coriell.13330
# Select one chromosome per subject for a tractable benchmark:
subs <- c("Coriell.05296", "Coriell.13330")

```

```

for (chr in c(1,4,11,17)) {
  block <- coriell[coriell$Chromosome==chr, c("Position", subs)]
  block <- block[complete.cases(block),]
  write.csv(block, paste0("cgh_chr", chr, ".csv"), row.names=FALSE)
}

```

Per-probe precision estimation. For each sample s and chromosome, fit a robust local-variance estimate using a rolling MAD on residuals from a piecewise-constant smoother; invert to get $w_t^{(s)} = 1/\widehat{\sigma}_t^{2,(s)}$.

BayesBreak settings. Heteroscedastic Gaussian block with known $\sigma^2 \equiv 1$ absorbed into w_t . Multi-subject pooling with shared boundaries (Assumption 8.1). Subject-specific priors $\mu \sim \mathcal{N}(0, 100)$. $k_{\max} = 20$ per chromosome. Index-uniform prior.

Outputs for Figure 12 and Table 11. Per-subject exported segmentations vs. pooled shared-boundary segmentation; boundary F1 against Snijders 2001 annotations; pooled log-evidence for independent vs. shared-boundary models. Expected diagnostic outcome: shared-boundary pooling yields higher boundary F1 when the true boundaries are shared.

A.13.3 S&P 500 volatility regimes

Source. Daily closing prices from Yahoo Finance via `yfinance` or `pandas-datareader`.

Fetch & preprocess (Python).

```

pip install yfinance pandas numpy
python - <<'PY'
import yfinance as yf, pandas as pd, numpy as np
df = yf.download("^GSPC", start="2018-01-01", end="2022-12-31",
                auto_adjust=True, progress=False)
df = df[["Close"]].dropna()
log_ret = np.log(df["Close"]).diff().dropna()
log_sq = np.log(log_ret**2 + 1e-12) # log-squared returns
log_sq.to_csv("spx_logsq.csv")
# Alternative: threshold crossings at tau = 1 daily sd
tau = log_ret.std()
crossings = (log_ret.abs() > tau).astype(int)
crossings.to_csv("spx_crossings.csv")
PY

```

BayesBreak settings. (A) Gaussian-with-known-variance block on $z_t = \log r_t^2$ after centering; $w_t \equiv 1$. Known σ^2 from a stable subperiod. Weak Gaussian prior on μ . (B) Poisson-with-exposure block on daily threshold-crossing counts aggregated into weekly bins. Both calls use index-uniform prior, $k_{\max} = 15$.

Outputs for Figure 13 and Table 12. Exported joint MAP with boundary marginals for each block model; qualitative alignment of top-5 posterior boundaries with annotated macro events (COVID-19 onset in March 2020, Feb 2022).

A.13.4 CpG atlas DNA methylation

Source. The DNA methylation atlas of Loyfer et al. (2023) provides processed methylation calls with per-CpG coverage. The raw whole-genome bisulfite sequencing data are deposited at NCBI GEO under accession GSE186458. The companion software is split across two repositories from the same group: nloyfer/wgbs_tools (the wgbstools suite used to build the atlas) and nloyfer/UXM_deconv (the UXM fragment-level deconvolution tool whose README cites Loyfer et al. (2023) directly). The older repository nloyfer/meth_atlas implements the array-based deconvolution method of Moss et al. (2018), *Nature Communications* 9:5068, and is *not* the code base for the 2023 atlas paper. *Author-verification task:* the finalized pipeline should re-verify the GEO accession and the current repository paths against the paper’s data-availability statement at run time.

Fetch & preprocess (bash + Python).

```
# Option A: GEO accession (canonical raw source for the 2023 atlas)
# Replace <ftp_path> with the directory listed at:
# https://www.ncbi.nlm.nih.gov/geo/query/acc.cgi?acc=GSE186458
# wget "https://ftp.ncbi.nlm.nih.gov/geo/series/GSE186nnn/GSE186458/..."

# Option B: Build/process atlas tables using wgbstools from the authors:
git clone https://github.com/nloyfer/wgbs_tools.git
# Follow the README to install and to assemble per-CpG beta and coverage
# matrices from the GSE186458 samples. The atlas tables produced will be:
# atlas.tsv.gz (rows = CpGs; cols = cell types; values = beta)
# coverage.tsv.gz (rows = CpGs; cols = cell types; values = counts)
```

Then subset to a single chromosome-arm region and two cell types:

```
python - <<'PY'
import pandas as pd, numpy as np, gzip
atlas = pd.read_csv("atlas.tsv.gz", sep="\t", index_col=0, compression="gzip")
cov = pd.read_csv("coverage.tsv.gz", sep="\t", index_col=0, compression="gzip")
region = "chr1:200000000-300000000" # pick any atlas-annotated region
mask = atlas.index.str.contains(region.split(":")[0]+"::")
A = atlas.loc[mask]; C = cov.loc[mask]
# clamp beta away from 0,1 for numerical stability:
eps = 1e-4
A = A.clip(eps, 1-eps)
# per-CpG precision: use coverage directly (phi proportional to coverage)
for ct in ["T_cell", "Hepatocyte"]:
    out = pd.DataFrame({"beta": A[ct], "phi": C[ct]}).dropna()
    out.to_csv(f"methyl_{ct}.csv")
```

PY

BayesBreak settings. Beta-response block (§11) with per-CpG precision ϕ_t from coverage. Quadrature order $G = 64$ Gauss–Legendre nodes on $\mu \in [\epsilon, 1 - \epsilon]$. Prior $\mu \sim \text{Beta}(1, 1)$. $k_{\max} = 25$. Index-uniform partition prior.

Outputs for Figure 14 and Table 13. Per-region-per-cell-type exported segmentation, marginal boundary posterior, held-out log-predictive on a random 20% held-out CpG subset within each region, and boundary F1 against the atlas block annotations. Expected diagnostic outcome: sharp transitions aligned with known methylation block boundaries in housekeeping regions; more diffuse posterior in variable regions.

Reproducibility note. The four pipelines are designed to be lightweight; report runtime from the finalized environment. Version or hash intermediate data caches to detect upstream changes in atlas or yfinance releases.